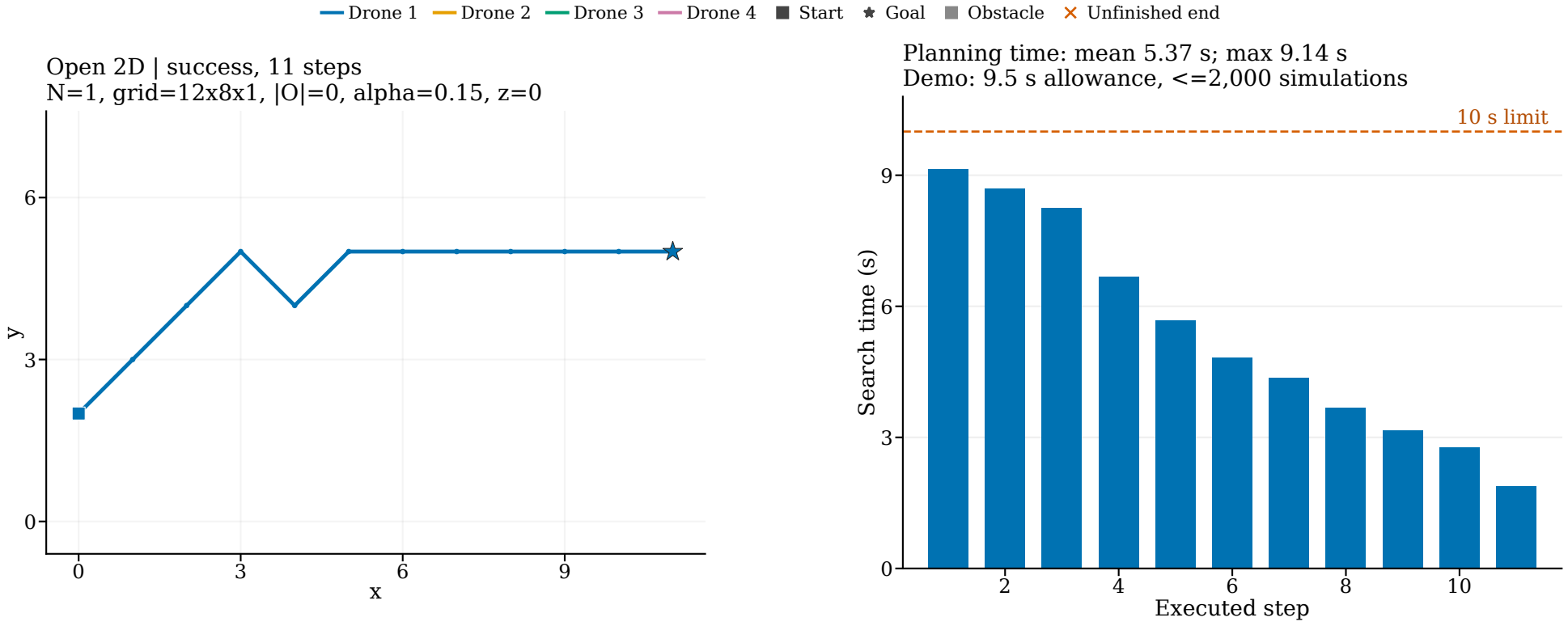
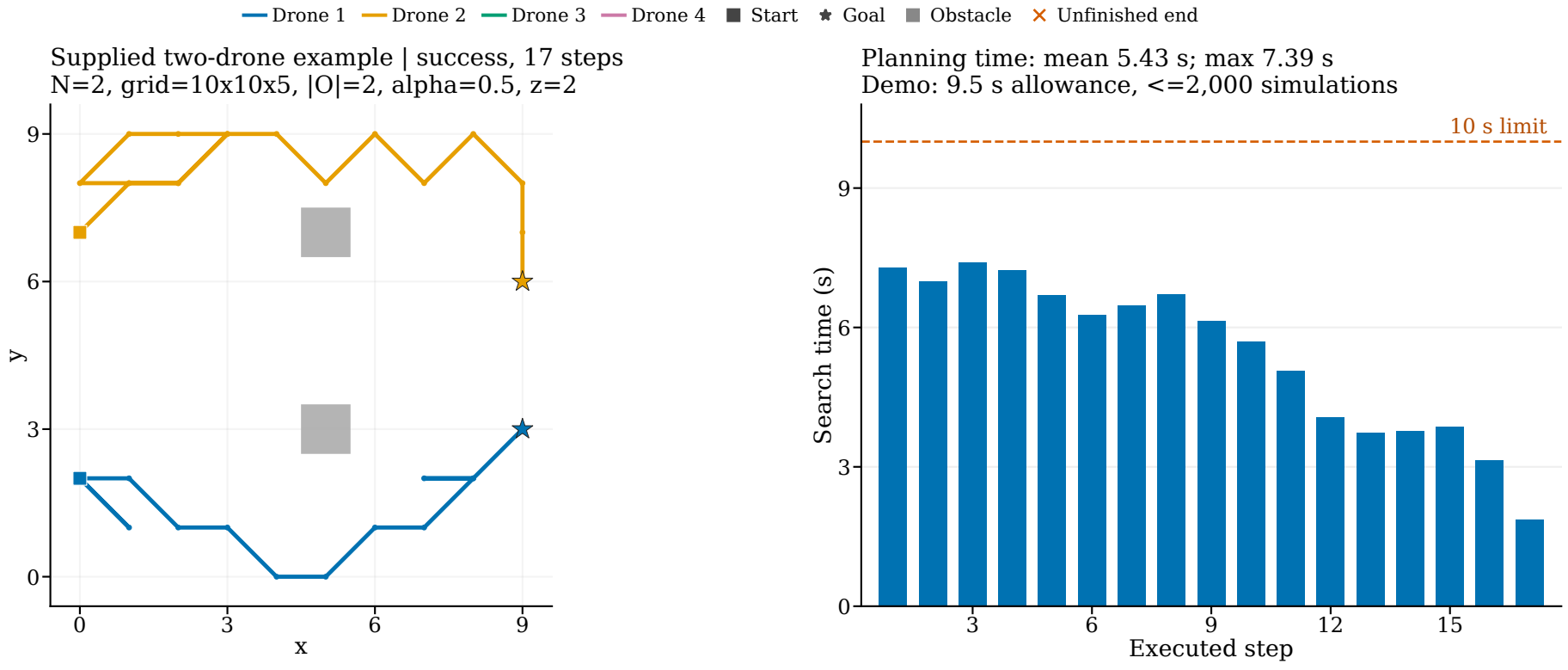
demo open 2d



Saved full-solver trace, seed 0: results/demo_open_2d_default.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

example config

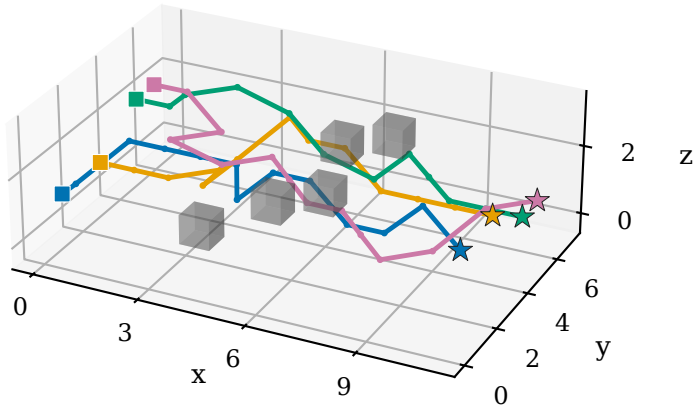


Saved full-solver trace, seed 0: results/example_config_default.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

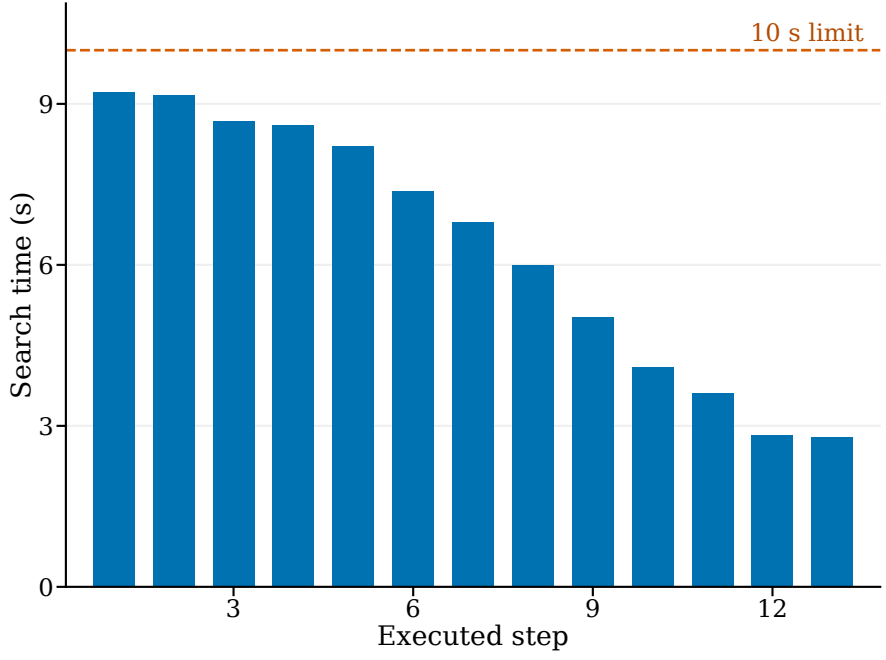
demo four 3d

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

Four drones in 3D | success, 13 steps
N=4, grid=12x8x4, |O|=5, alpha=0.2

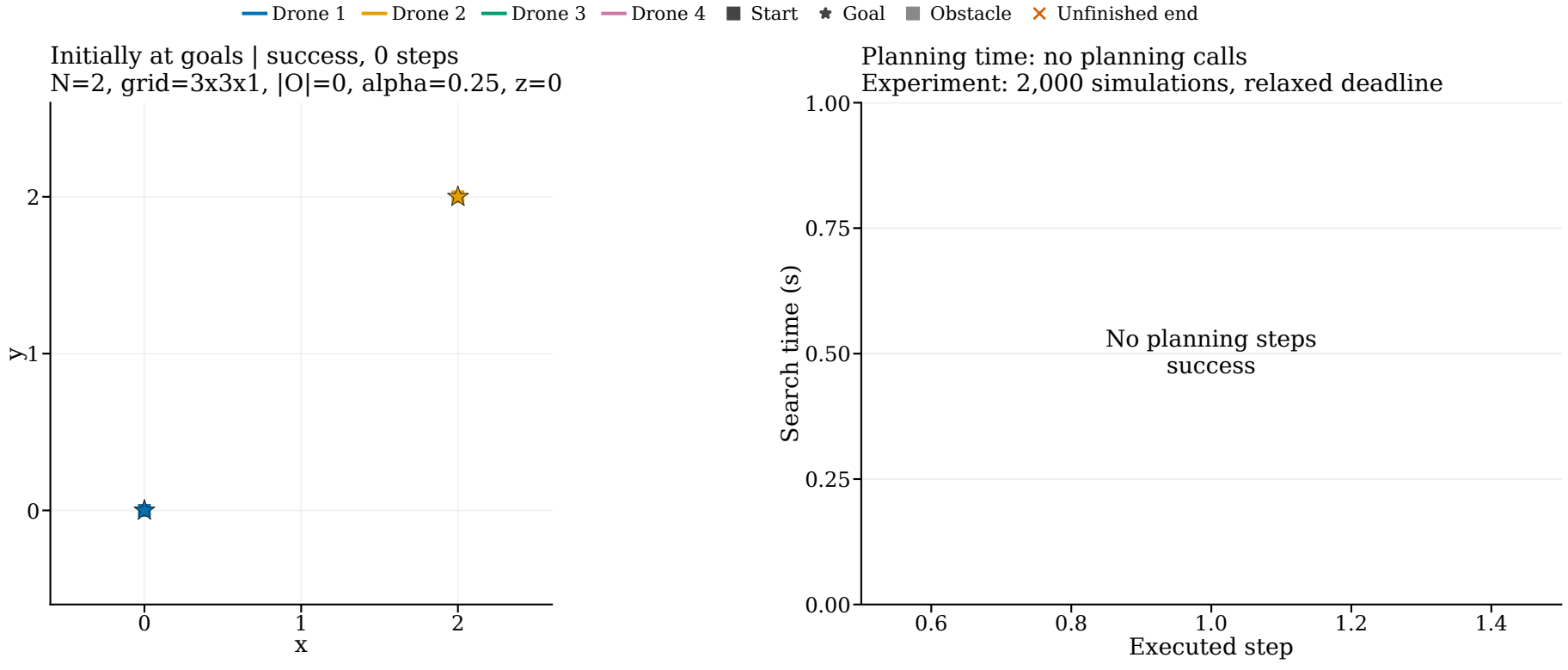


Planning time: mean 6.34 s; max 9.21 s
Demo: 9.5 s allowance, <=2,000 simulations

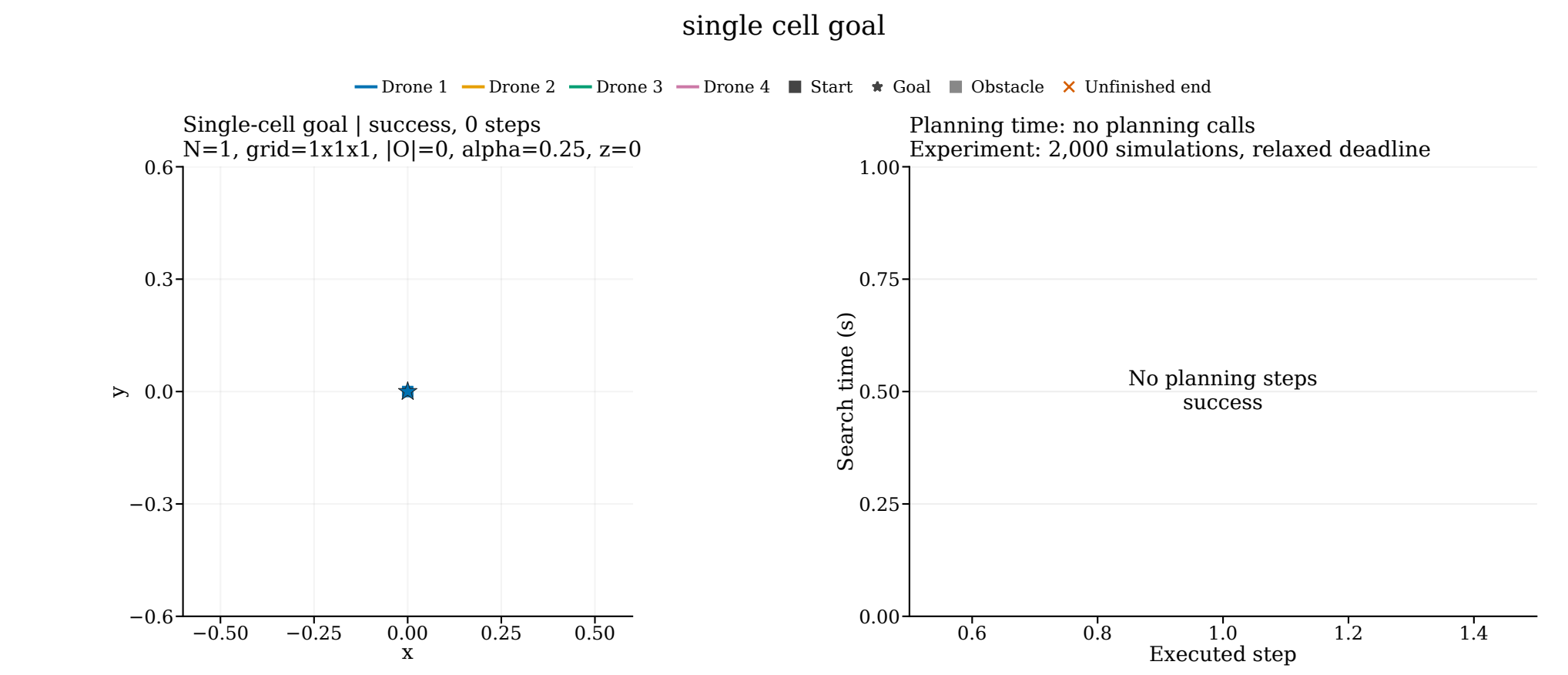


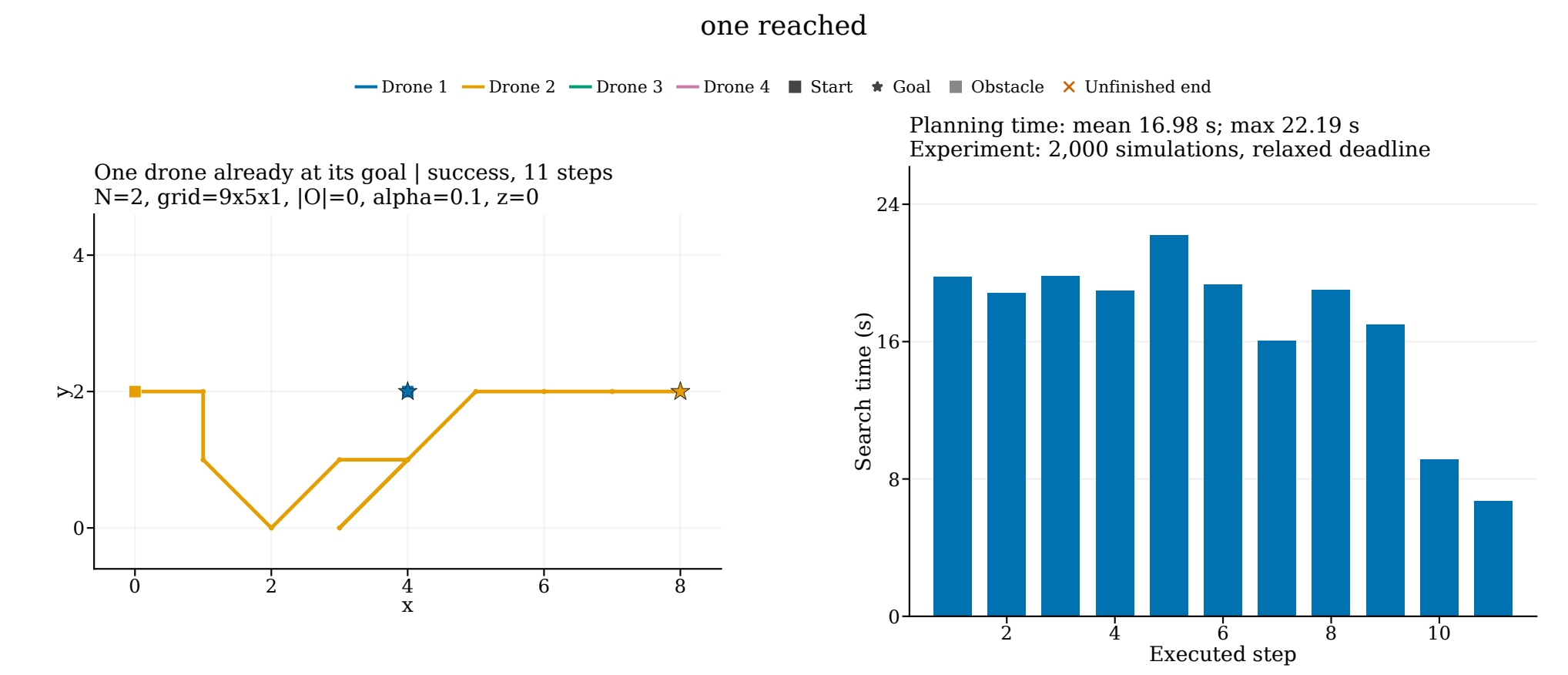
Saved full-solver trace, seed 0: results/demo_four_3d_default.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

already at goals

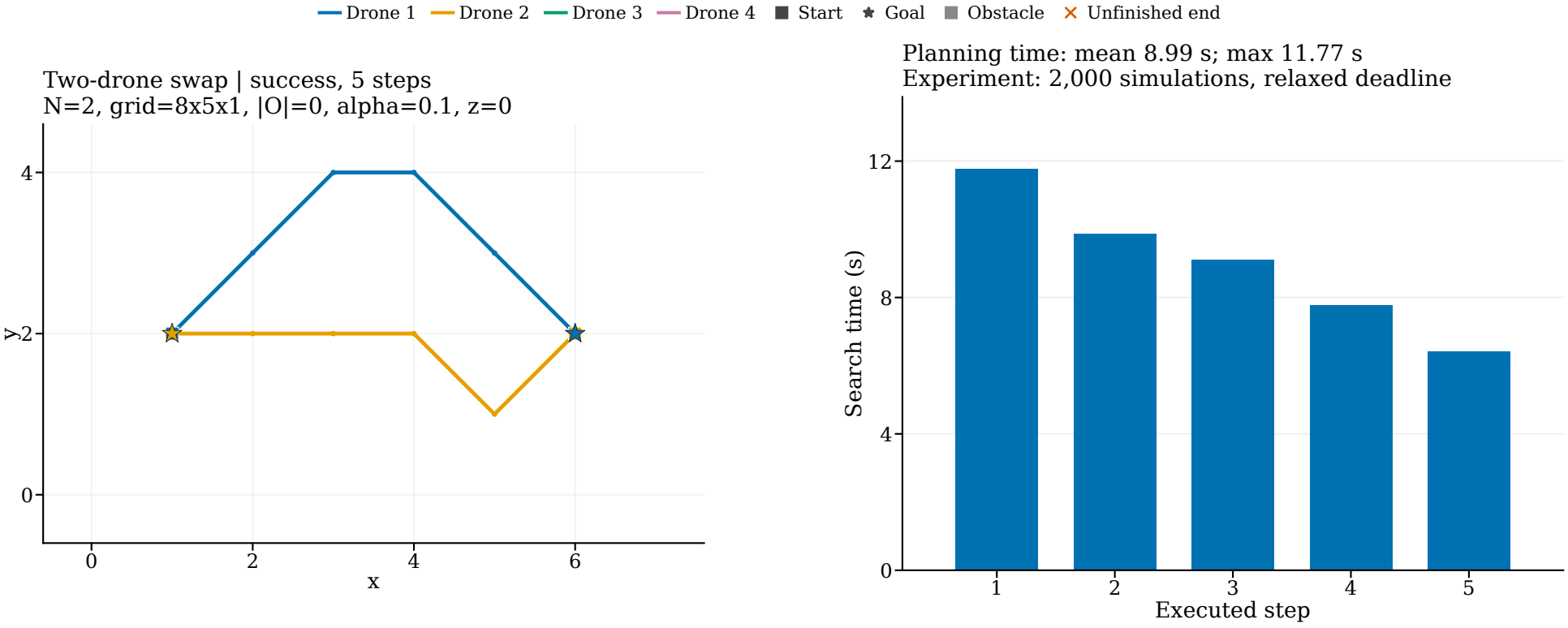


Saved full-solver trace, seed 0: results/trajectories/regression_already_at_goals_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.


single cell goal
Drone 1
Drone 2
Drone 3
Drone 4
Start
Goal
Obstacle
Unfinished end
Single-cell goal | success, 0 steps
N=1, grid=1x1x1, |O|=0, alpha=0.25, z=0
y
x
0.6
0.3
0.0
−0.3
−0.6
−0.50
−0.25
0.00
0.25
0.50
Planning time: no planning calls
Experiment: 2,000 simulations, relaxed deadline
Search time (s)
Executed step
1.00
0.75
0.50
0.25
0.00
0.6
0.8
1.0
1.2
1.4
No planning steps
success
Saved full-solver trace, seed 0: results/trajectories/regression_single_cell_goal_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

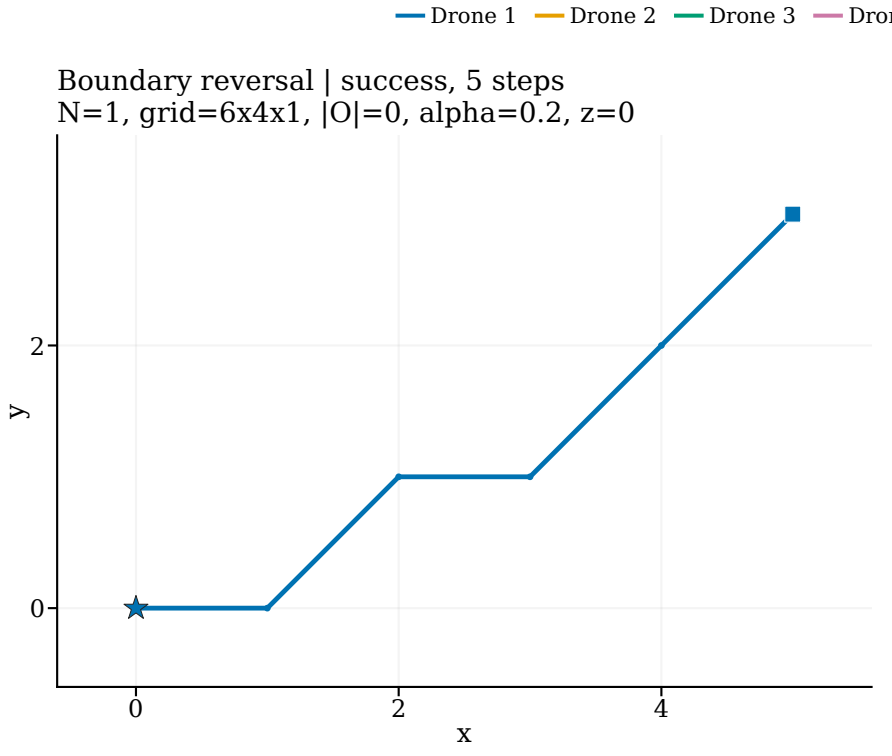

one reached
Drone 1
Drone 2
Drone 3
Drone 4
Start
Goal
Obstacle
Unfinished end
One drone already at its goal | success, 11 steps
N=2, grid=9x5x1, |O|=0, alpha=0.1, z=0
y
x
0
2
4
6
8
Planning time: mean 16.98 s; max 22.19 s
Experiment: 2,000 simulations, relaxed deadline
Search time (s)
Executed step
0
8
16
24
2
4
6
8
10
Saved full-solver trace, seed 0: results/trajectories/regression_one_reached_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

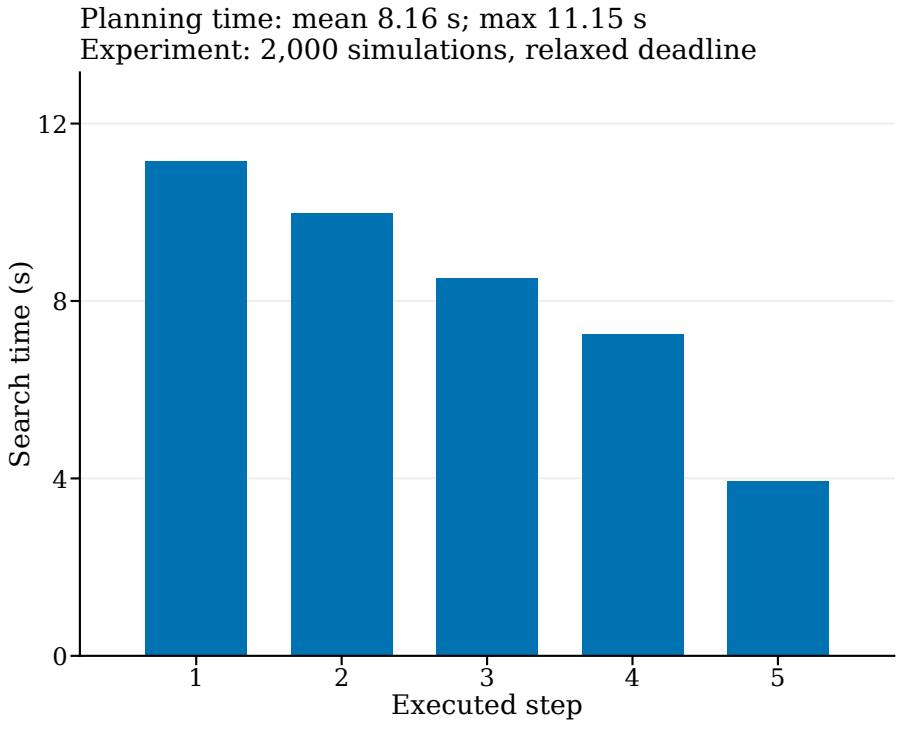
swap 2d



Saved full-solver trace, seed 0: results/trajectories/regression_swap_2d_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

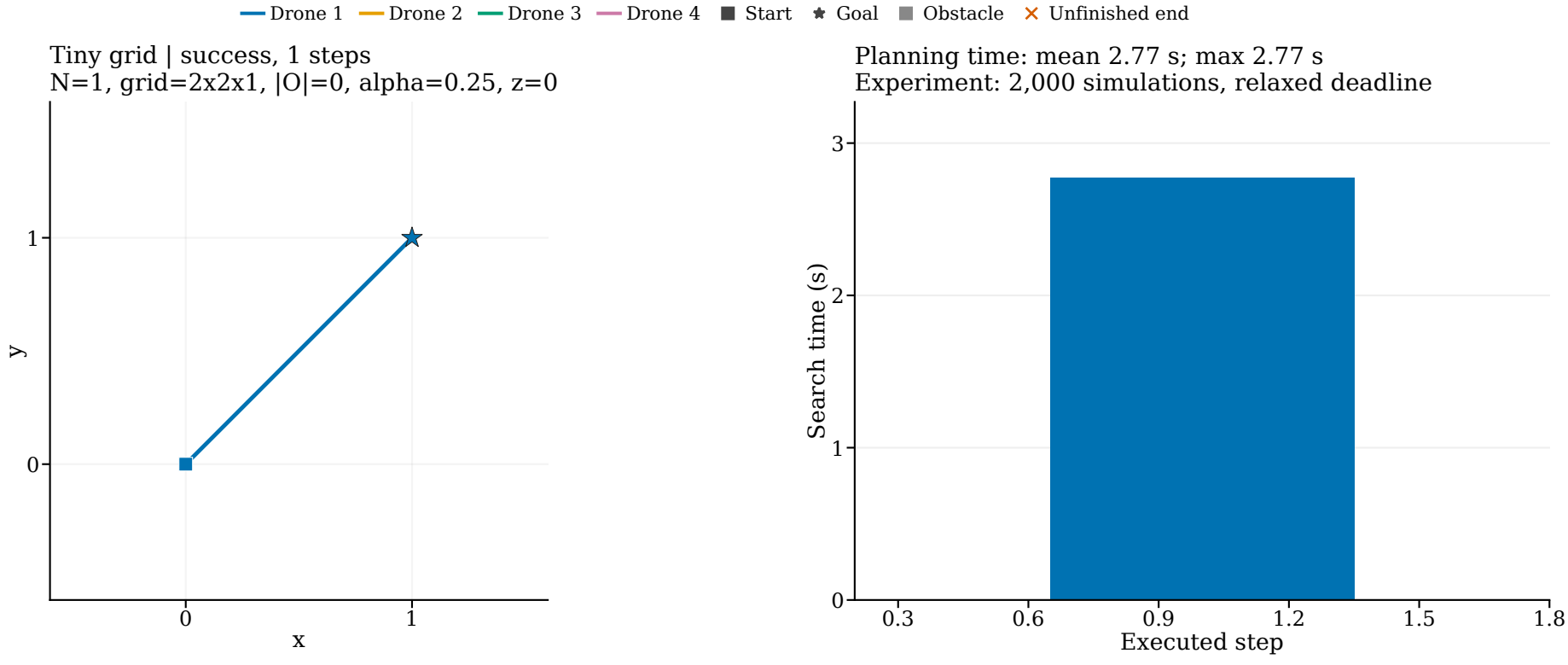
boundary reverse





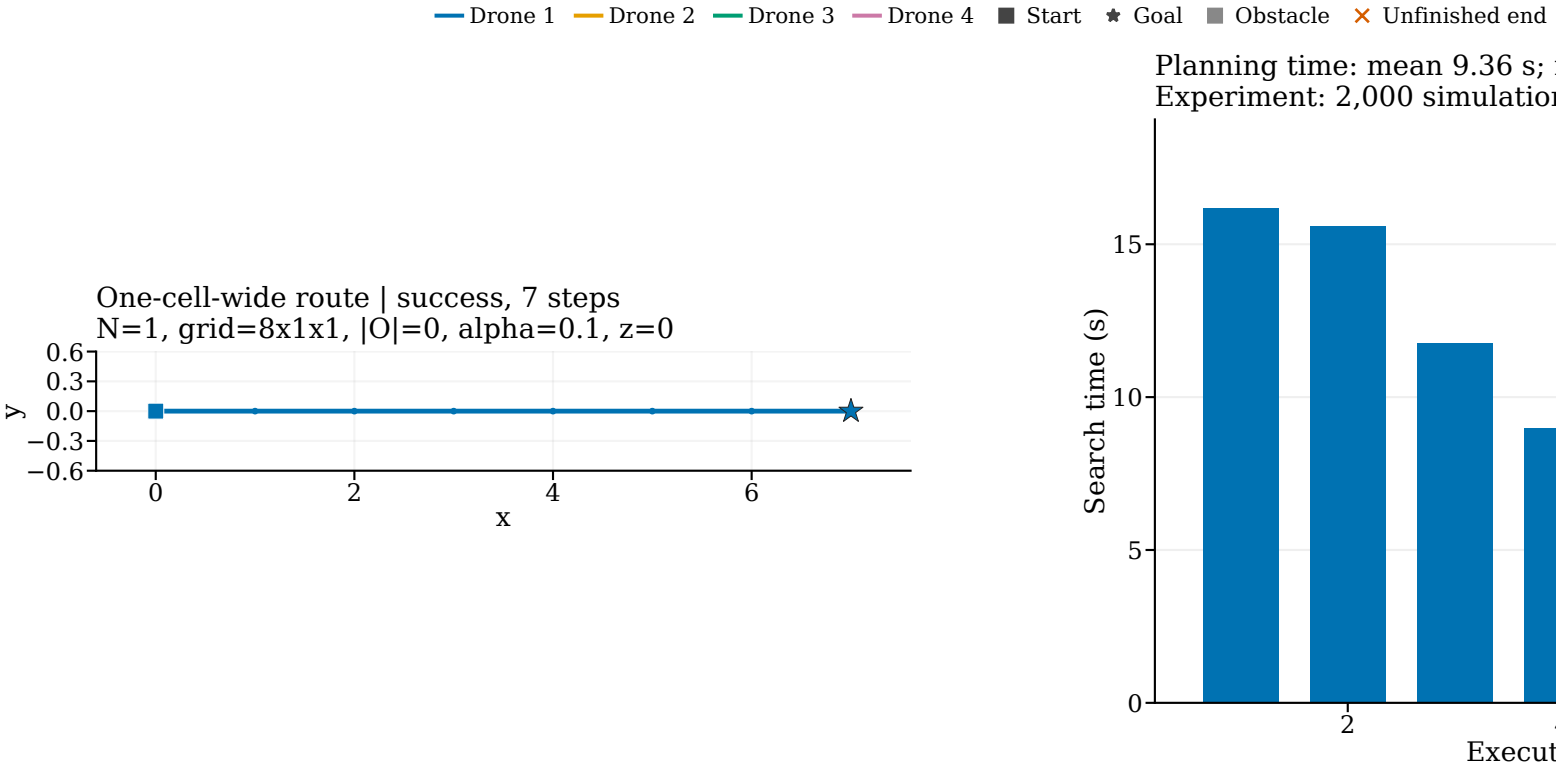
Saved full-solver trace, seed 0: results/trajectories/regression_boundary_reverse_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

tiny 2d



Saved full-solver trace, seed 0: results/trajectories/regression_tiny_2d_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

thin 1d

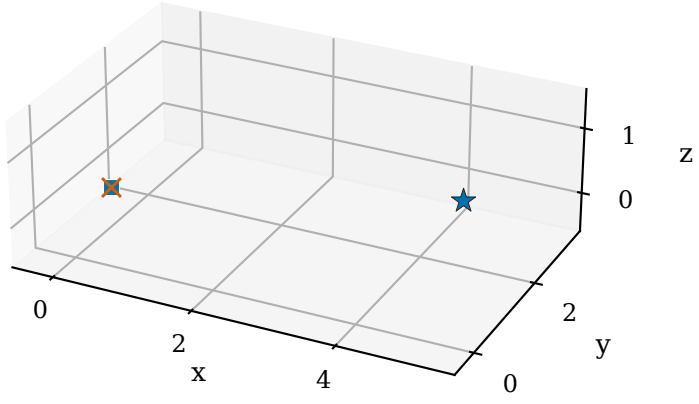


Saved full-solver trace, seed 0: results/trajectories/regression_thin_1d_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

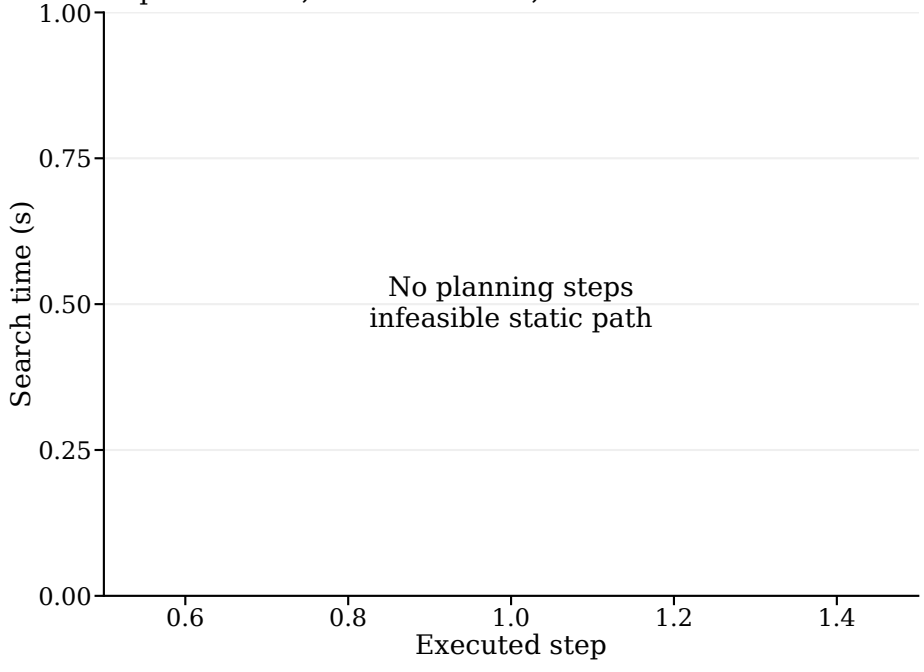
infeasible altitude

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

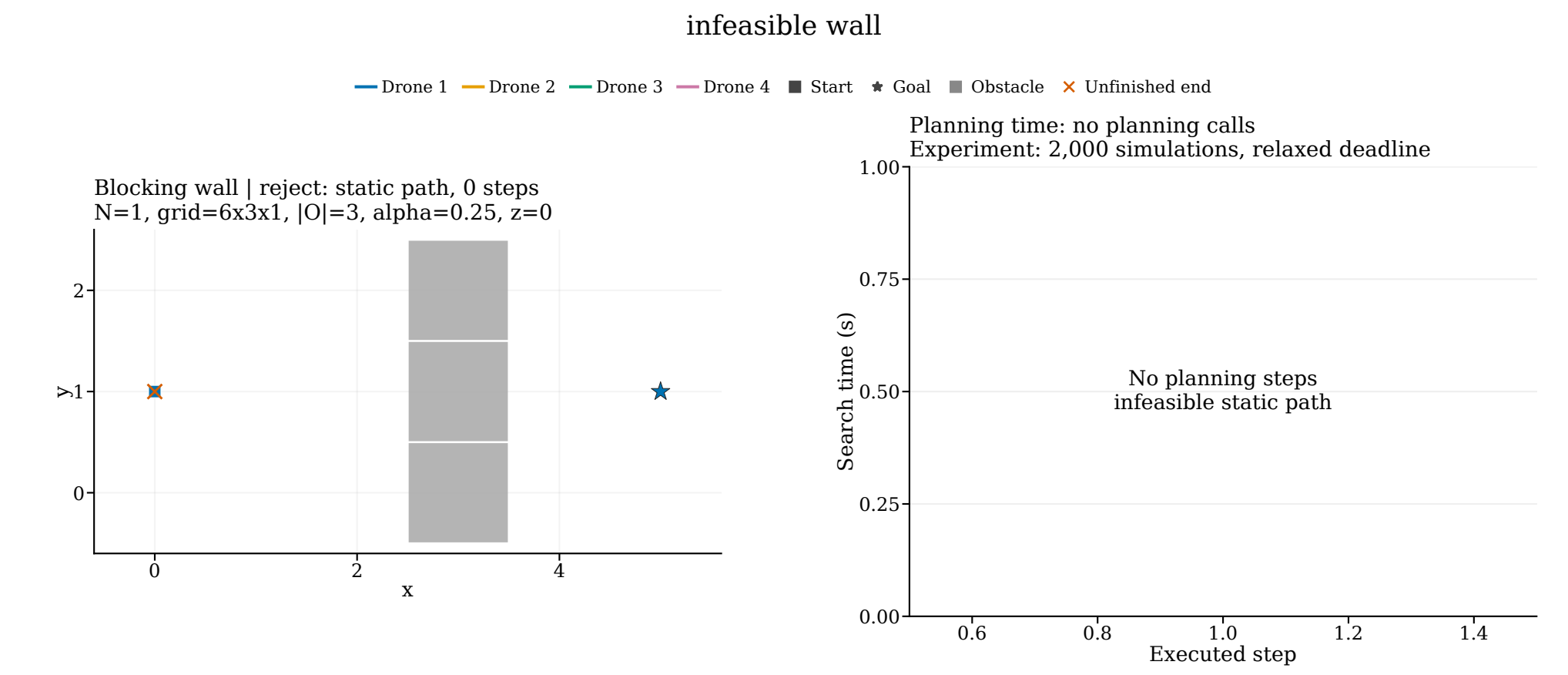
Unreachable altitude | reject: static path, 0 steps
N=1, grid=6x4x2, |O|=0, alpha=0.25



Planning time: no planning calls
Experiment: 2,000 simulations, relaxed deadline

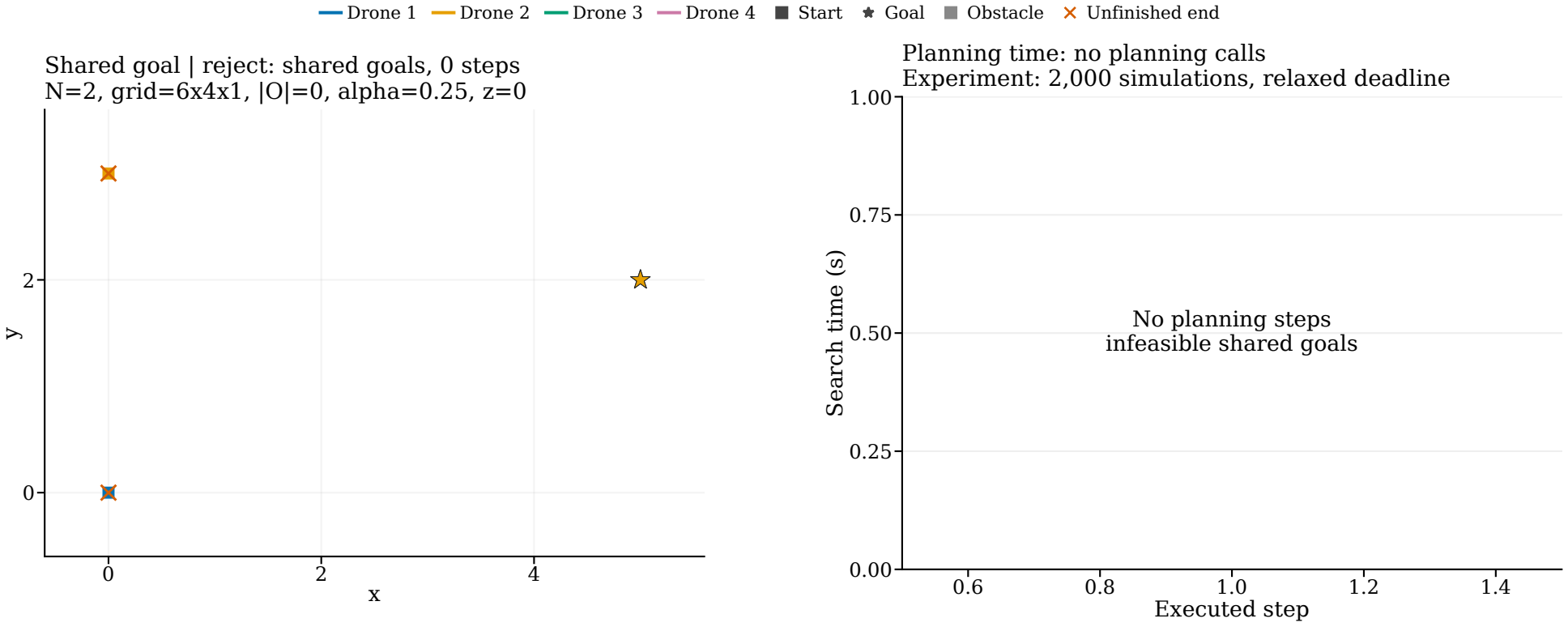


Saved full-solver trace, seed 0: results/trajectories/regression_infeasible_altitude_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.



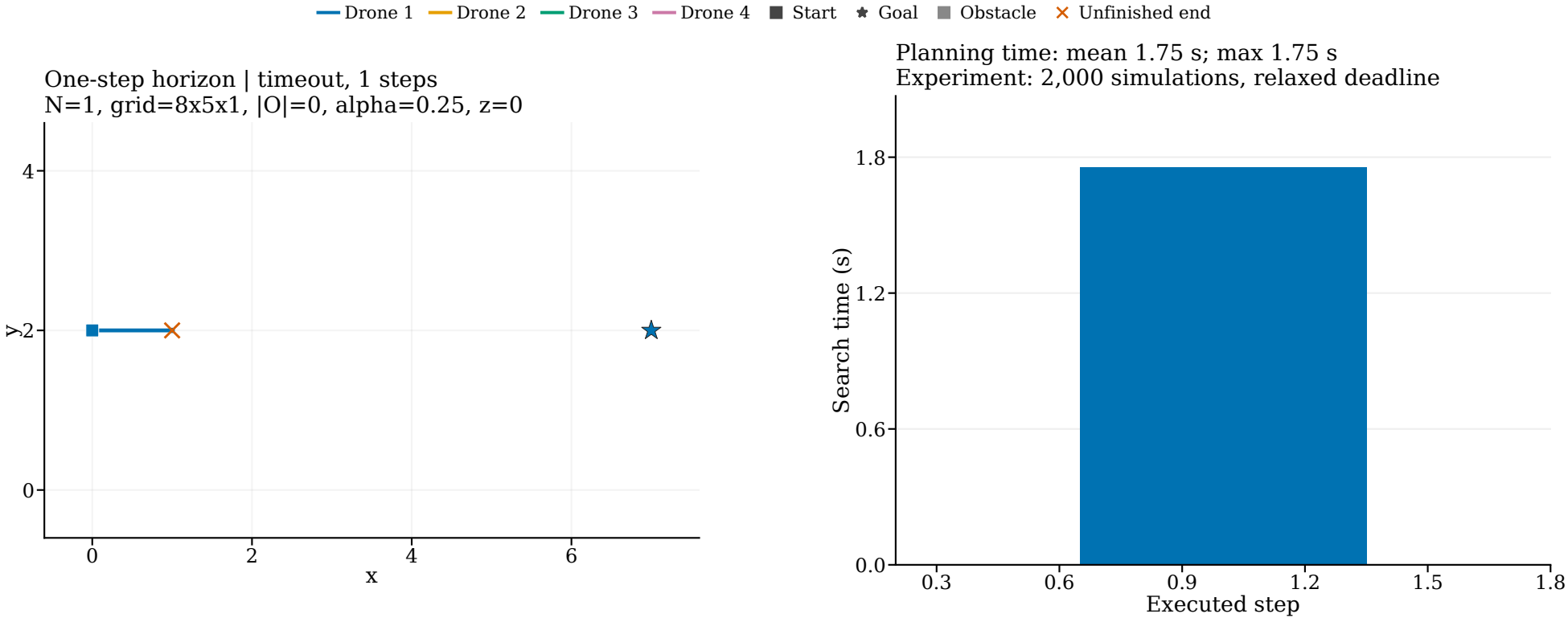
Saved full-solver trace, seed 0: results/trajectories/regression_infeasible_wall_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

infeasible shared goal



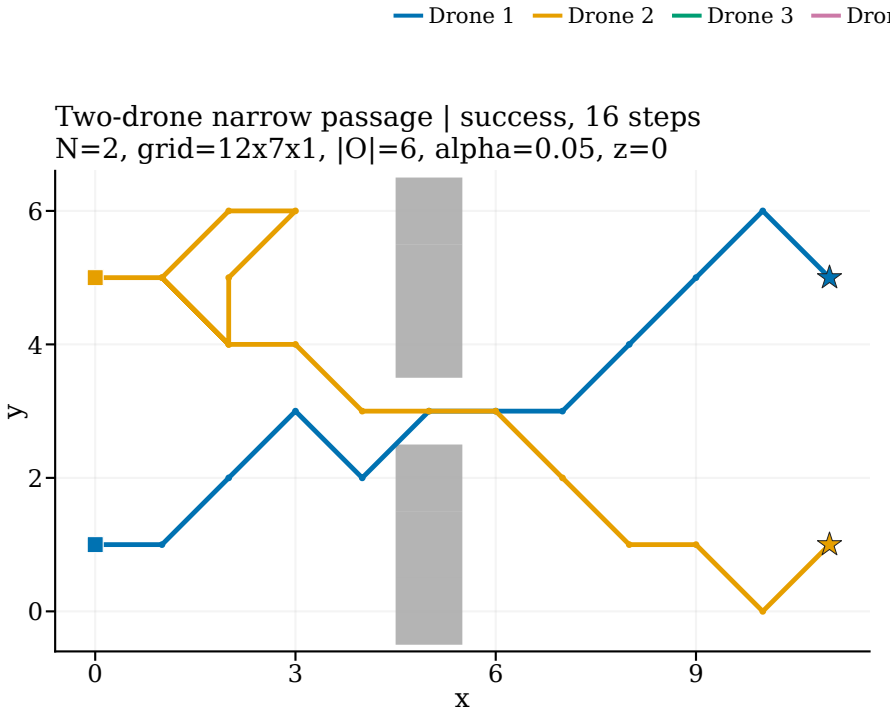
Saved full-solver trace, seed 0: results/trajectories/regression_infeasible_shared_goal_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

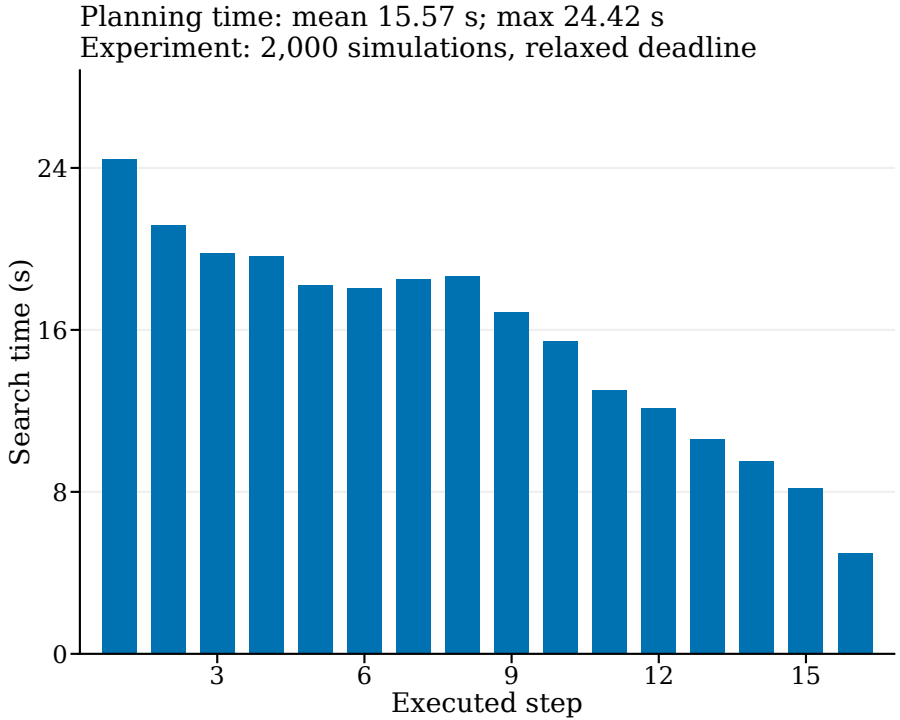
horizon one



Saved full-solver trace, seed 0: results/trajectories/regression_horizon_one_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

narrow passage



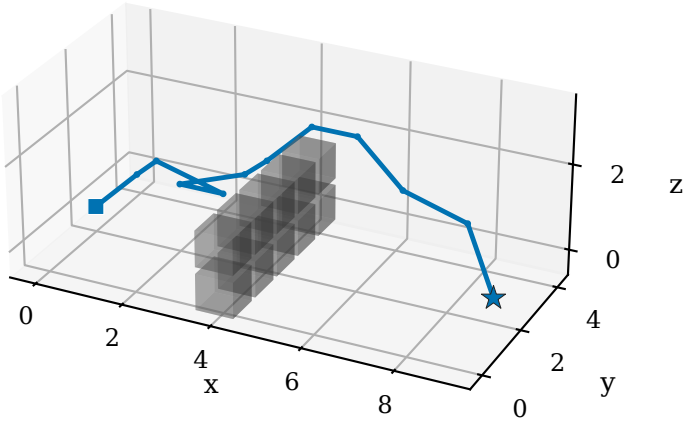


Saved full-solver trace, seed 0: results/trajectories/regression_narrow_passage_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

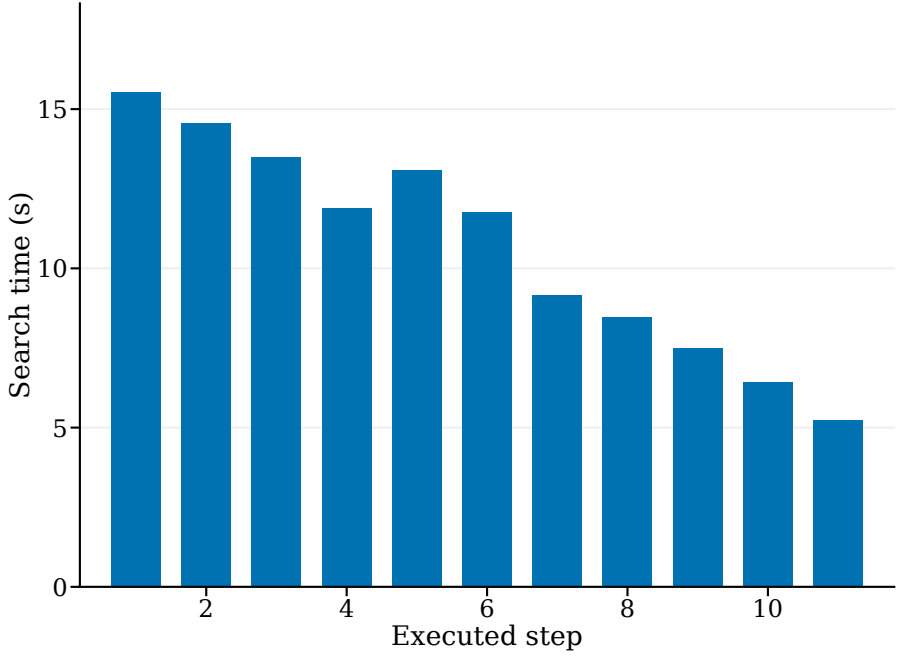
altitude detour

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

Altitude detour | success, 11 steps
N=1, grid=10x5x4, |O|=10, alpha=0.1

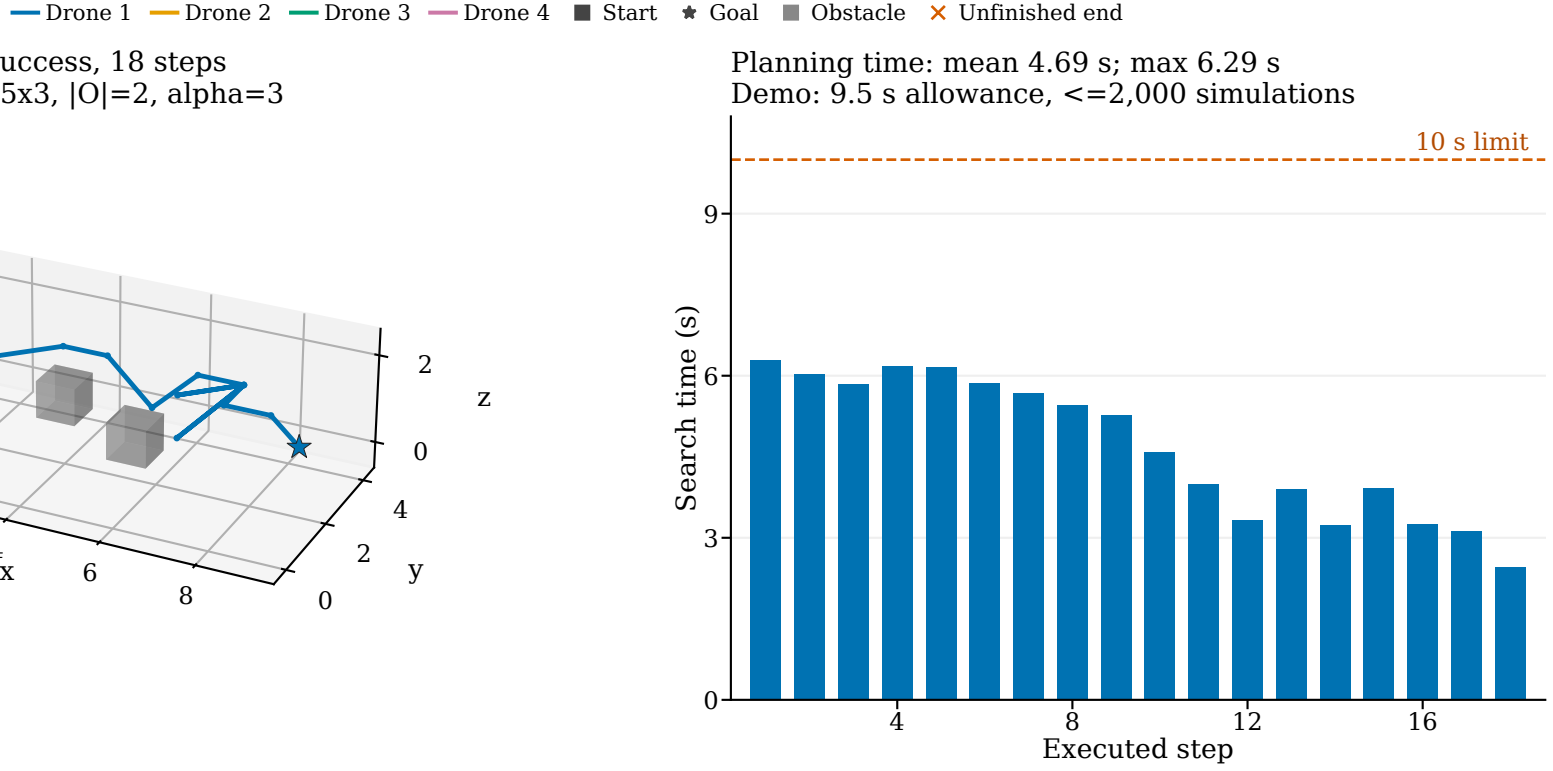


Planning time: mean 10.64 s; max 15.52 s
Experiment: 2,000 simulations, relaxed deadline



Saved full-solver trace, seed 0: results/trajectories/regression_altitude_detour_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

strong wind

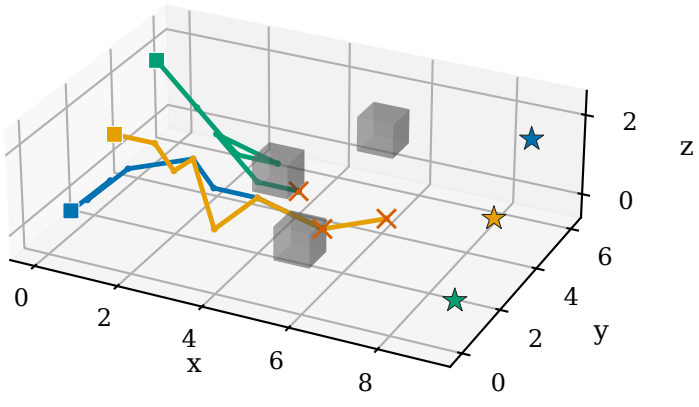


Saved full-solver trace, seed 0: results/strong_wind_default.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

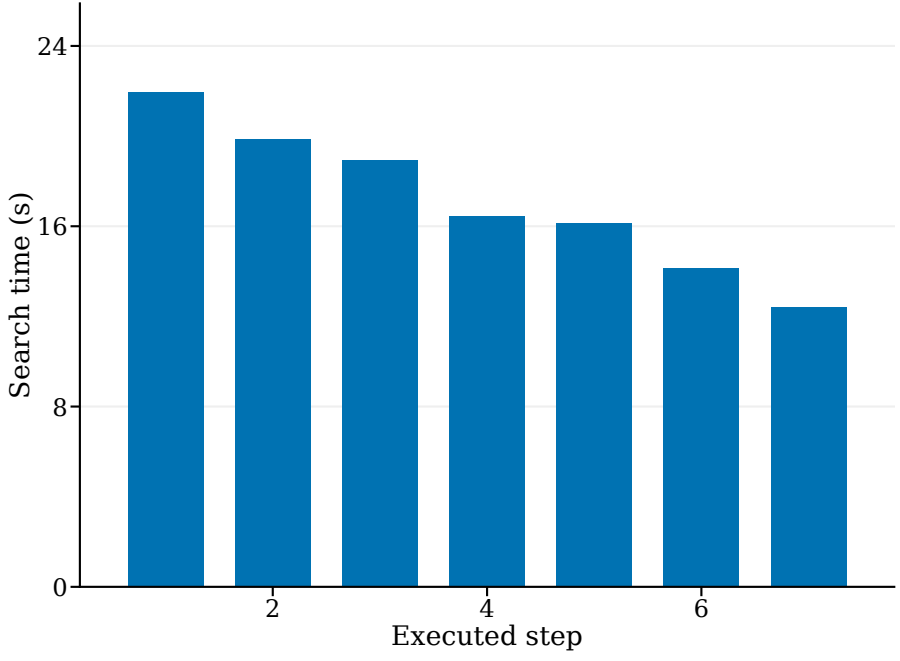
three 3d

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

Three drones: collision | collision, 7 steps
N=3, grid=10x7x3, |O|=3, alpha=0.15

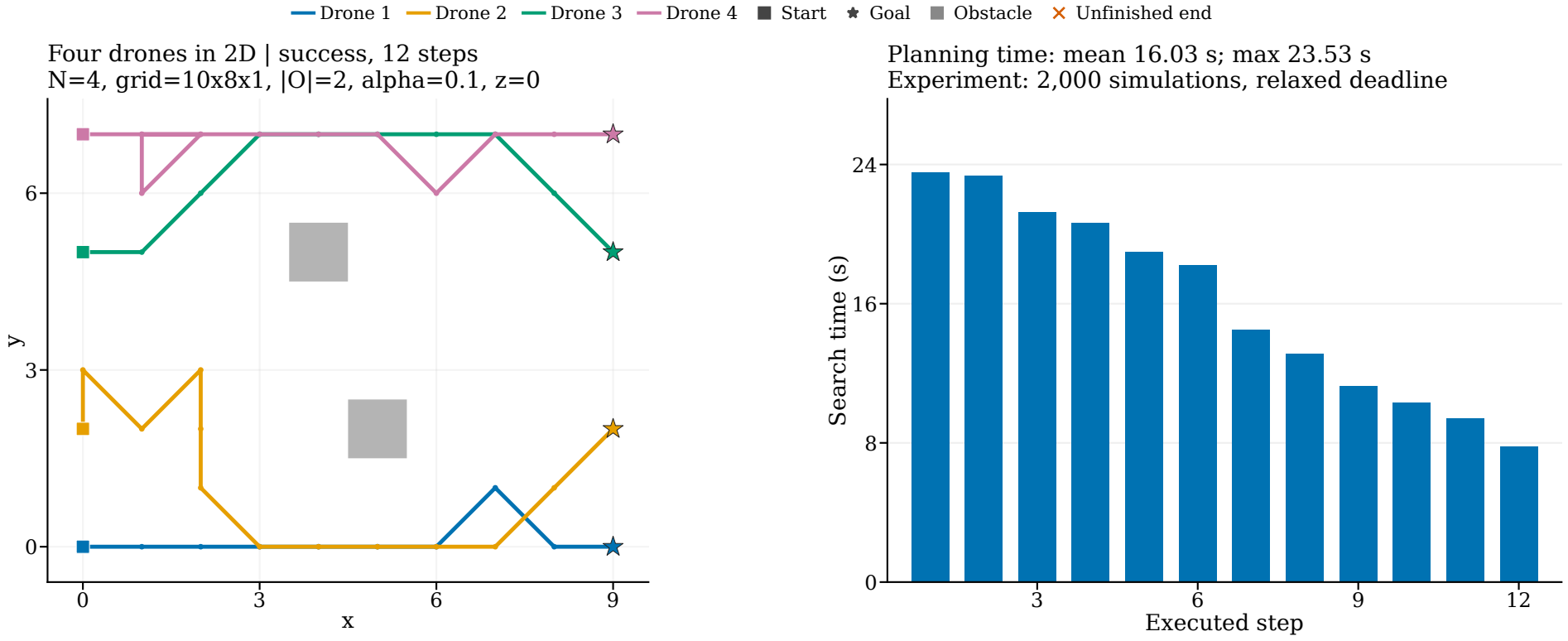


Planning time: mean 17.12 s; max 21.94 s
Experiment: 2,000 simulations, relaxed deadline



Saved full-solver trace, seed 0: results/trajectories/regression_three_3d_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

four 2d

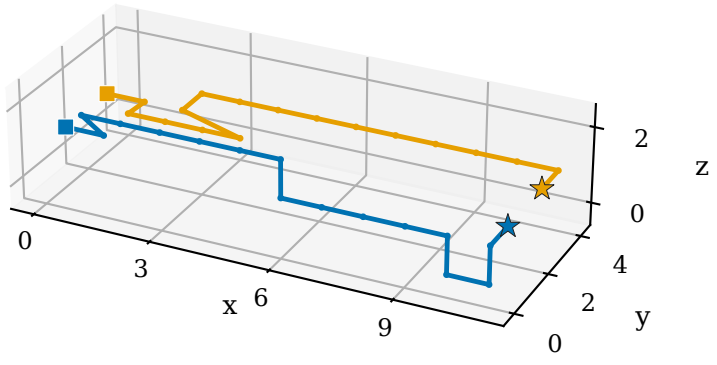


Saved full-solver trace, seed 0: results/trajectories/regression_four_2d_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

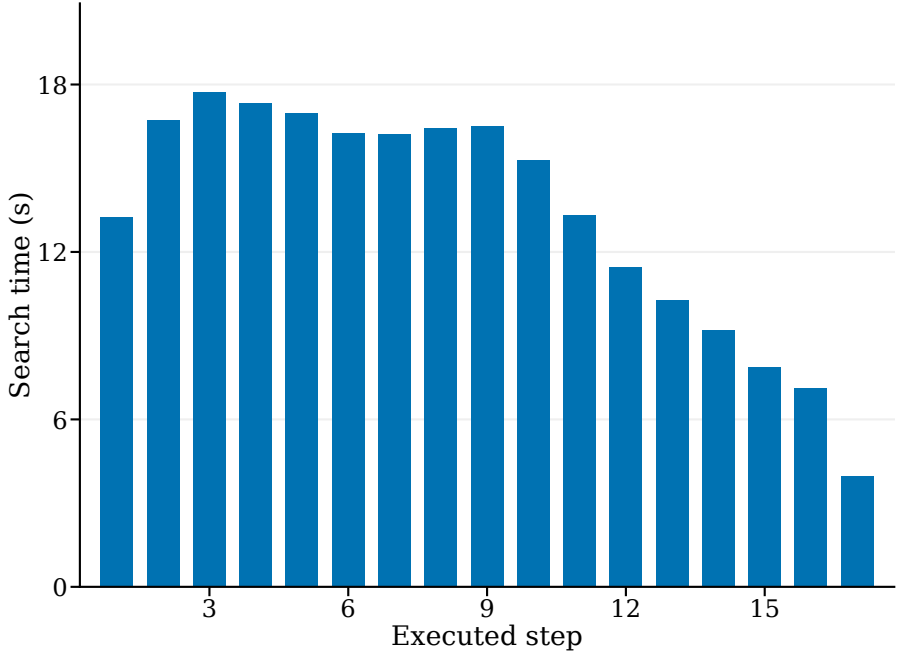
factor S5 A6 C0

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

factor S5 A6 C0 | success, 17 steps
N=2, grid=12x5x3, |O|=0, alpha=0.25



Planning time: mean 13.28 s; max 17.72 s
Experiment: 2,000 simulations, relaxed deadline

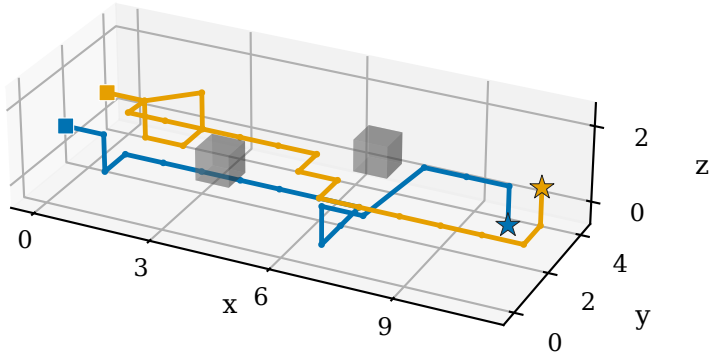


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S5_A6_C0_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

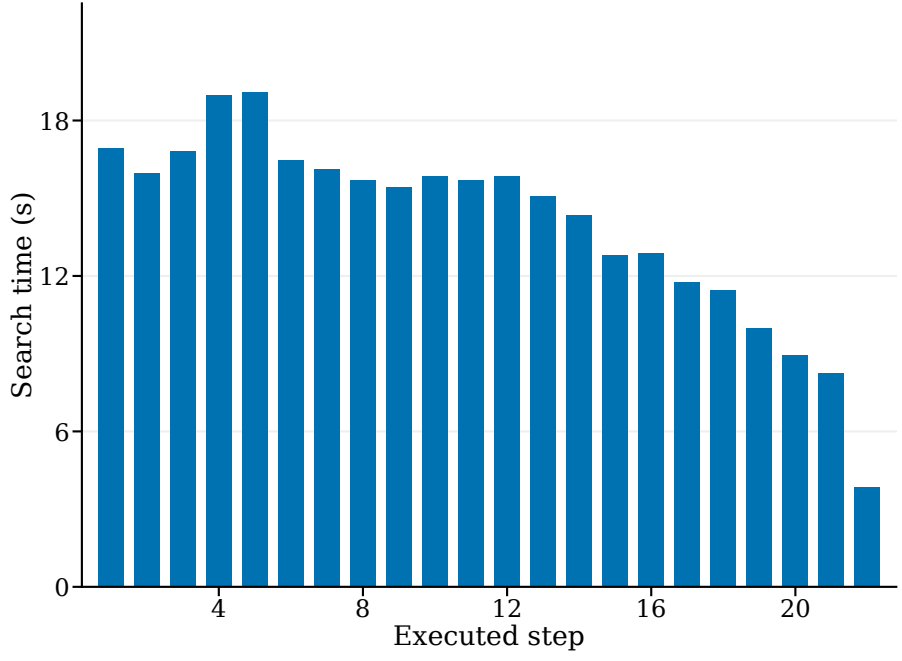
factor S5 A6 C2

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

factor S5 A6 C2 | success, 22 steps
N=2, grid=12x5x3, |O|=2, alpha=0.25



Planning time: mean 14.00 s; max 19.08 s
Experiment: 2,000 simulations, relaxed deadline

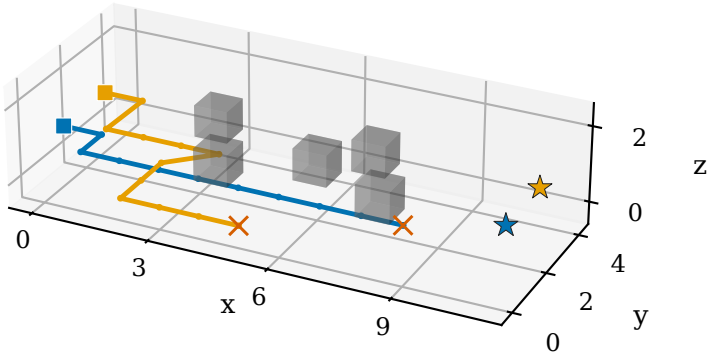


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S5_A6_C2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

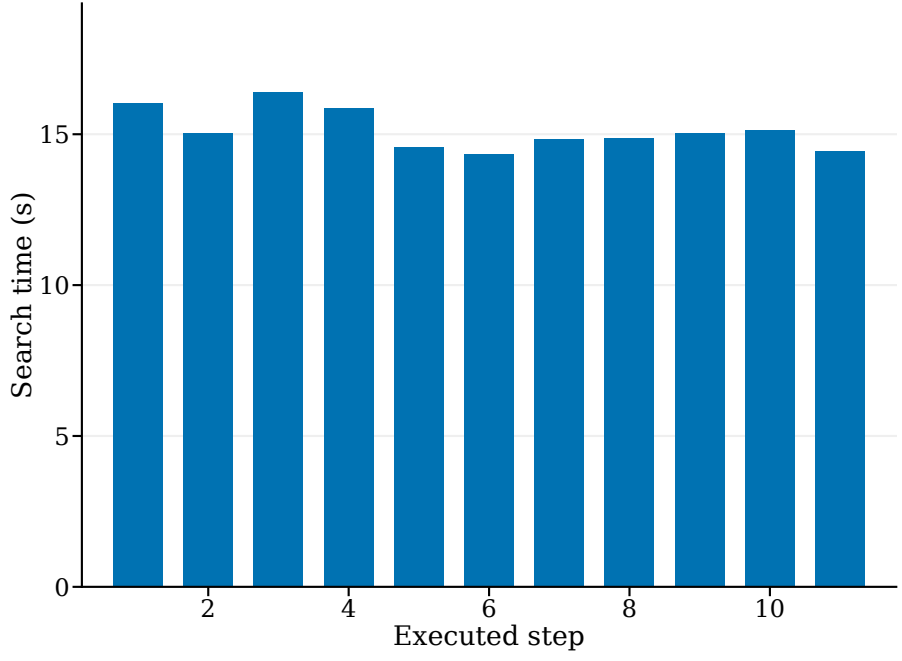
factor S5 A6 C5

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

factor S5 A6 C5 | collision, 11 steps
N=2, grid=12x5x3, |O|=5, alpha=0.25

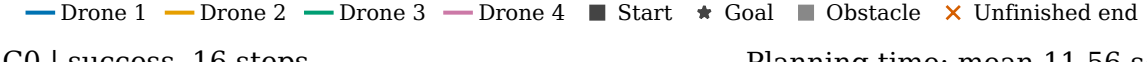


Planning time: mean 15.13 s; max 16.39 s
Experiment: 2,000 simulations, relaxed deadline

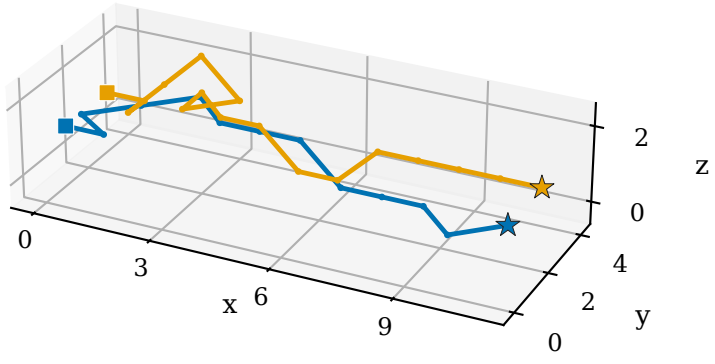


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S5_A6_C5_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

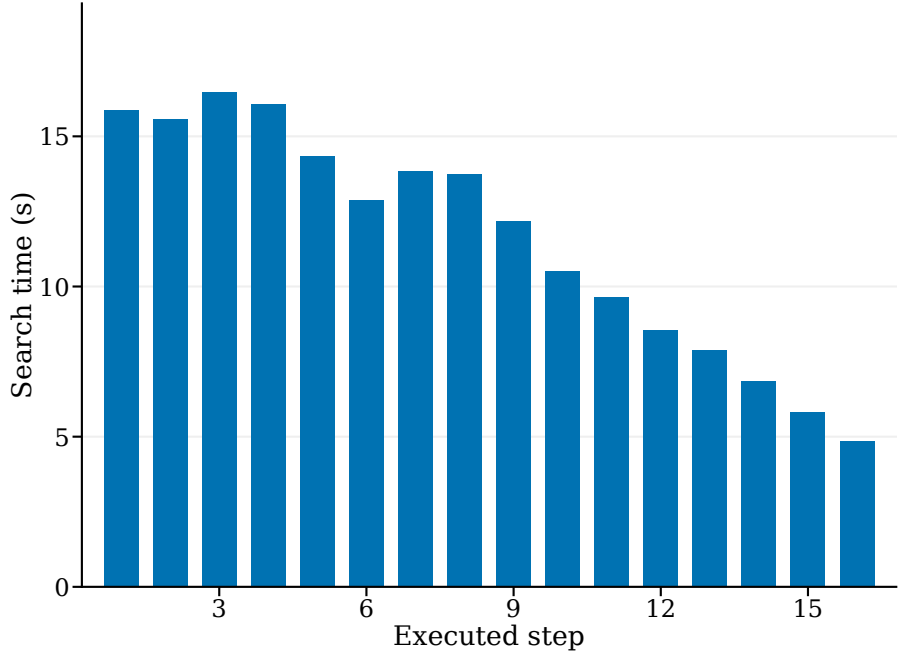
factor S5 A18 C0



factor S5 A18 C0 | success, 16 steps
N=2, grid=12x5x3, |O|=0, alpha=0.25

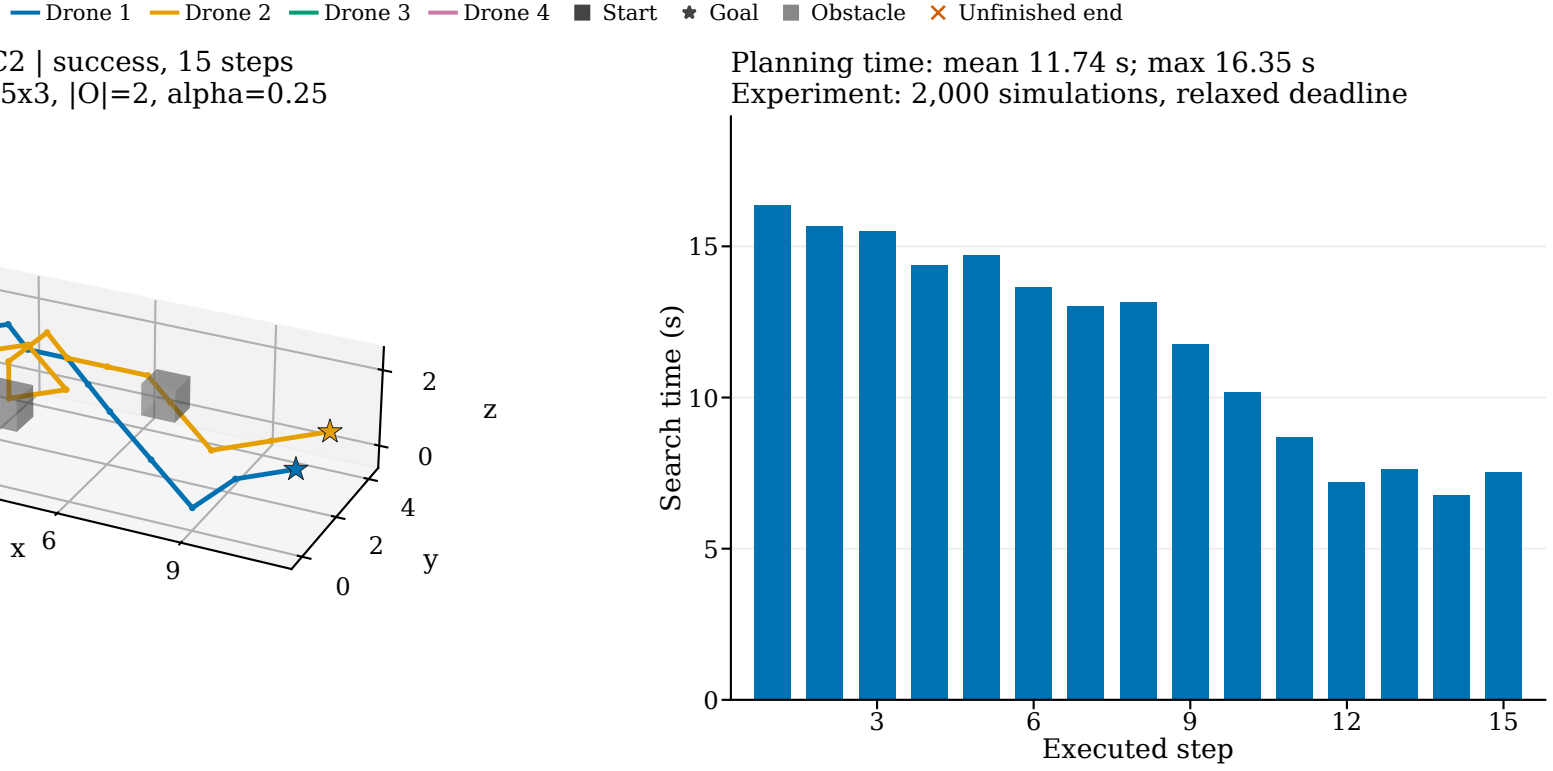


Planning time: mean 11.56 s; max 16.46 s
Experiment: 2,000 simulations, relaxed deadline



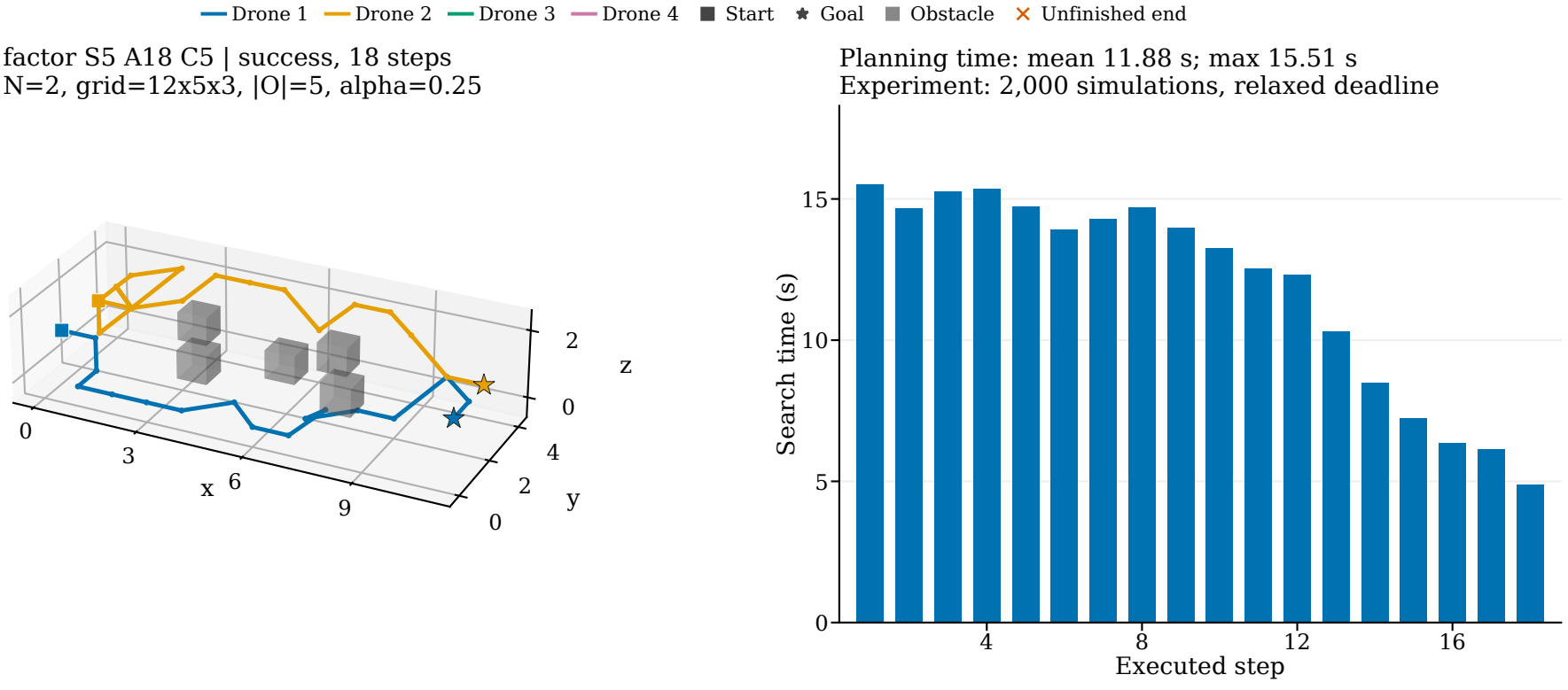
Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S5_A18_C0_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

factor S5 A18 C2



Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S5_A18_C2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

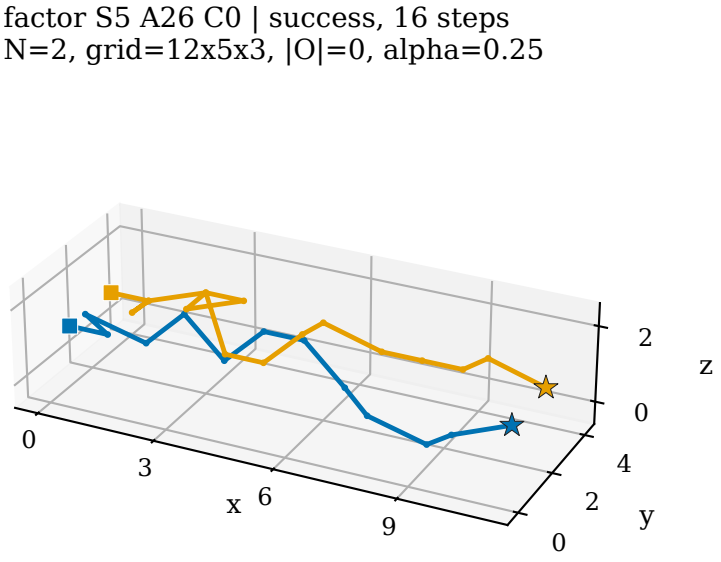
factor S5 A18 C5

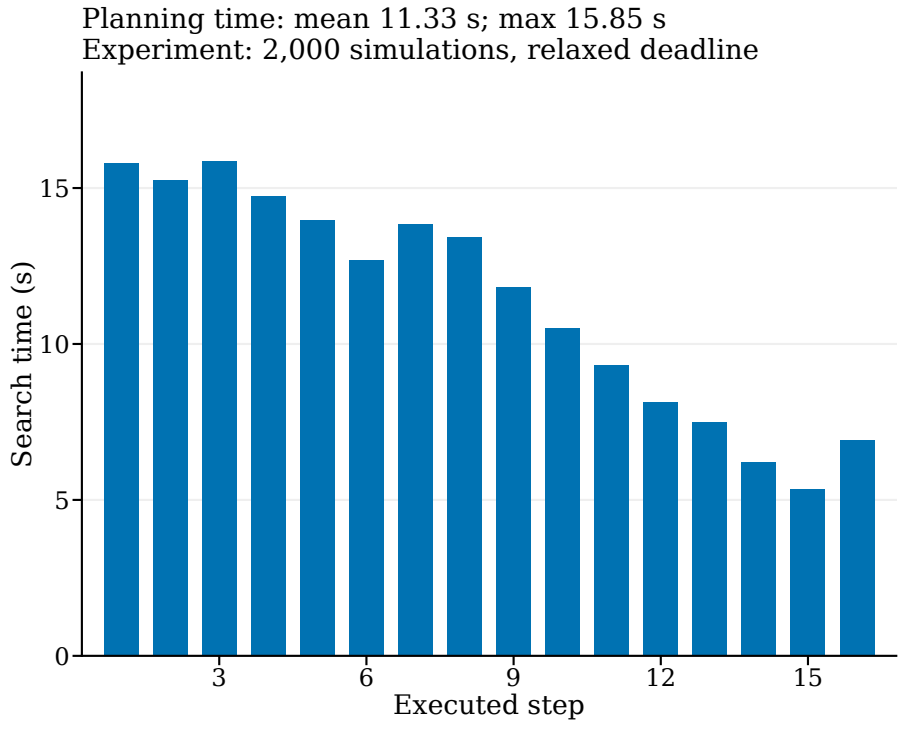


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S5_A18_C5_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

factor S5 A26 C0

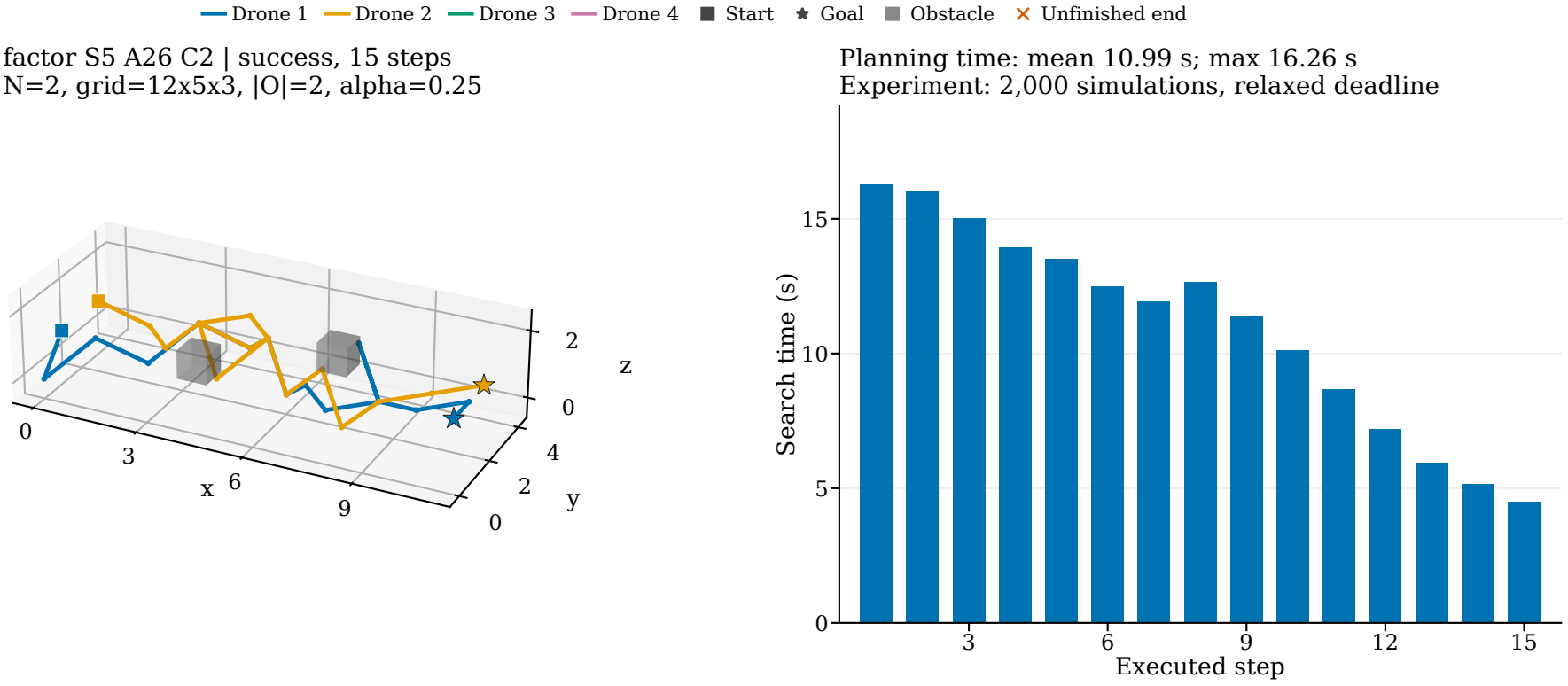






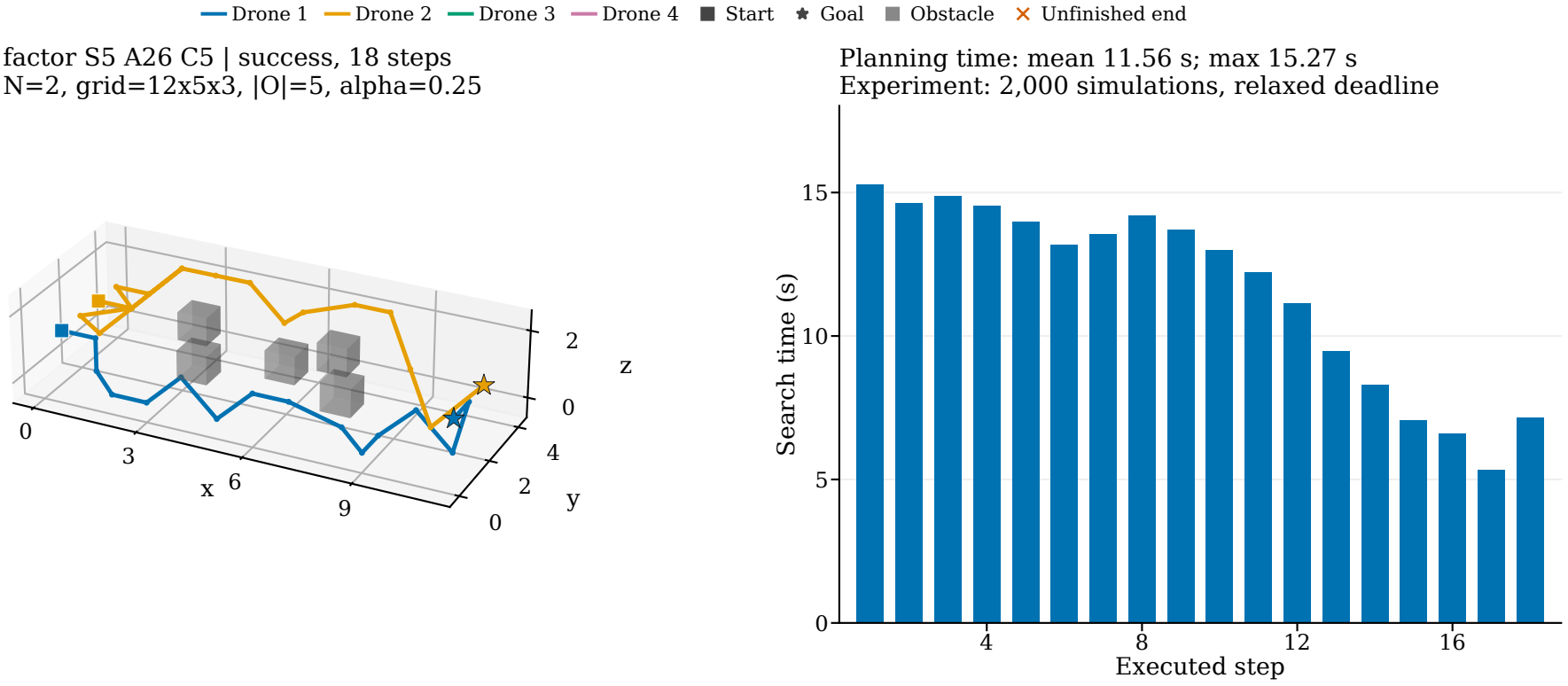
Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S5_A26_C0_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

factor S5 A26 C2



Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S5_A26_C2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

factor S5 A26 C5

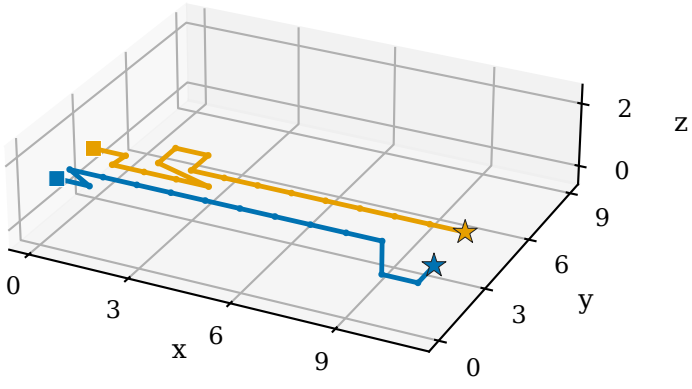


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S5_A26_C5_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

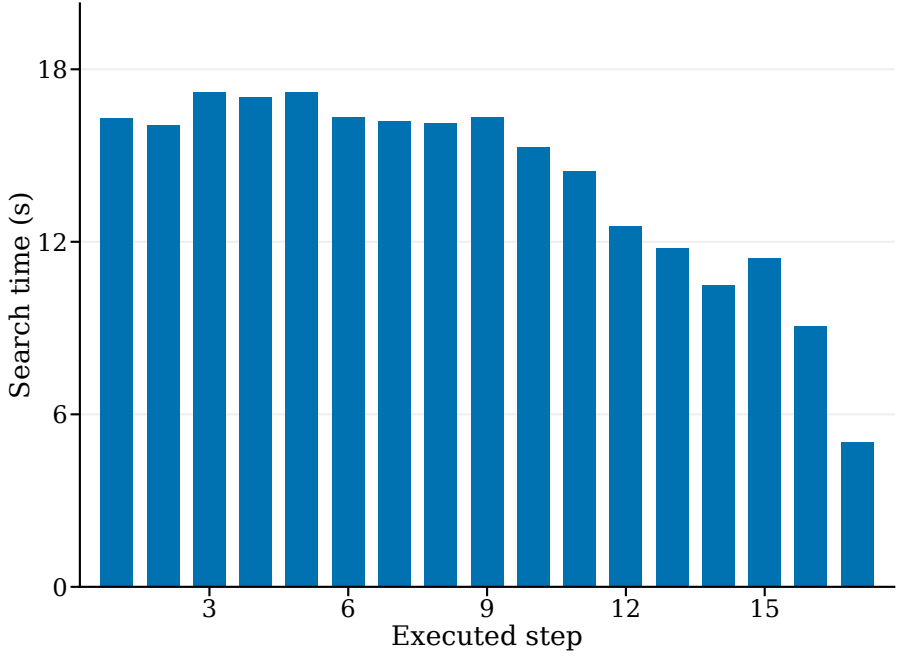
factor S10 A6 C0

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

factor S10 A6 C0 | success, 17 steps
N=2, grid=12x10x3, |O|=0, alpha=0.25



Planning time: mean 14.04 s; max 17.20 s
Experiment: 2,000 simulations, relaxed deadline

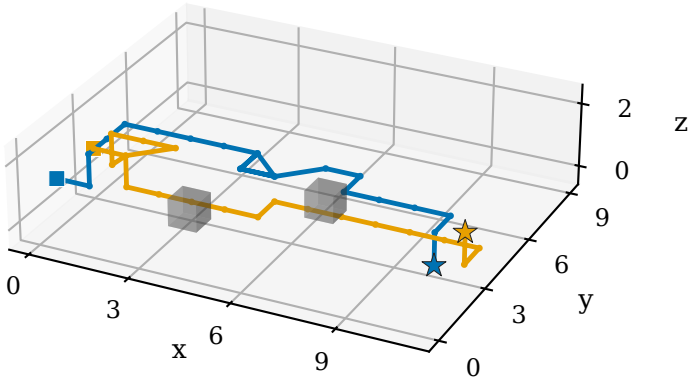


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S10_A6_C0_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

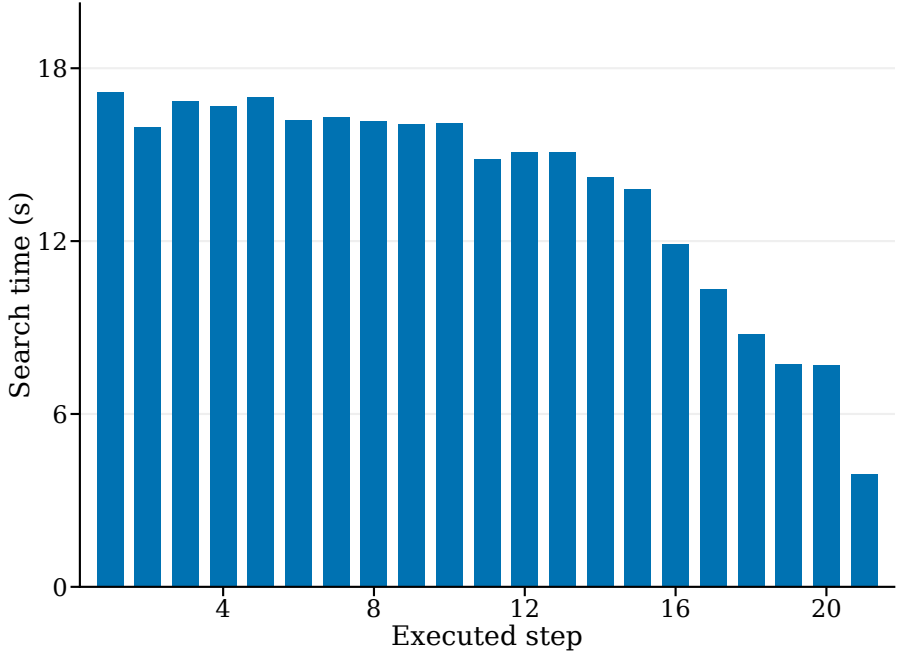
factor S10 A6 C2



factor S10 A6 C2 | success, 21 steps
N=2, grid=12x10x3, |O|=2, alpha=0.25



Planning time: mean 13.70 s; max 17.16 s
Experiment: 2,000 simulations, relaxed deadline

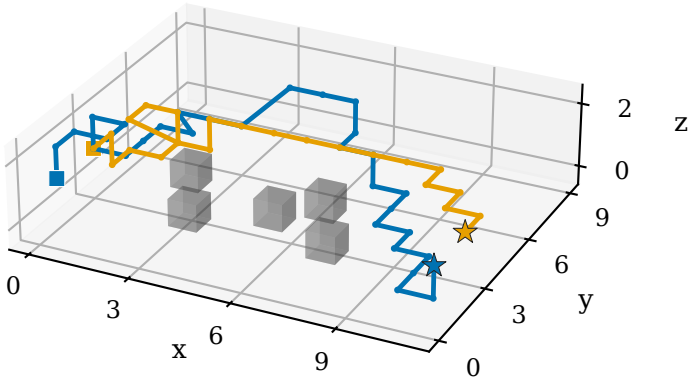


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S10_A6_C2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

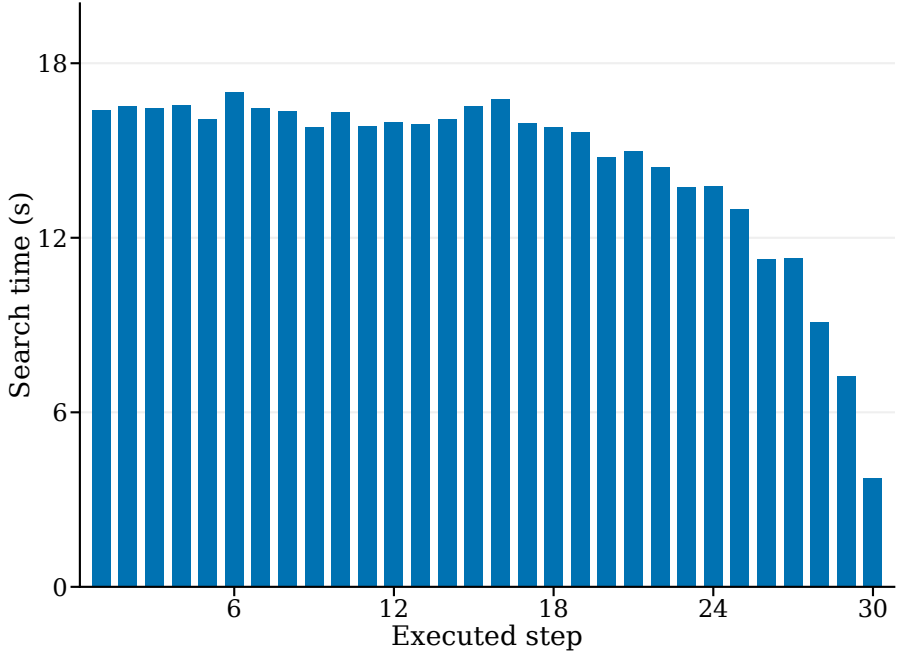
factor S10 A6 C5



factor S10 A6 C5 | success, 30 steps
N=2, grid=12x10x3, |O|=5, alpha=0.25



Planning time: mean 14.52 s; max 17.00 s
Experiment: 2,000 simulations, relaxed deadline

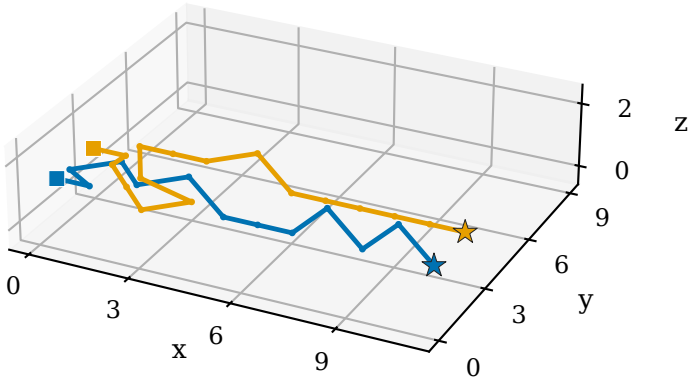


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S10_A6_C5_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

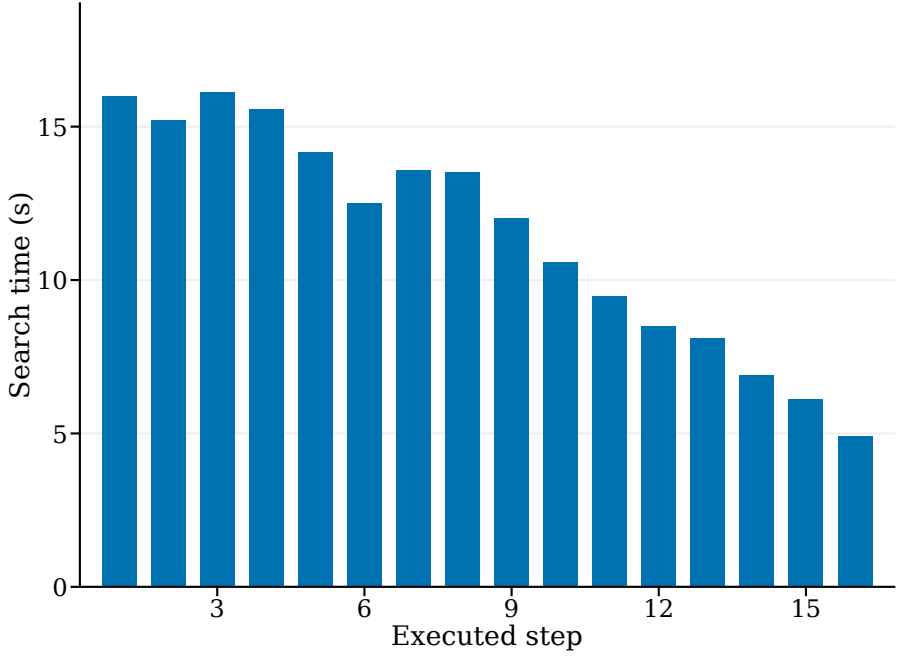
factor S10 A18 C0

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

factor S10 A18 C0 | success, 16 steps
N=2, grid=12x10x3, |O|=0, alpha=0.25



Planning time: mean 11.45 s; max 16.12 s
Experiment: 2,000 simulations, relaxed deadline

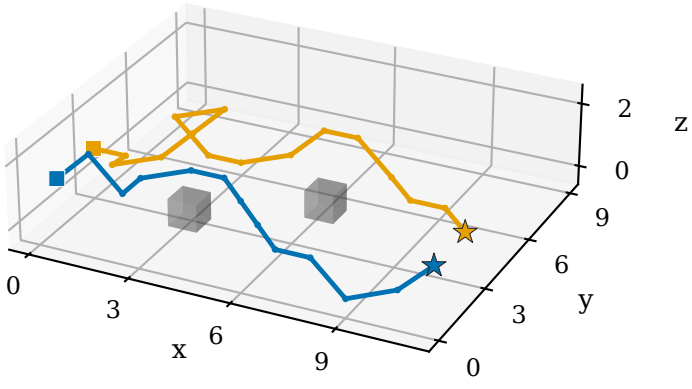


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S10_A18_C0_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

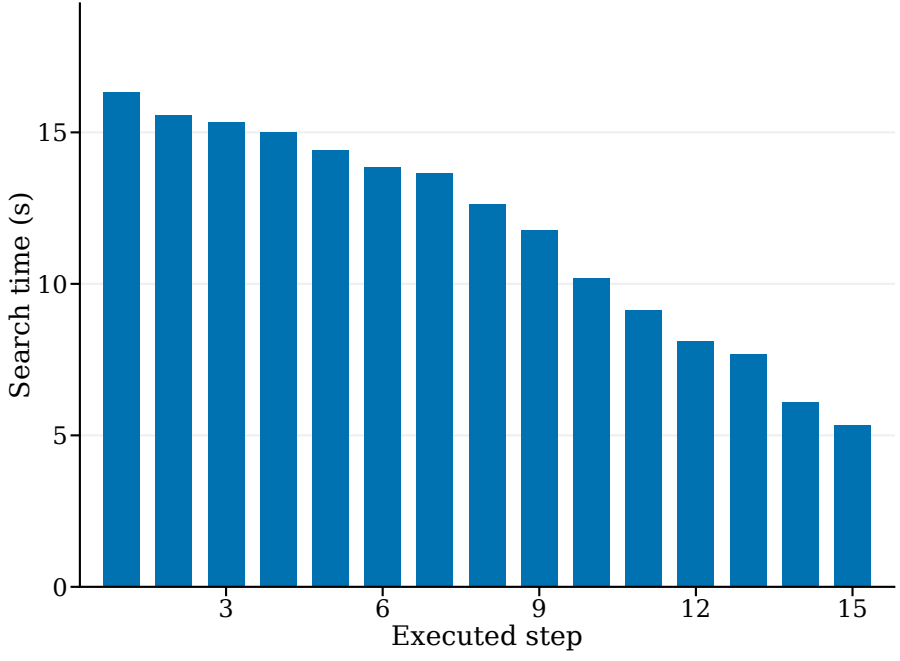
factor S10 A18 C2

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

factor S10 A18 C2 | success, 15 steps
N=2, grid=12x10x3, |O|=2, alpha=0.25



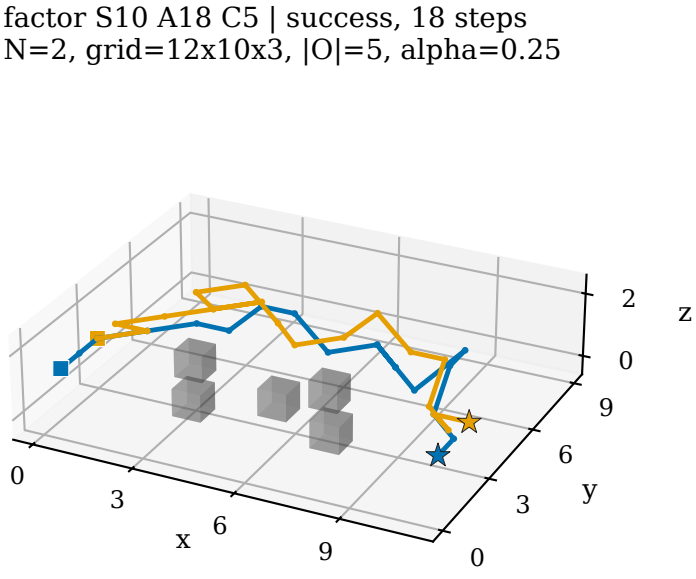
Planning time: mean 11.67 s; max 16.32 s
Experiment: 2,000 simulations, relaxed deadline

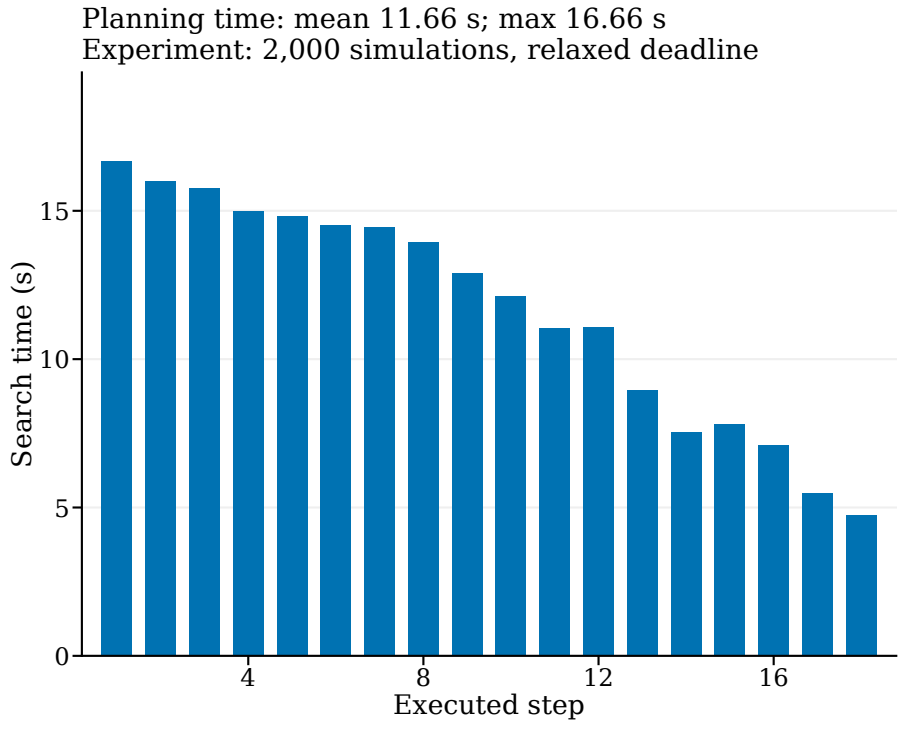


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S10_A18_C2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

factor S10 A18 C5





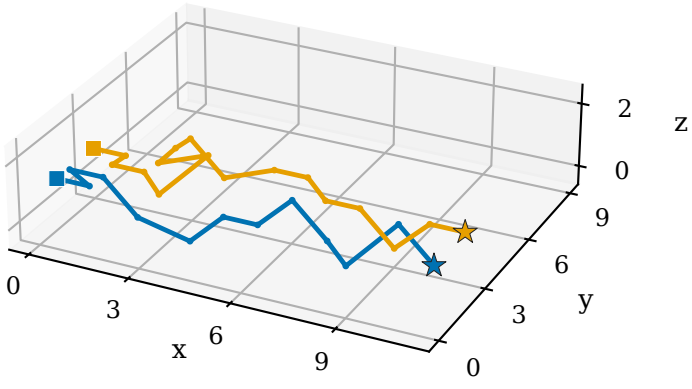


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S10_A18_C5_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

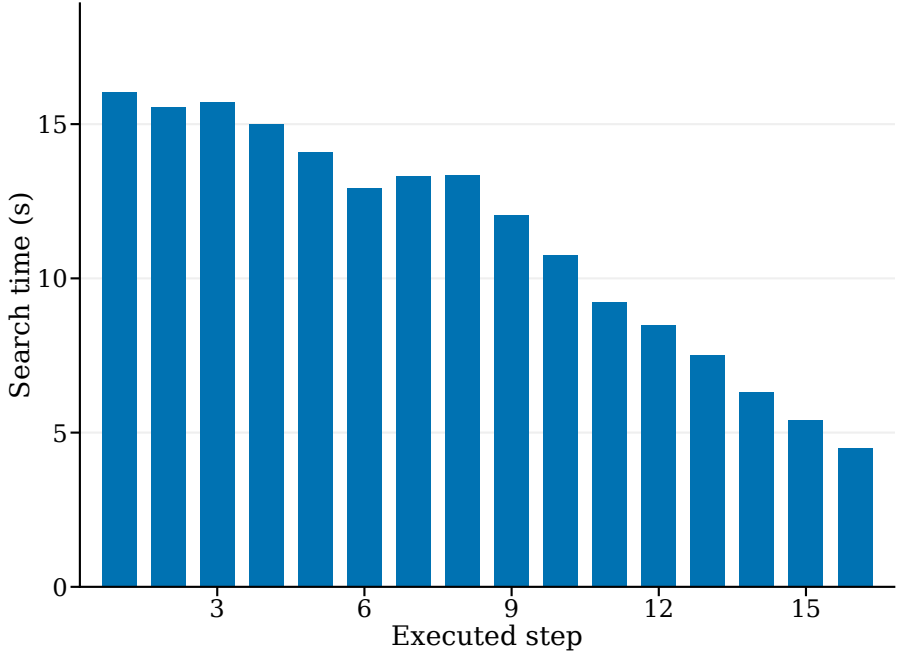
factor S10 A26 C0

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

factor S10 A26 C0 | success, 16 steps
N=2, grid=12x10x3, |O|=0, alpha=0.25



Planning time: mean 11.26 s; max 16.03 s
Experiment: 2,000 simulations, relaxed deadline

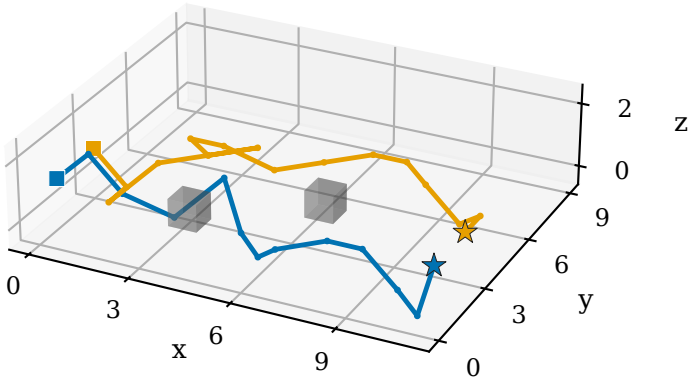


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S10_A26_C0_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

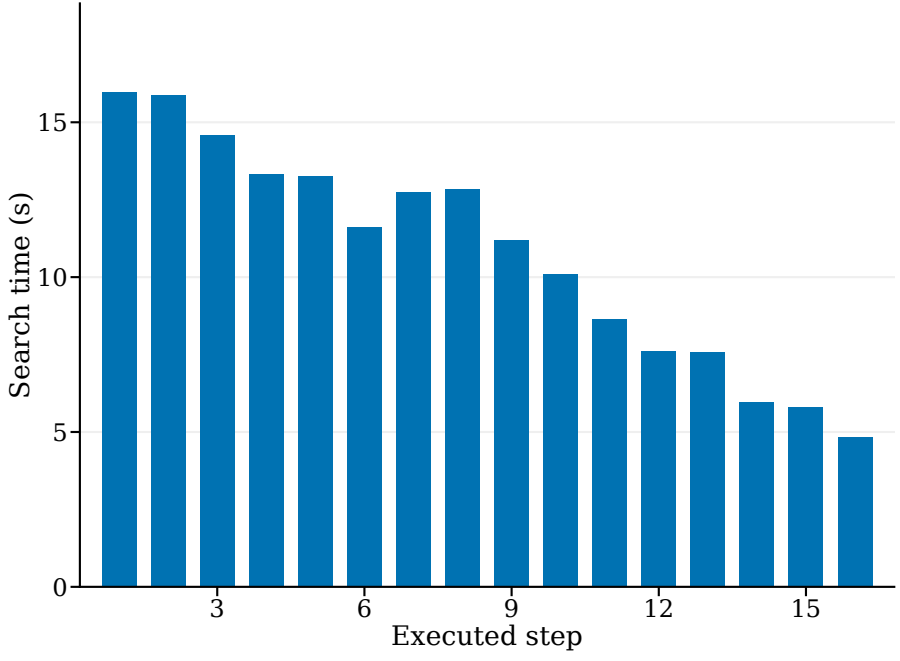
factor S10 A26 C2

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

factor S10 A26 C2 | success, 16 steps
N=2, grid=12x10x3, |O|=2, alpha=0.25



Planning time: mean 10.74 s; max 15.96 s
Experiment: 2,000 simulations, relaxed deadline

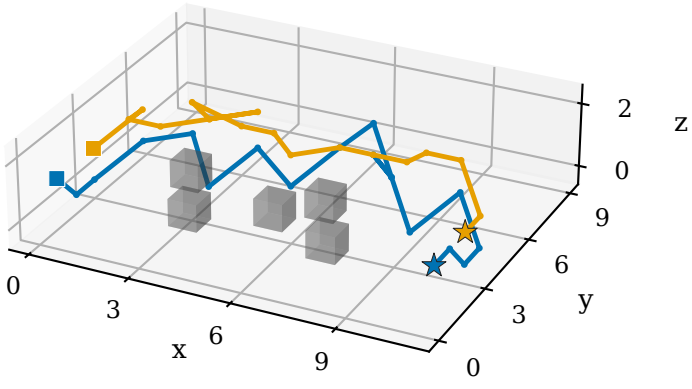


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S10_A26_C2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

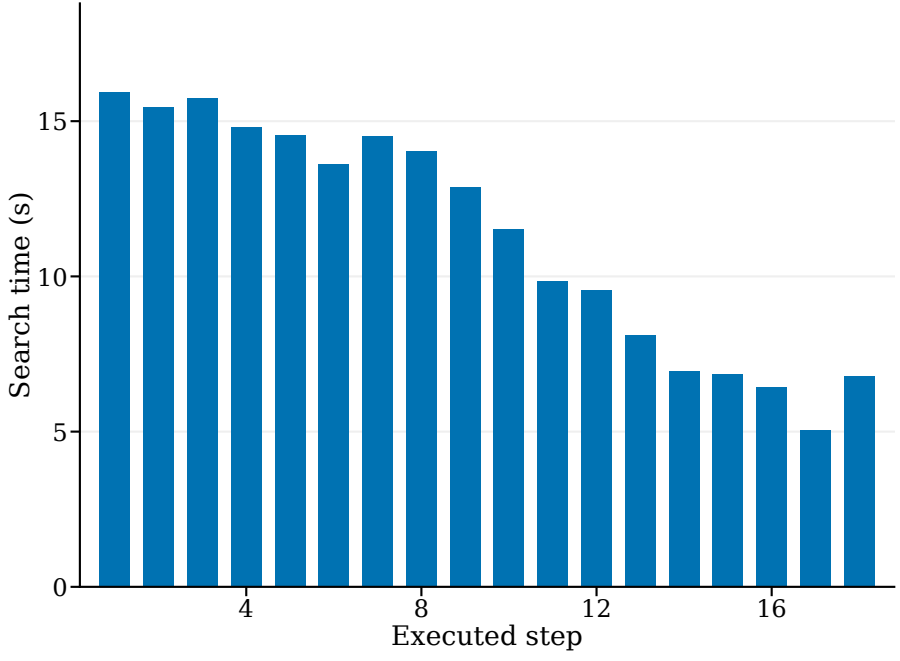
factor S10 A26 C5

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

factor S10 A26 C5 | success, 18 steps
N=2, grid=12x10x3, |O|=5, alpha=0.25



Planning time: mean 11.25 s; max 15.93 s
Experiment: 2,000 simulations, relaxed deadline

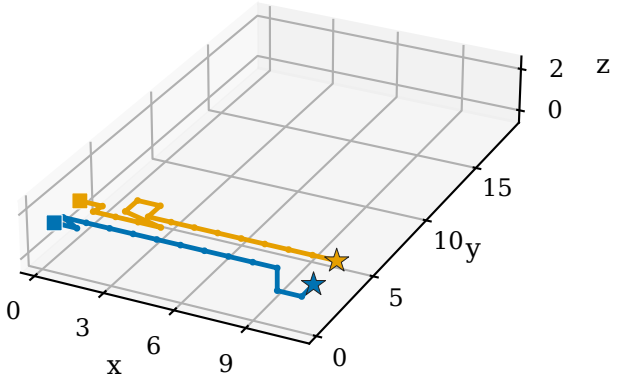


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S10_A26_C5_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

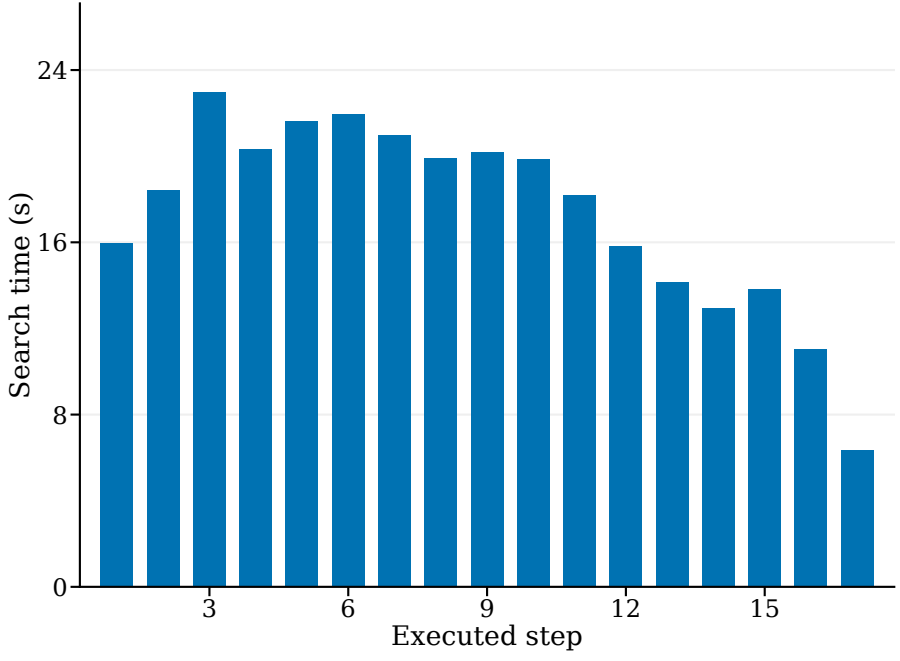
factor S20 A6 C0

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

factor S20 A6 C0 | success, 17 steps
N=2, grid=12x20x3, |O|=0, alpha=0.25



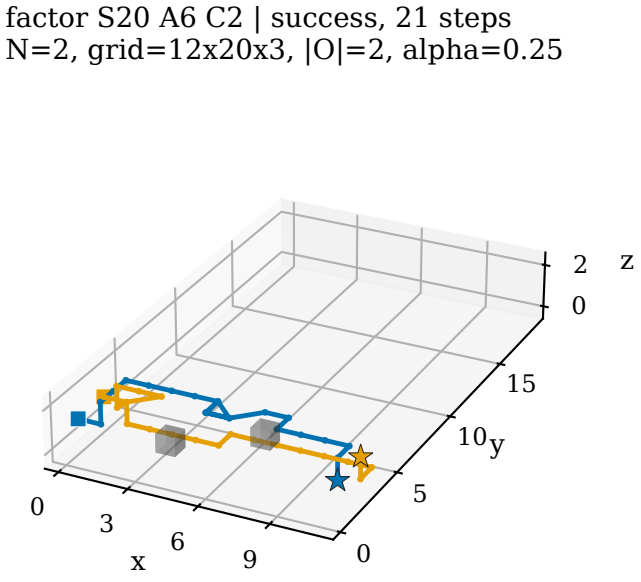
Planning time: mean 17.31 s; max 22.96 s
Experiment: 2,000 simulations, relaxed deadline

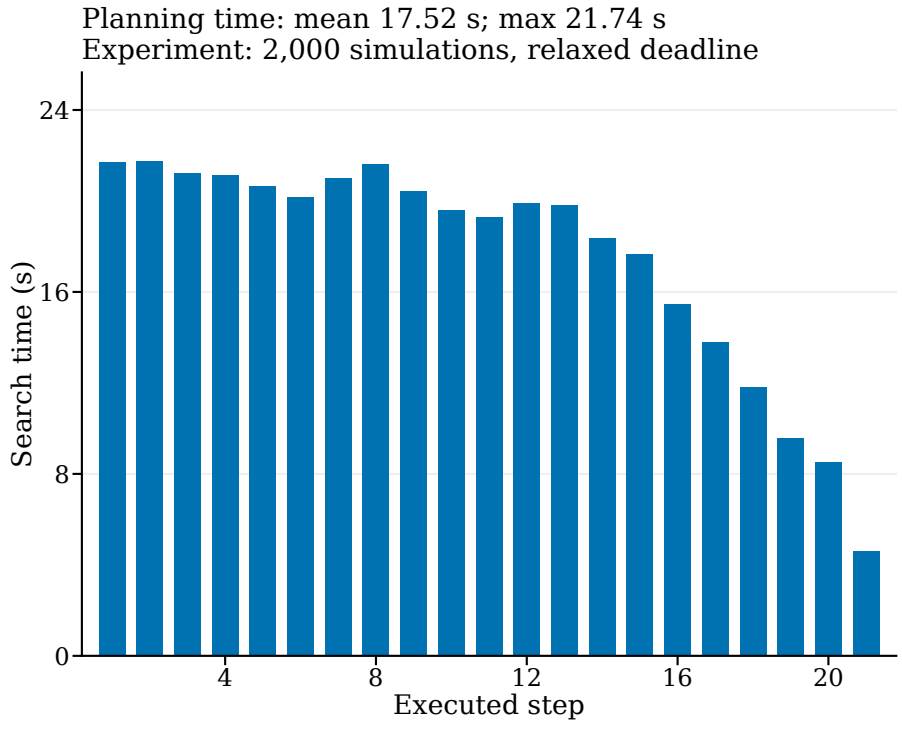


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S20_A6_C0_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

factor S20 A6 C2





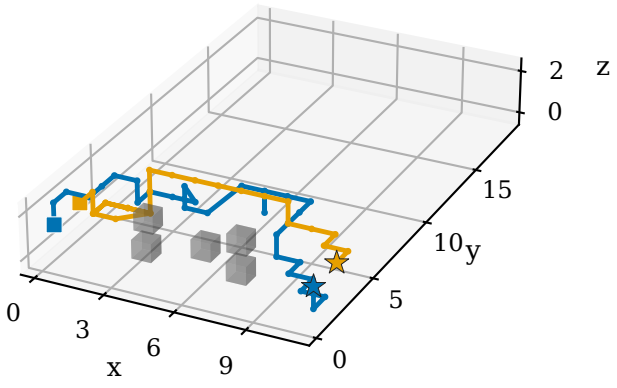


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S20_A6_C2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

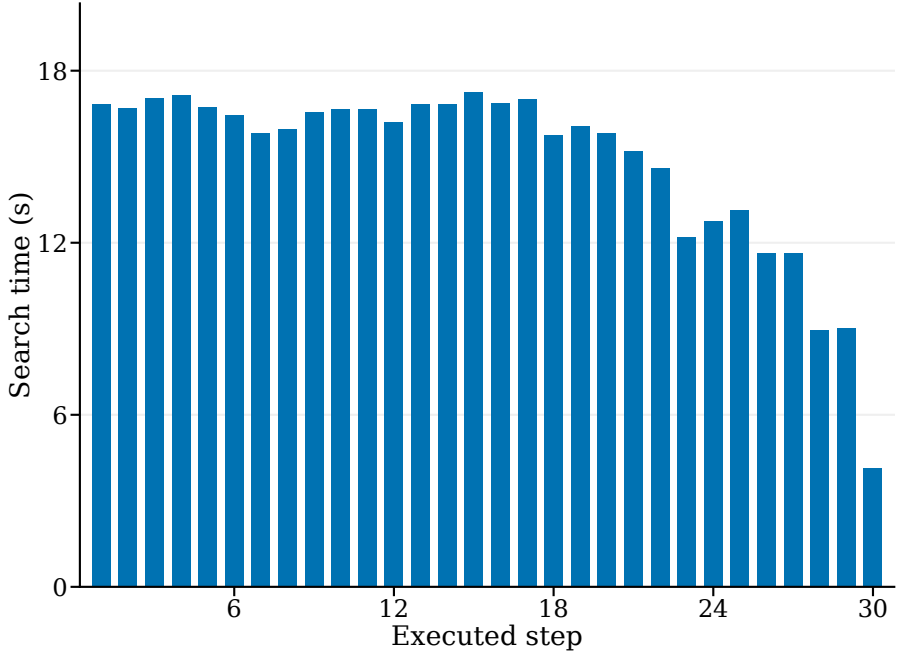
factor S20 A6 C5



factor S20 A6 C5 | success, 30 steps
N=2, grid=12x20x3, |O|=5, alpha=0.25



Planning time: mean 14.81 s; max 17.24 s
Experiment: 2,000 simulations, relaxed deadline

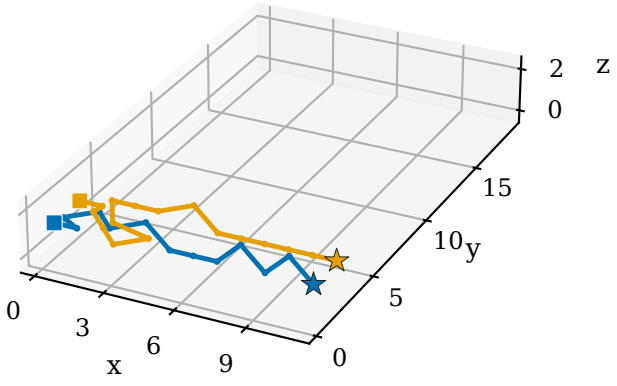


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S20_A6_C5_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

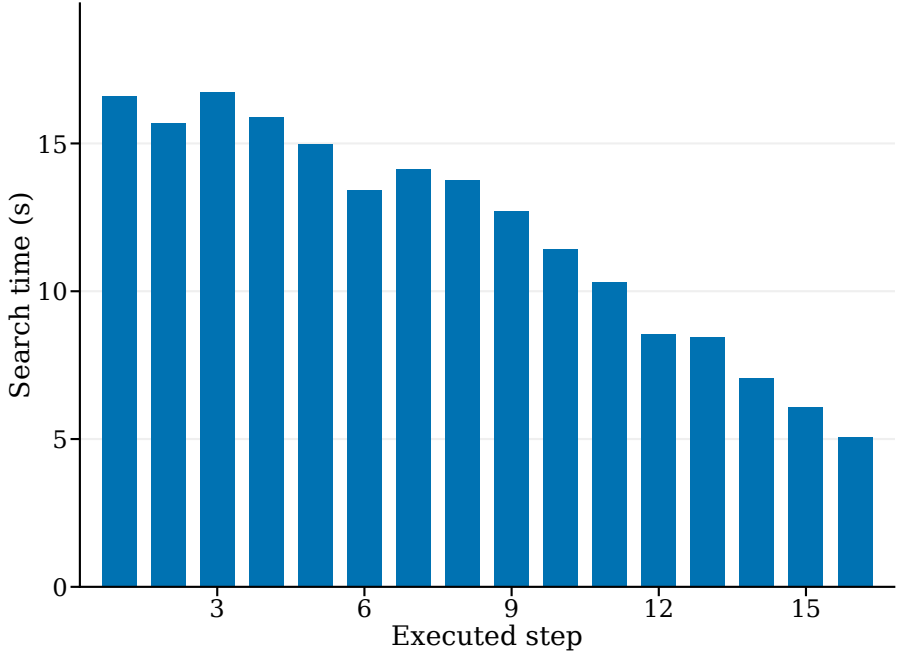
factor S20 A18 C0

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

factor S20 A18 C0 | success, 16 steps
N=2, grid=12x20x3, |O|=0, alpha=0.25



Planning time: mean 11.92 s; max 16.73 s
Experiment: 2,000 simulations, relaxed deadline

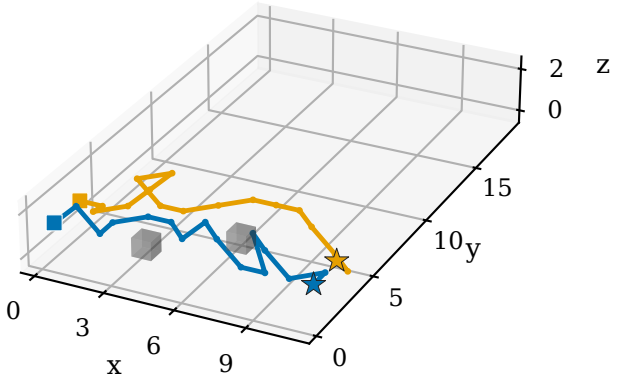


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S20_A18_C0_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

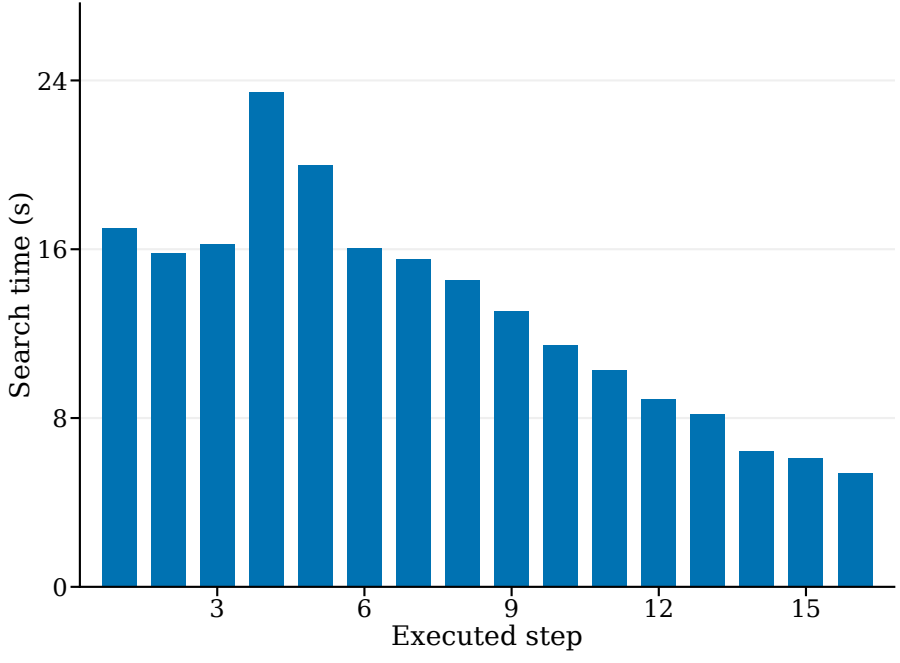
factor S20 A18 C2



factor S20 A18 C2 | success, 16 steps
N=2, grid=12x20x3, |O|=2, alpha=0.25



Planning time: mean 13.01 s; max 23.43 s
Experiment: 2,000 simulations, relaxed deadline

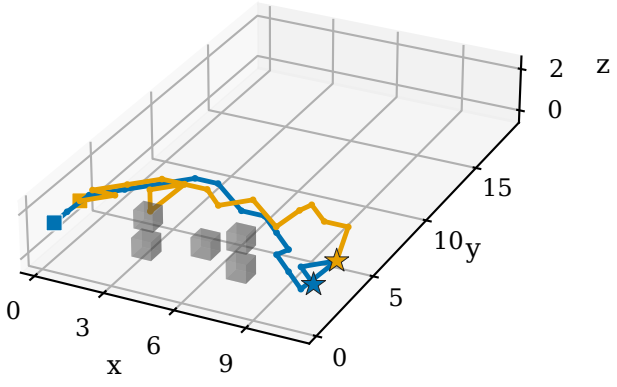


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S20_A18_C2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

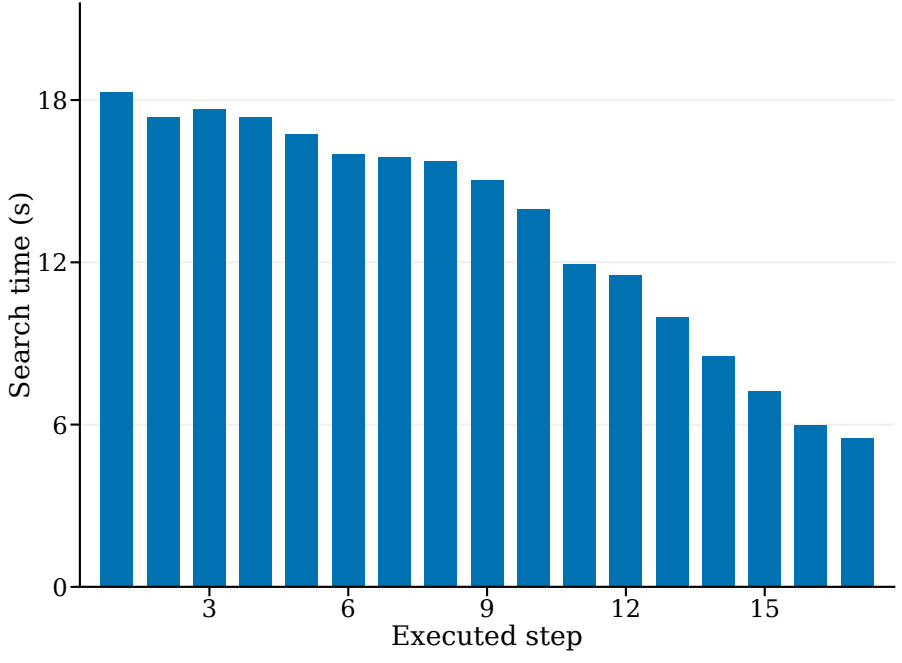
factor S20 A18 C5



factor S20 A18 C5 | success, 17 steps
N=2, grid=12x20x3, |O|=5, alpha=0.25



Planning time: mean 13.22 s; max 18.28 s
Experiment: 2,000 simulations, relaxed deadline

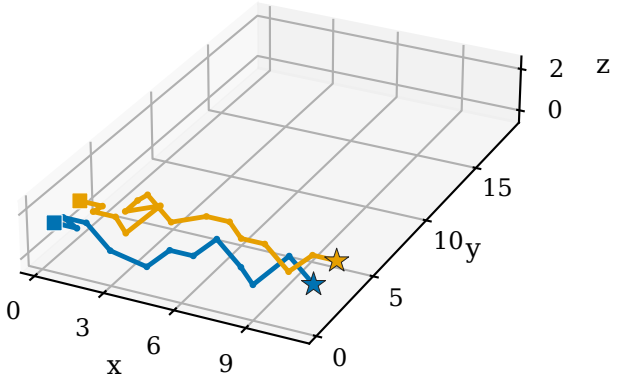


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S20_A18_C5_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

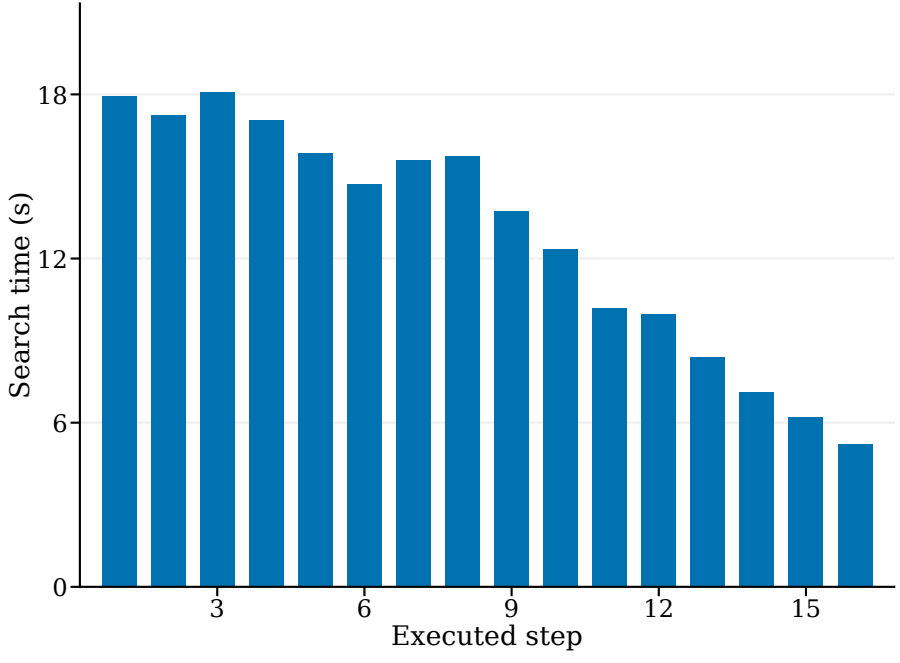
factor S20 A26 C0



factor S20 A26 C0 | success, 16 steps
N=2, grid=12x20x3, |O|=0, alpha=0.25



Planning time: mean 12.83 s; max 18.07 s
Experiment: 2,000 simulations, relaxed deadline

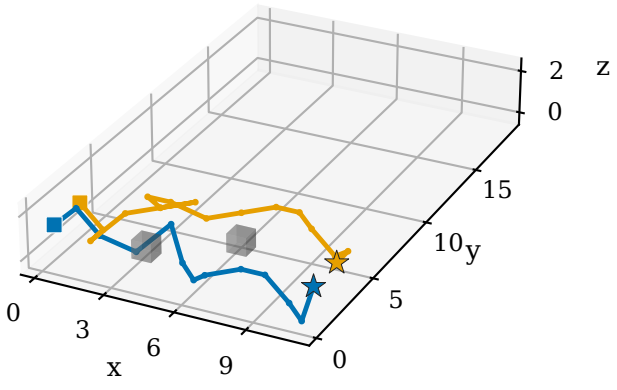


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S20_A26_C0_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

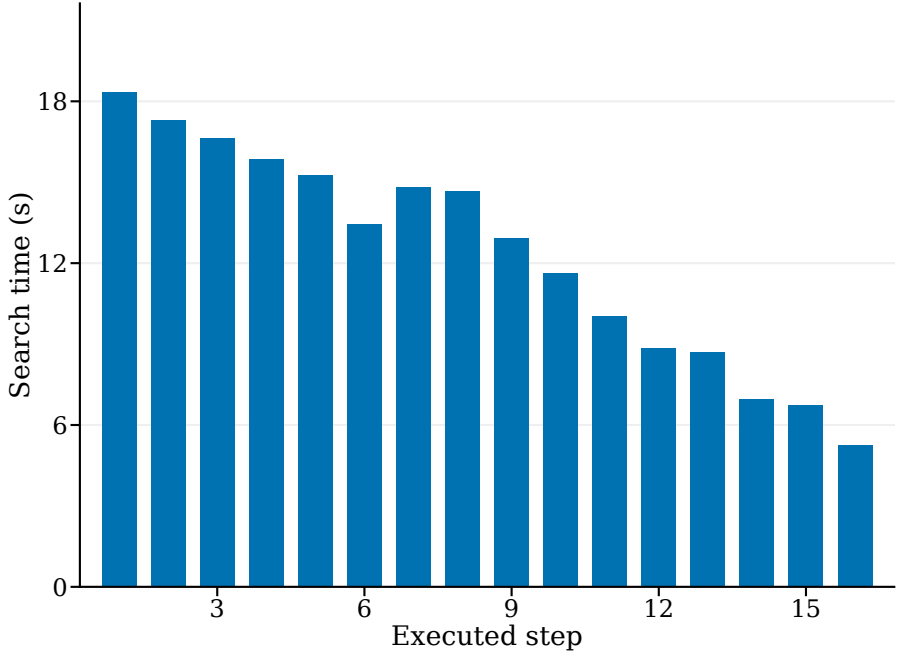
factor S20 A26 C2



factor S20 A26 C2 | success, 16 steps
N=2, grid=12x20x3, |O|=2, alpha=0.25



Planning time: mean 12.33 s; max 18.33 s
Experiment: 2,000 simulations, relaxed deadline

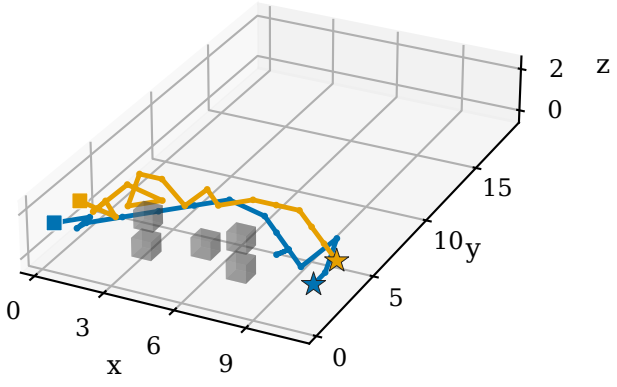


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S20_A26_C2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

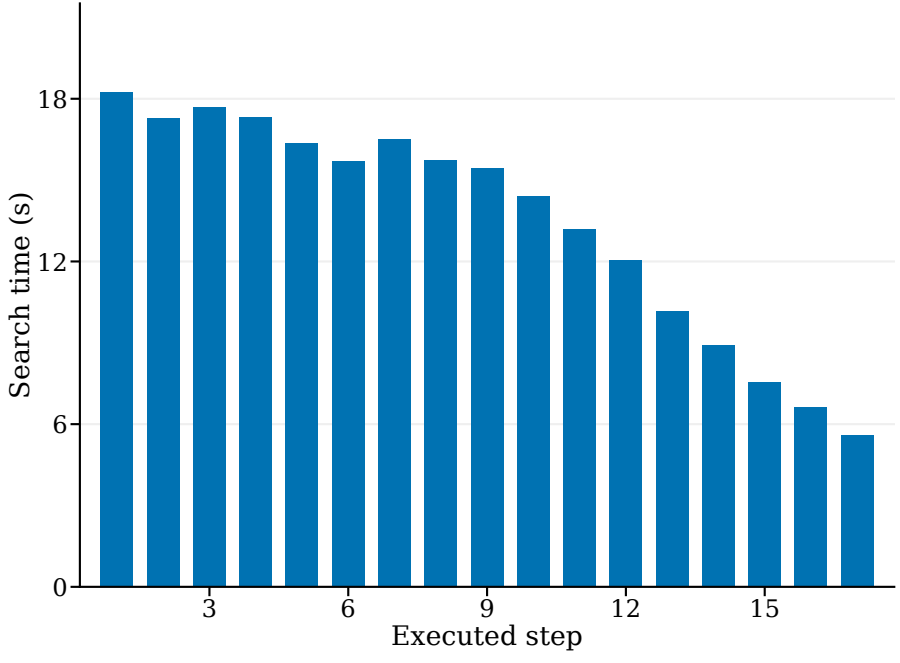
factor S20 A26 C5



factor S20 A26 C5 | success, 17 steps
N=2, grid=12x20x3, |O|=5, alpha=0.25



Planning time: mean 13.45 s; max 18.23 s
Experiment: 2,000 simulations, relaxed deadline

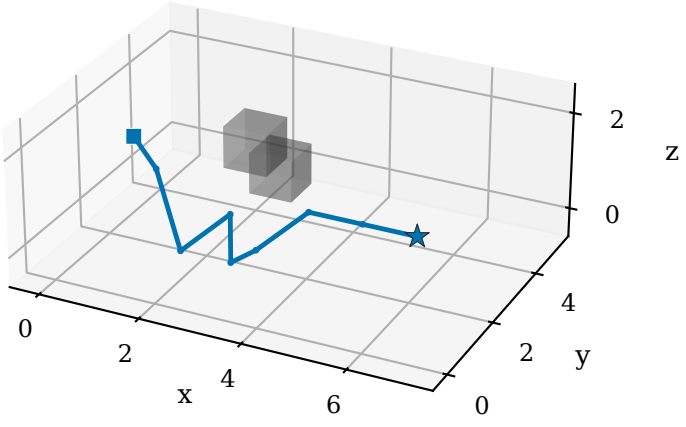


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S20_A26_C5_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

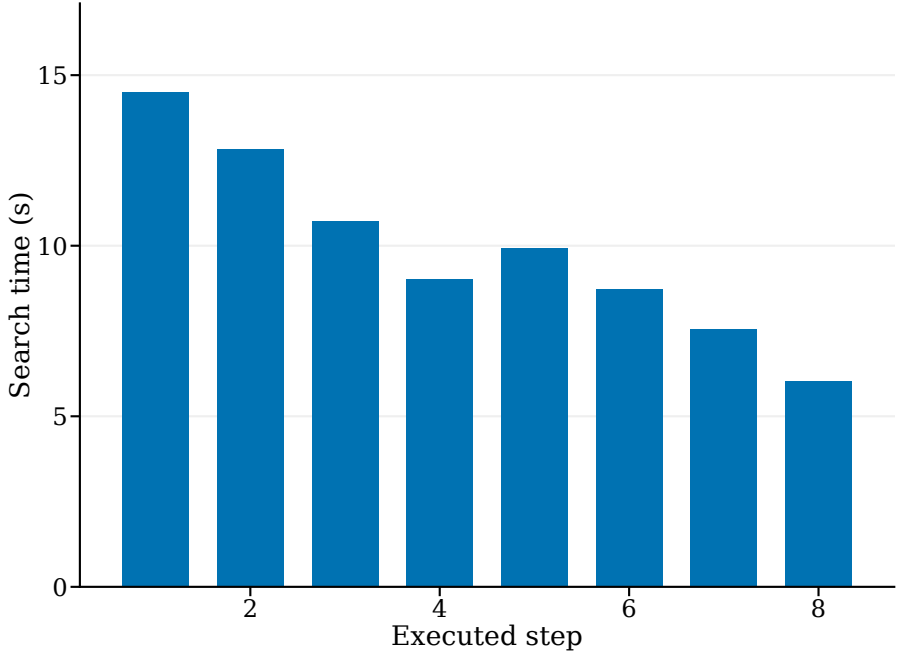
random 00 N1

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

random 00 N1 | success, 8 steps
N=1, grid=8x6x3, |O|=2, alpha=0.05

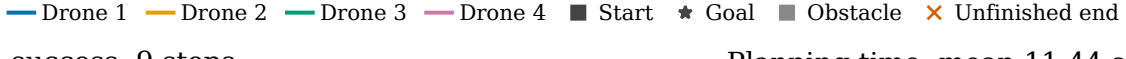


Planning time: mean 9.90 s; max 14.49 s
Experiment: 2,000 simulations, relaxed deadline

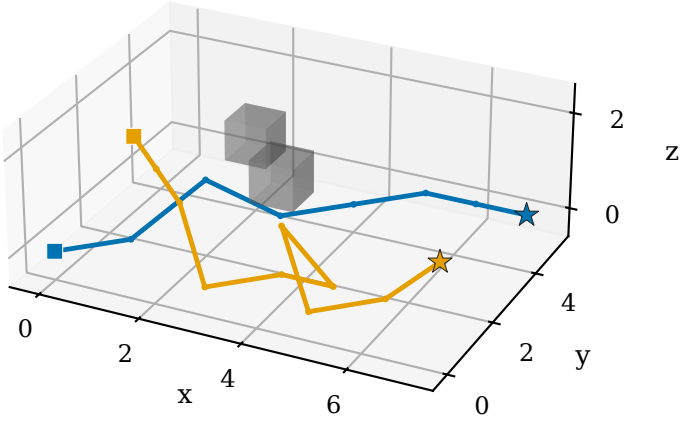


Saved full-solver trace, seed 0: results/trajectories/random_random_00_N1_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

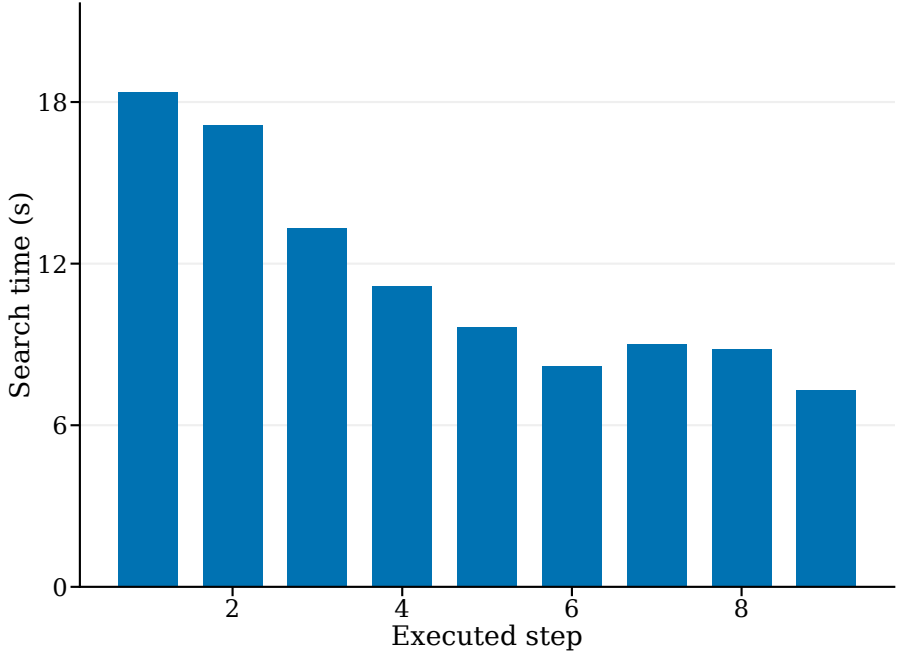
random 01 N2



random 01 N2 | success, 9 steps
N=2, grid=8x6x3, |O|=2, alpha=0.05



Planning time: mean 11.44 s; max 18.36 s
Experiment: 2,000 simulations, relaxed deadline

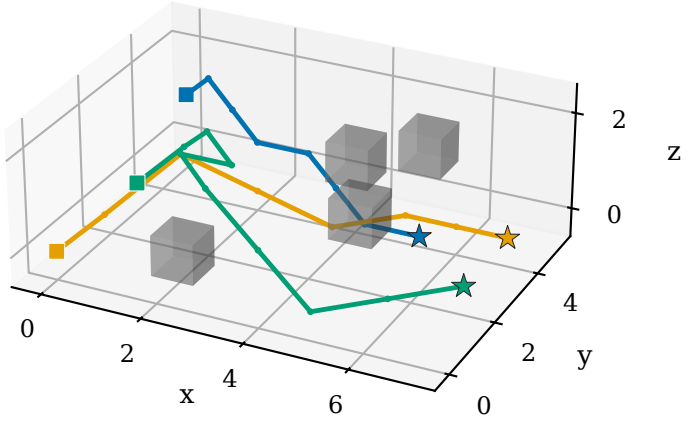


Saved full-solver trace, seed 0: results/trajectories/random_random_01_N2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

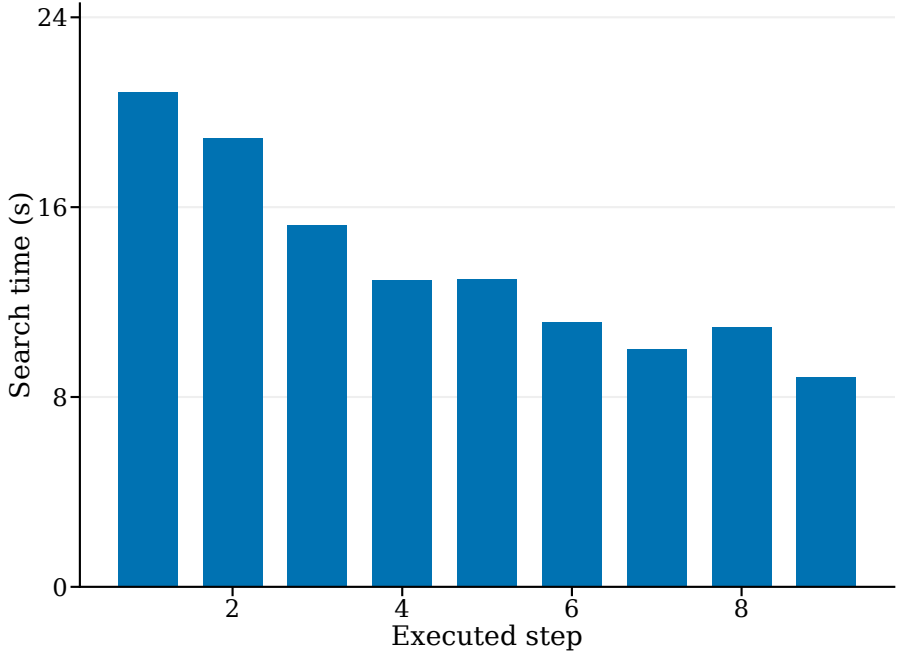
random 02 N3

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

random 02 N3 | success, 9 steps
N=3, grid=8x6x3, |O|=4, alpha=0.05



Planning time: mean 13.53 s; max 20.83 s
Experiment: 2,000 simulations, relaxed deadline

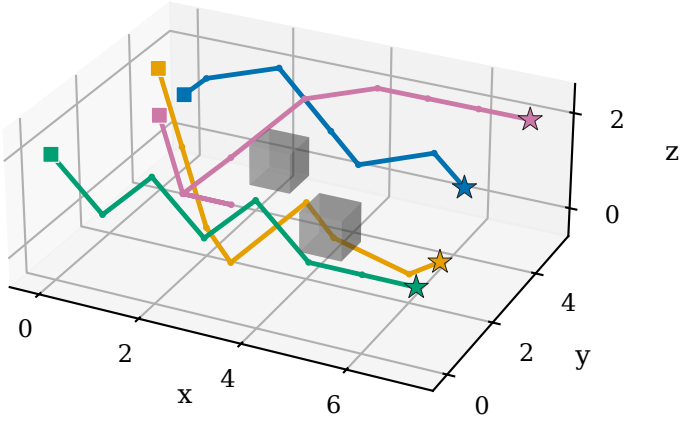


Saved full-solver trace, seed 0: results/trajectories/random_random_02_N3_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

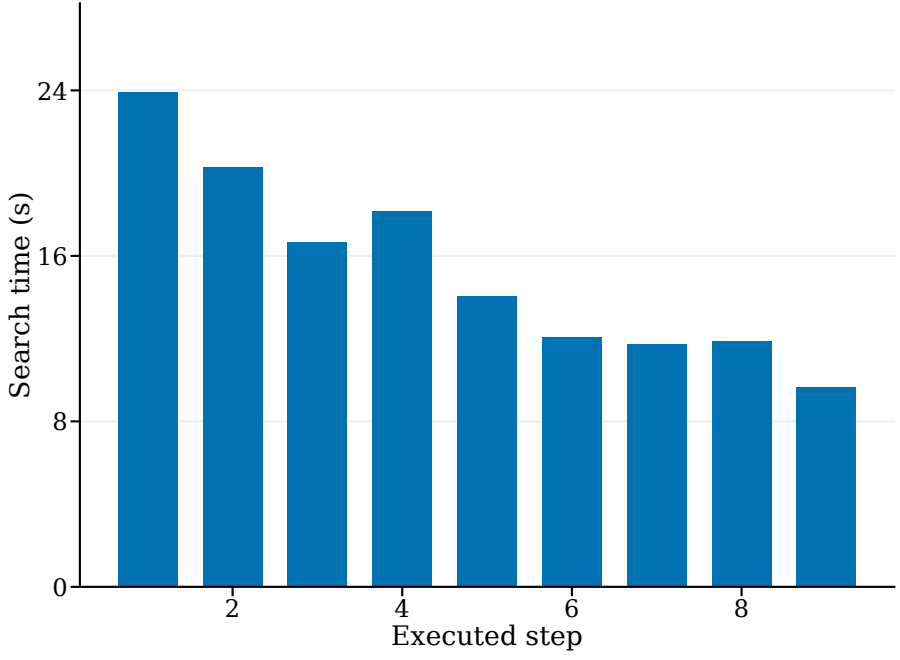
random 03 N4

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

random 03 N4 | success, 9 steps
N=4, grid=8x6x3, |O|=2, alpha=0.05



Planning time: mean 15.37 s; max 23.90 s
Experiment: 2,000 simulations, relaxed deadline

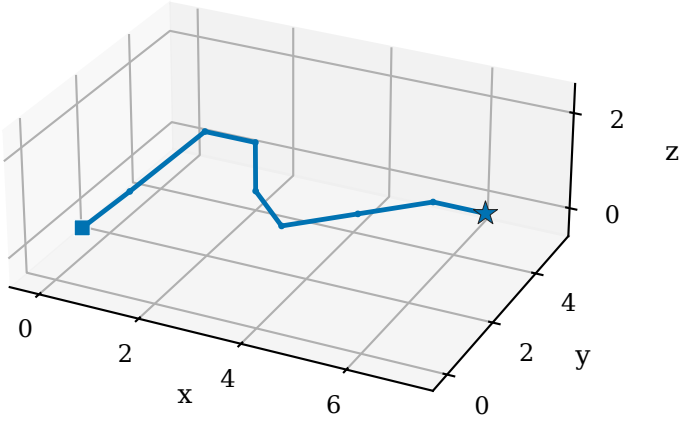


Saved full-solver trace, seed 0: results/trajectories/random_random_03_N4_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

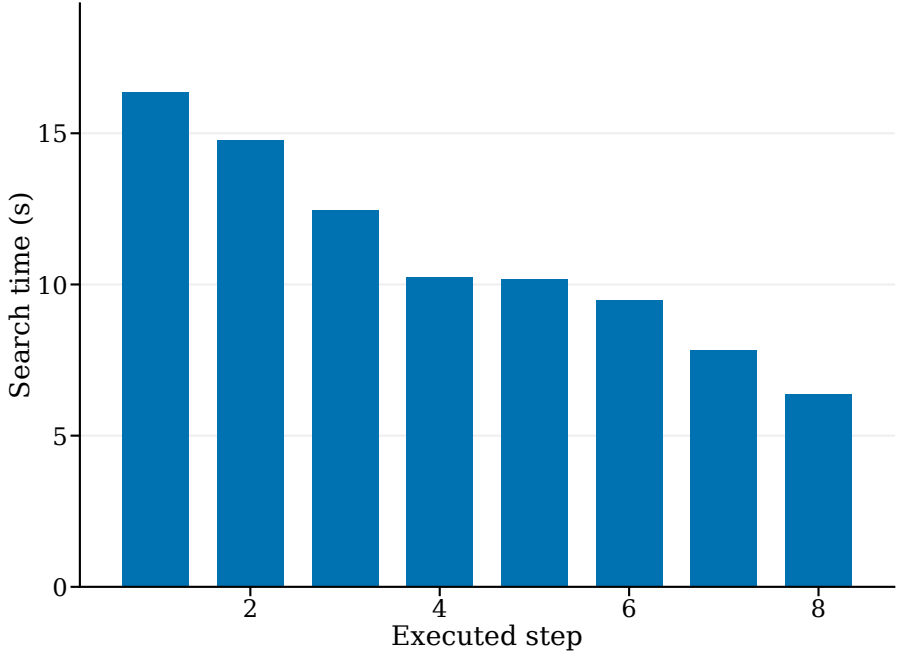
random 04 N1

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

random 04 N1 | success, 8 steps
N=1, grid=8x6x3, |O|=0, alpha=0.25



Planning time: mean 10.95 s; max 16.35 s
Experiment: 2,000 simulations, relaxed deadline

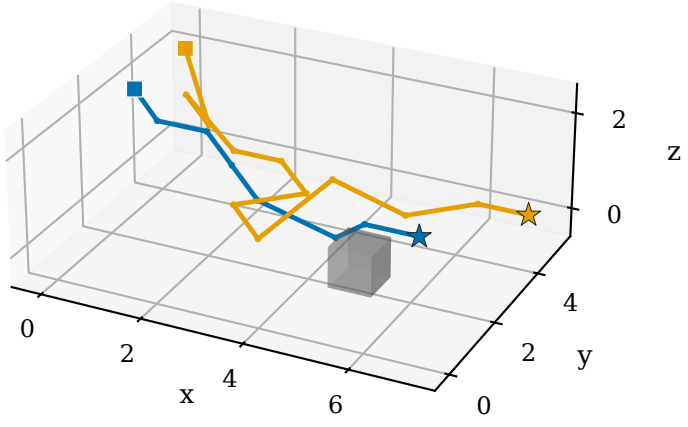


Saved full-solver trace, seed 0: results/trajectories/random_random_04_N1_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

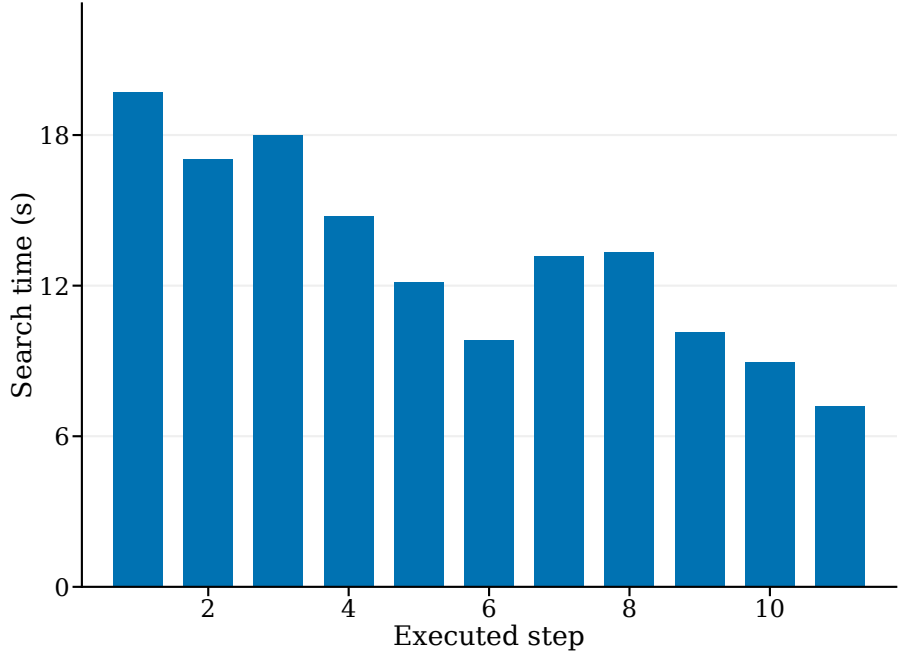
random 05 N2

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

random 05 N2 | success, 11 steps
N=2, grid=8x6x3, |O|=1, alpha=0.25



Planning time: mean 13.11 s; max 19.70 s
Experiment: 2,000 simulations, relaxed deadline

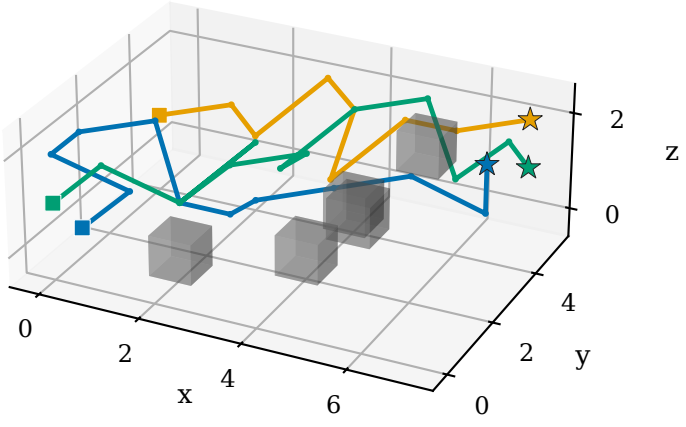


Saved full-solver trace, seed 0: results/trajectories/random_random_05_N2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

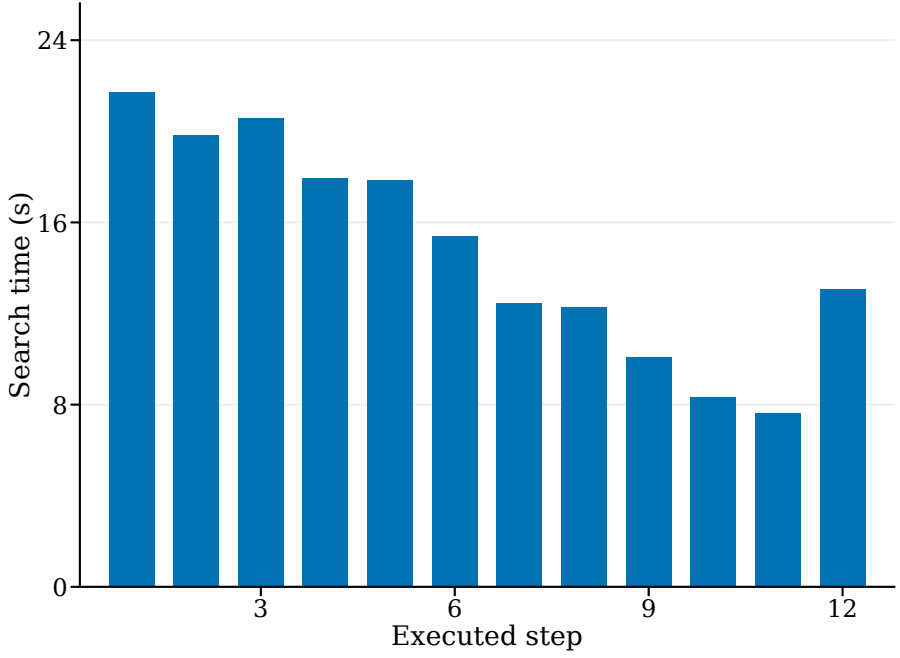
random 06 N3

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

random 06 N3 | success, 12 steps
N=3, grid=8x6x3, |O|=5, alpha=0.25



Planning time: mean 14.75 s; max 21.71 s
Experiment: 2,000 simulations, relaxed deadline

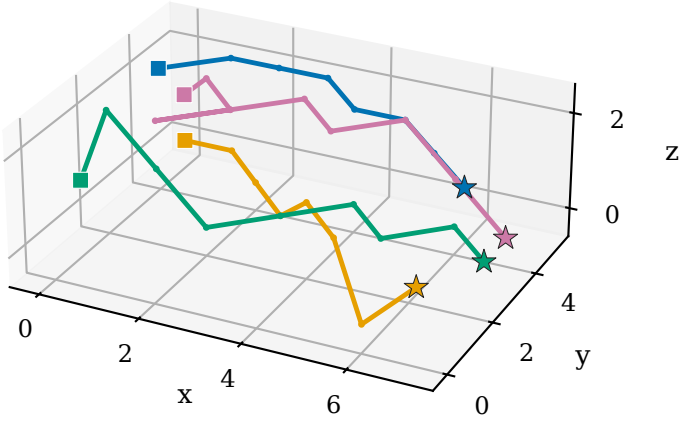


Saved full-solver trace, seed 0: results/trajectories/random_random_06_N3_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

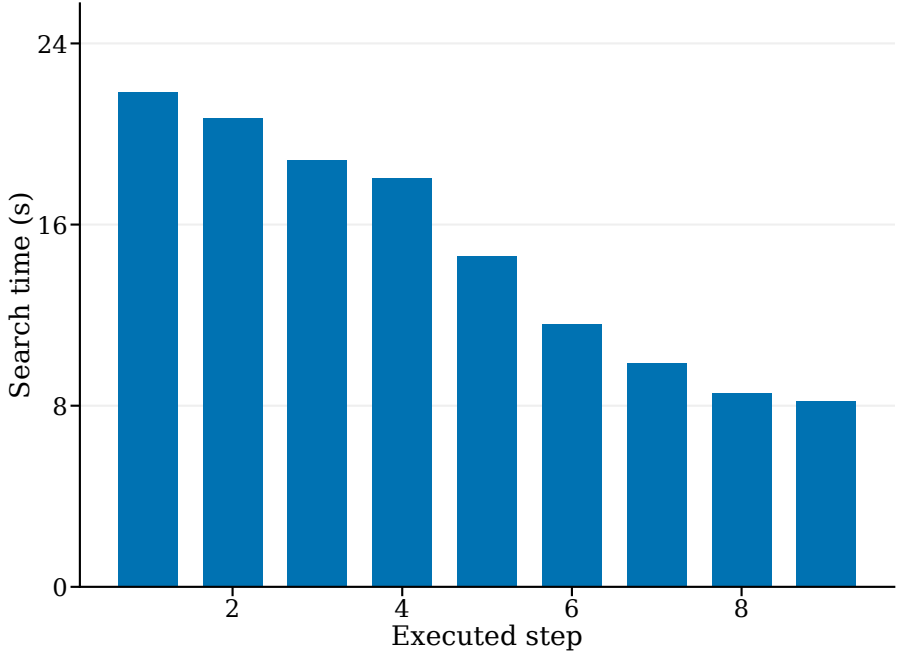
random 07 N4

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

random 07 N4 | success, 9 steps
N=4, grid=8x6x3, |O|=0, alpha=0.25

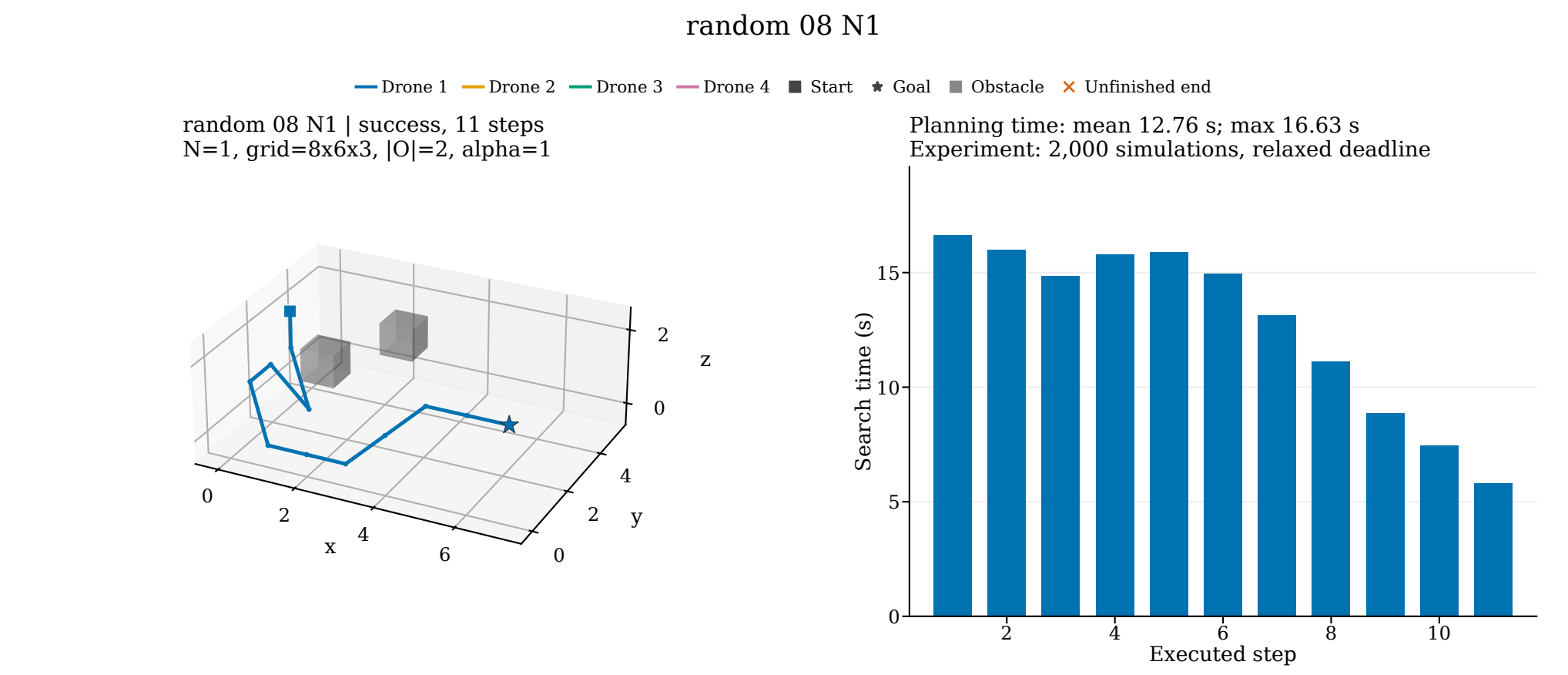


Planning time: mean 14.69 s; max 21.84 s
Experiment: 2,000 simulations, relaxed deadline



Saved full-solver trace, seed 0: results/trajectories/random_random_07_N4_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

random 08 N1

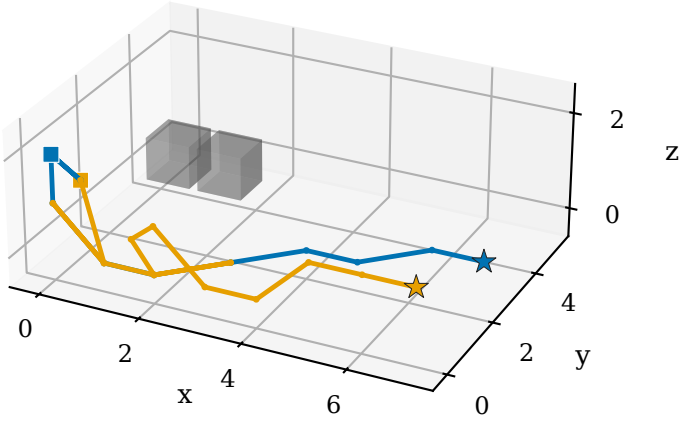


Saved full-solver trace, seed 0: results/trajectories/random_random_08_N1_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

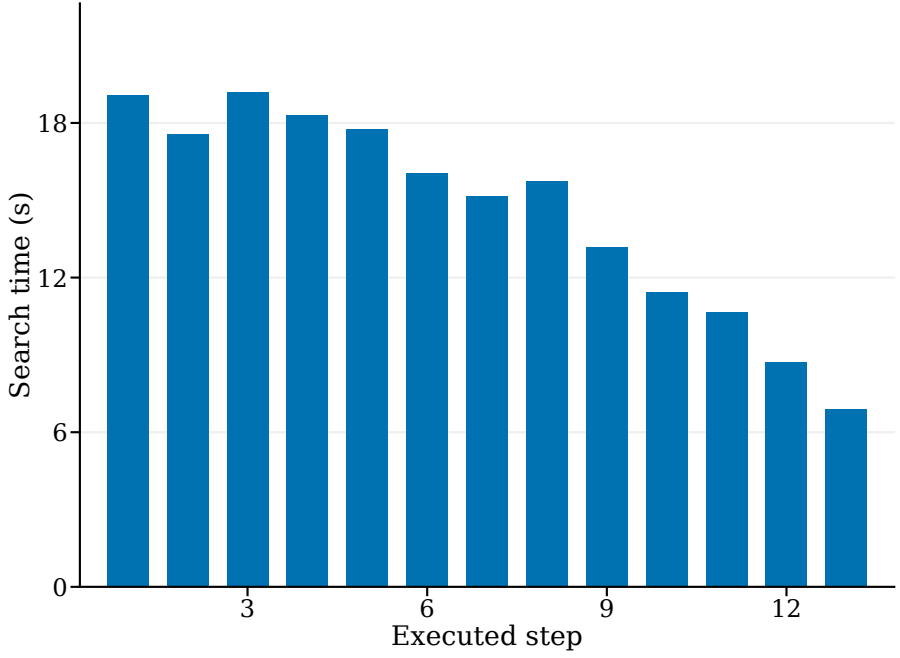
random 09 N2

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

random 09 N2 | success, 13 steps
N=2, grid=8x6x3, |O|=2, alpha=1



Planning time: mean 14.59 s; max 19.18 s
Experiment: 2,000 simulations, relaxed deadline

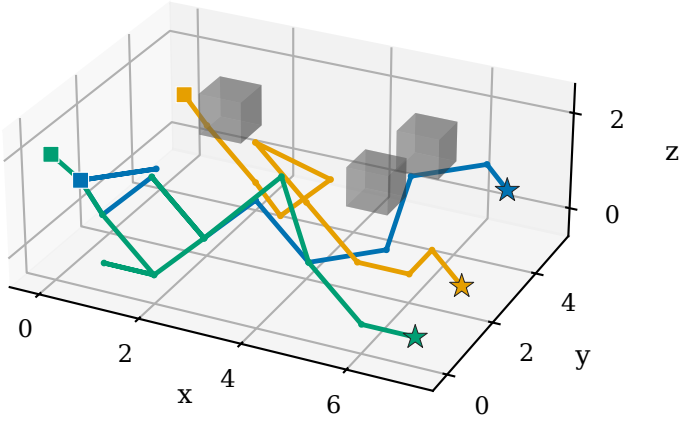


Saved full-solver trace, seed 0: results/trajectories/random_random_09_N2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

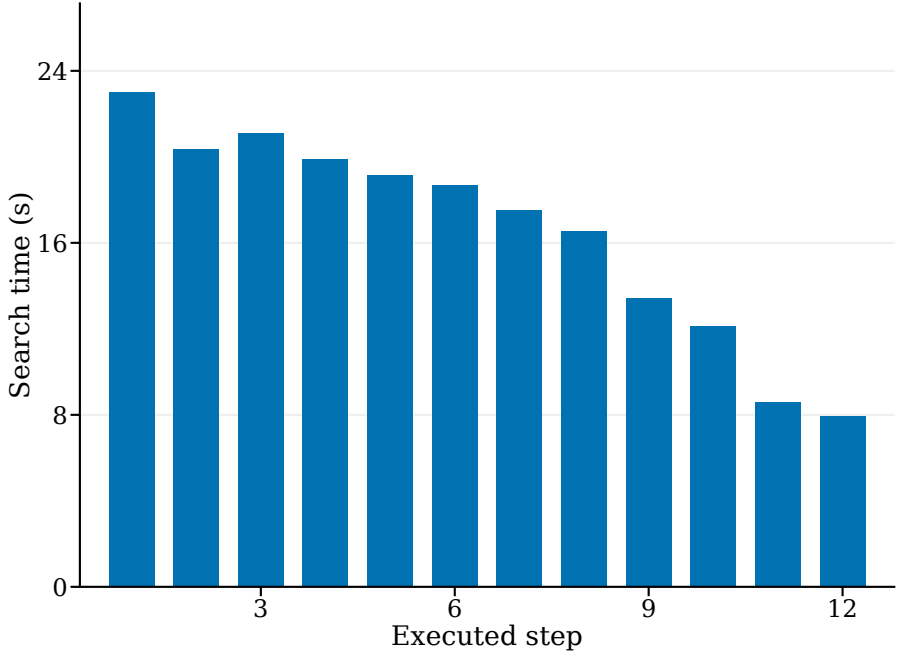
random 10 N3

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

random 10 N3 | success, 12 steps
N=3, grid=8x6x3, |O|=3, alpha=1



Planning time: mean 16.52 s; max 23.00 s
Experiment: 2,000 simulations, relaxed deadline

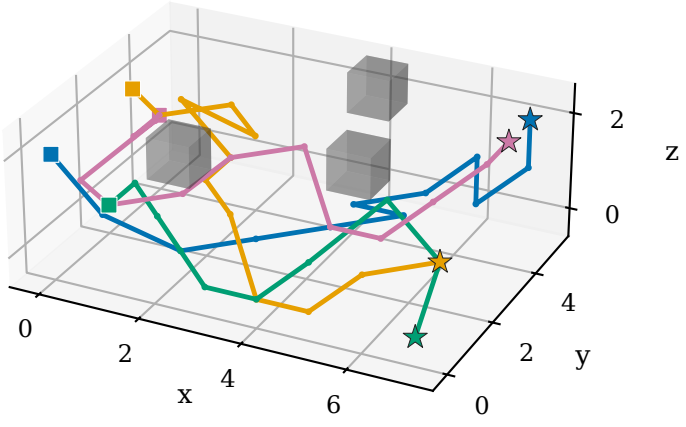


Saved full-solver trace, seed 0: results/trajectories/random_random_10_N3_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

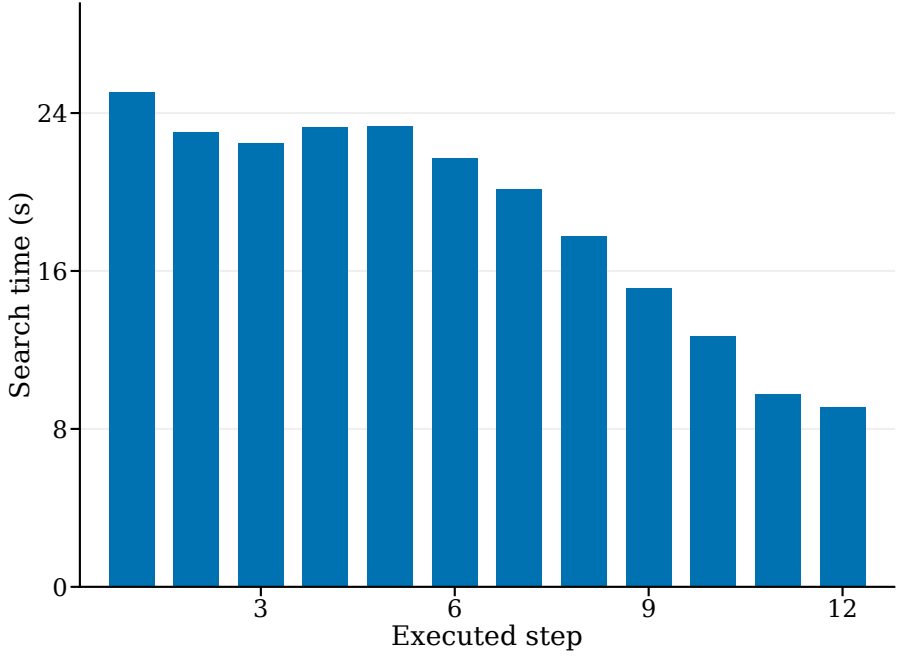
random 11 N4

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

random 11 N4 | success, 12 steps
N=4, grid=8x6x3, |O|=3, alpha=1



Planning time: mean 18.61 s; max 25.04 s
Experiment: 2,000 simulations, relaxed deadline

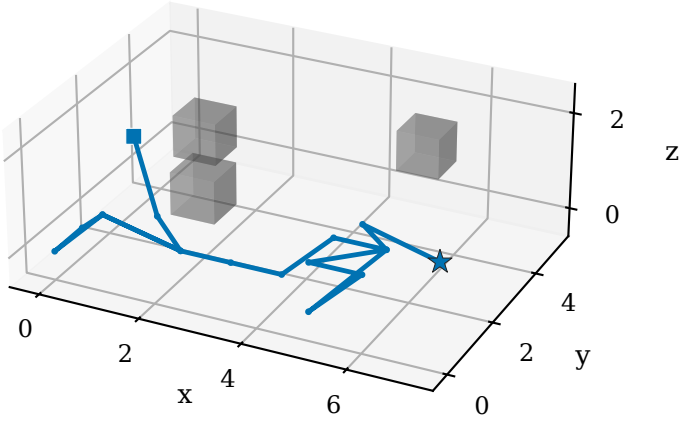


Saved full-solver trace, seed 0: results/trajectories/random_random_11_N4_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

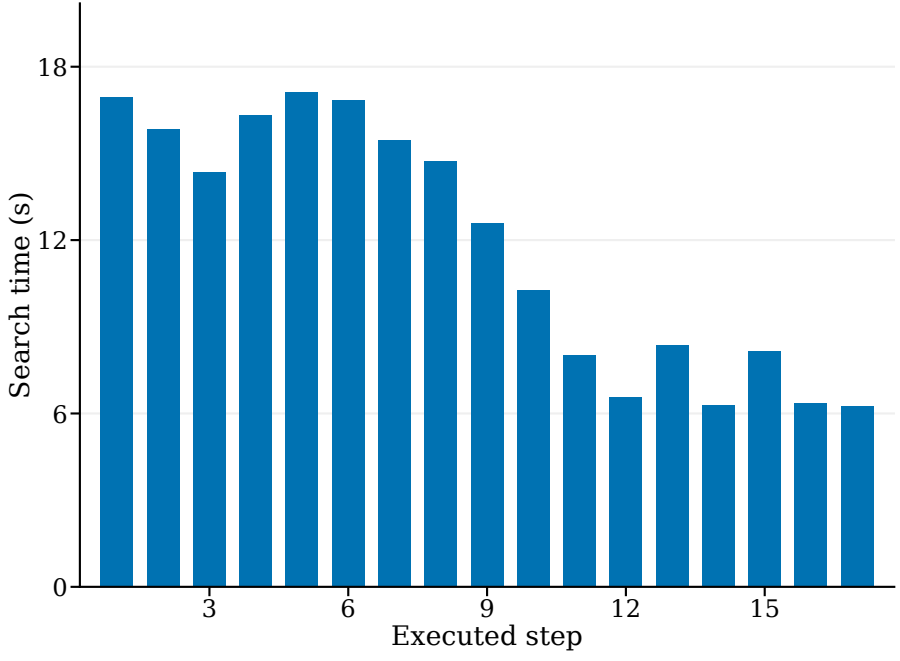
random 12 N1



random 12 N1 | success, 17 steps
N=1, grid=8x6x3, |O|=3, alpha=2



Planning time: mean 11.78 s; max 17.11 s
Experiment: 2,000 simulations, relaxed deadline

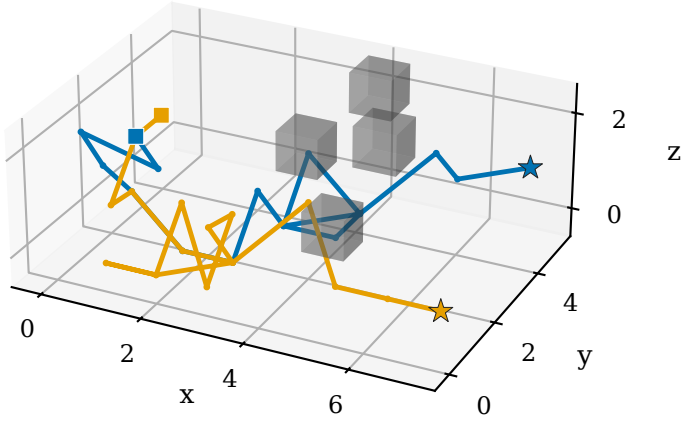


Saved full-solver trace, seed 0: results/trajectories/random_random_12_N1_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

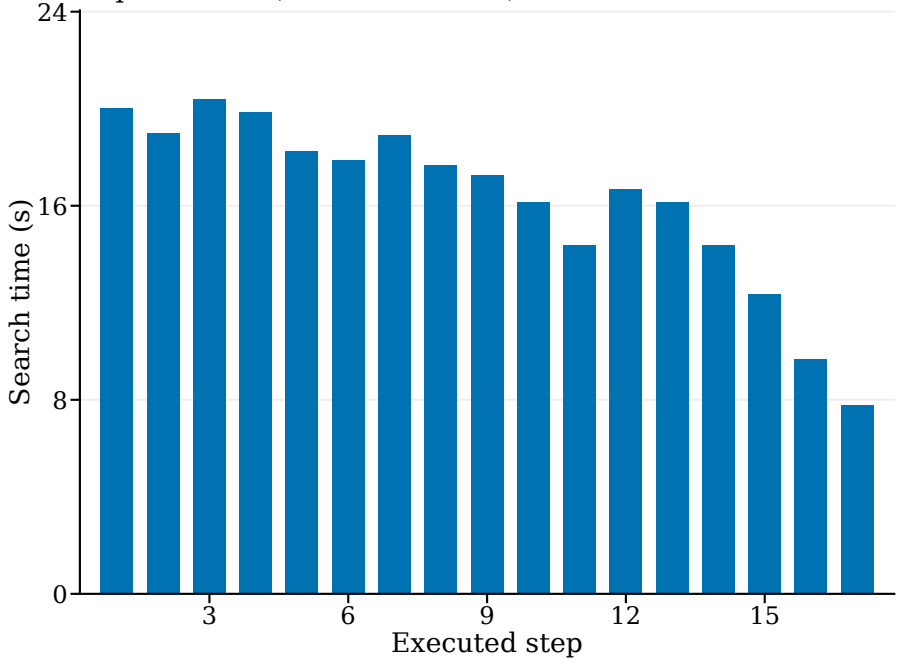
random 13 N2

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

random 13 N2 | success, 17 steps
N=2, grid=8x6x3, |O|=4, alpha=2



Planning time: mean 16.27 s; max 20.38 s
Experiment: 2,000 simulations, relaxed deadline

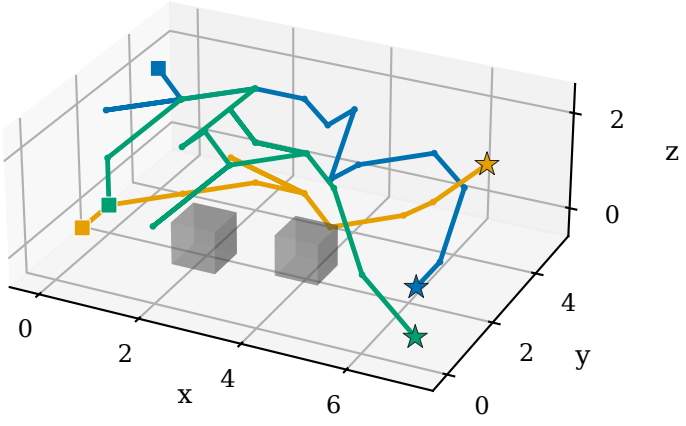


Saved full-solver trace, seed 0: results/trajectories/random_random_13_N2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

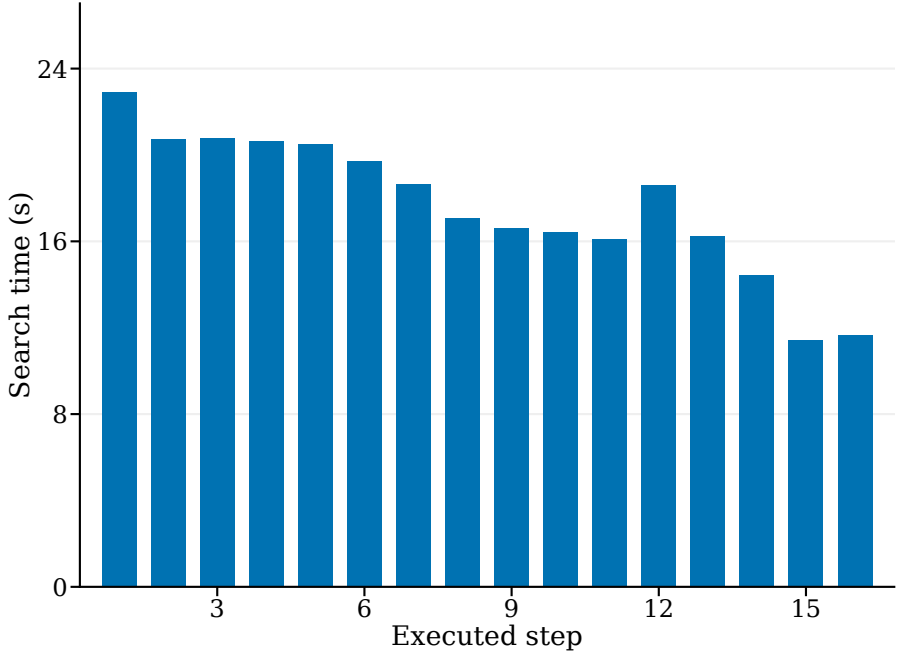
random 14 N3

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

random 14 N3 | success, 16 steps
N=3, grid=8x6x3, |O|=2, alpha=2



Planning time: mean 17.65 s; max 22.90 s
Experiment: 2,000 simulations, relaxed deadline

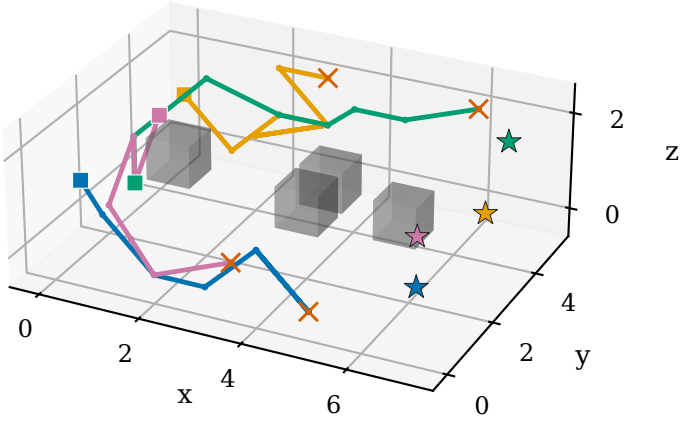


Saved full-solver trace, seed 0: results/trajectories/random_random_14_N3_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

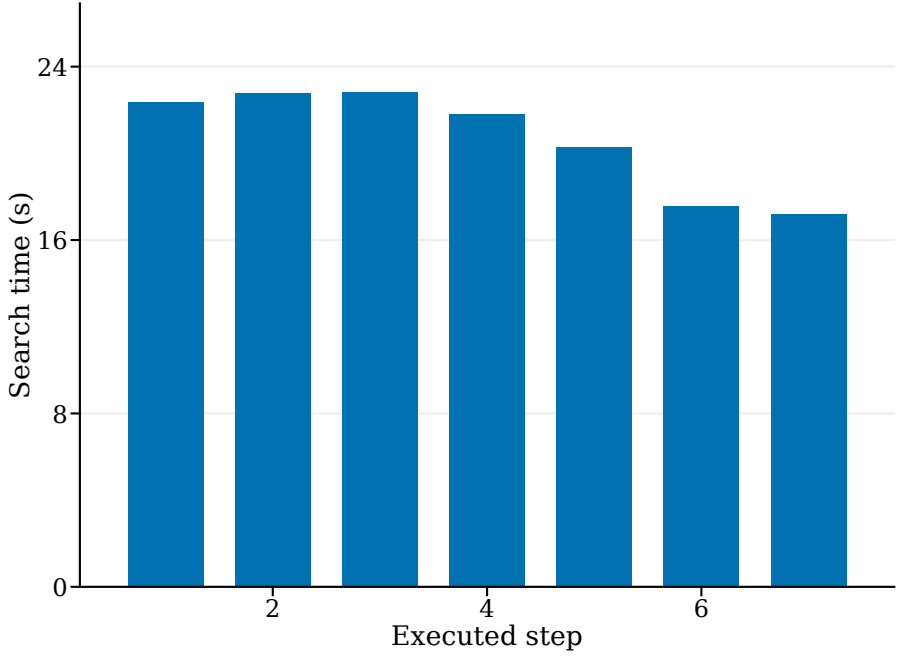
random 15 N4



random 15 N4 | collision, 7 steps
N=4, grid=8x6x3, |O|=4, alpha=2

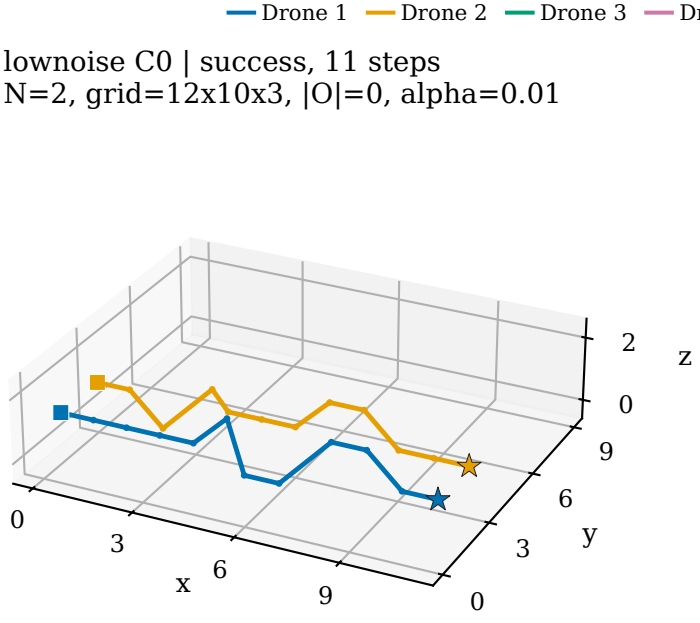


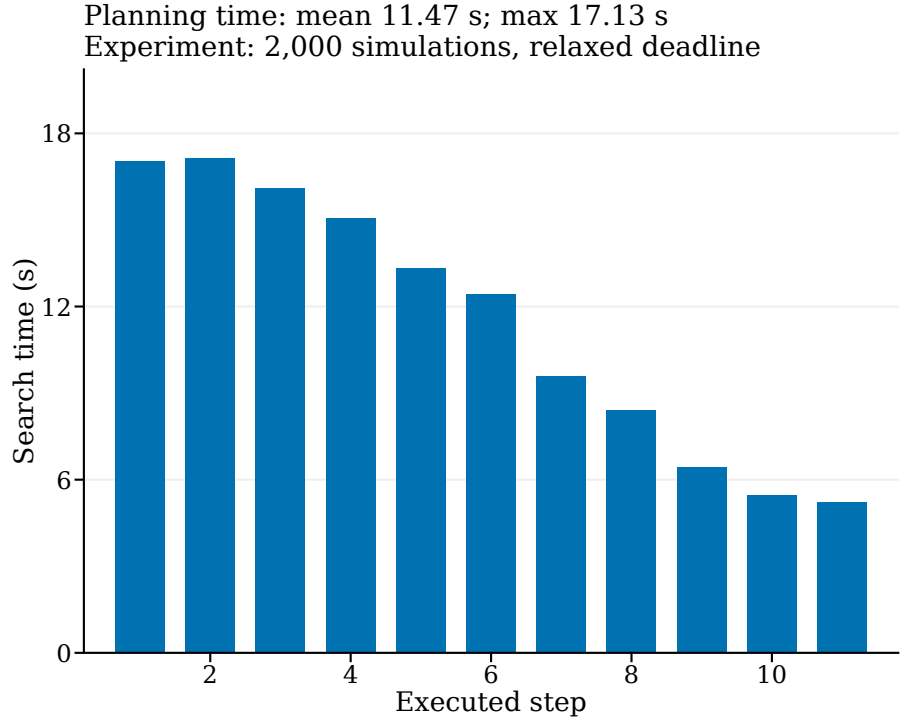
Planning time: mean 20.68 s; max 22.81 s
Experiment: 2,000 simulations, relaxed deadline



Saved full-solver trace, seed 0: results/trajectories/random_random_15_N4_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

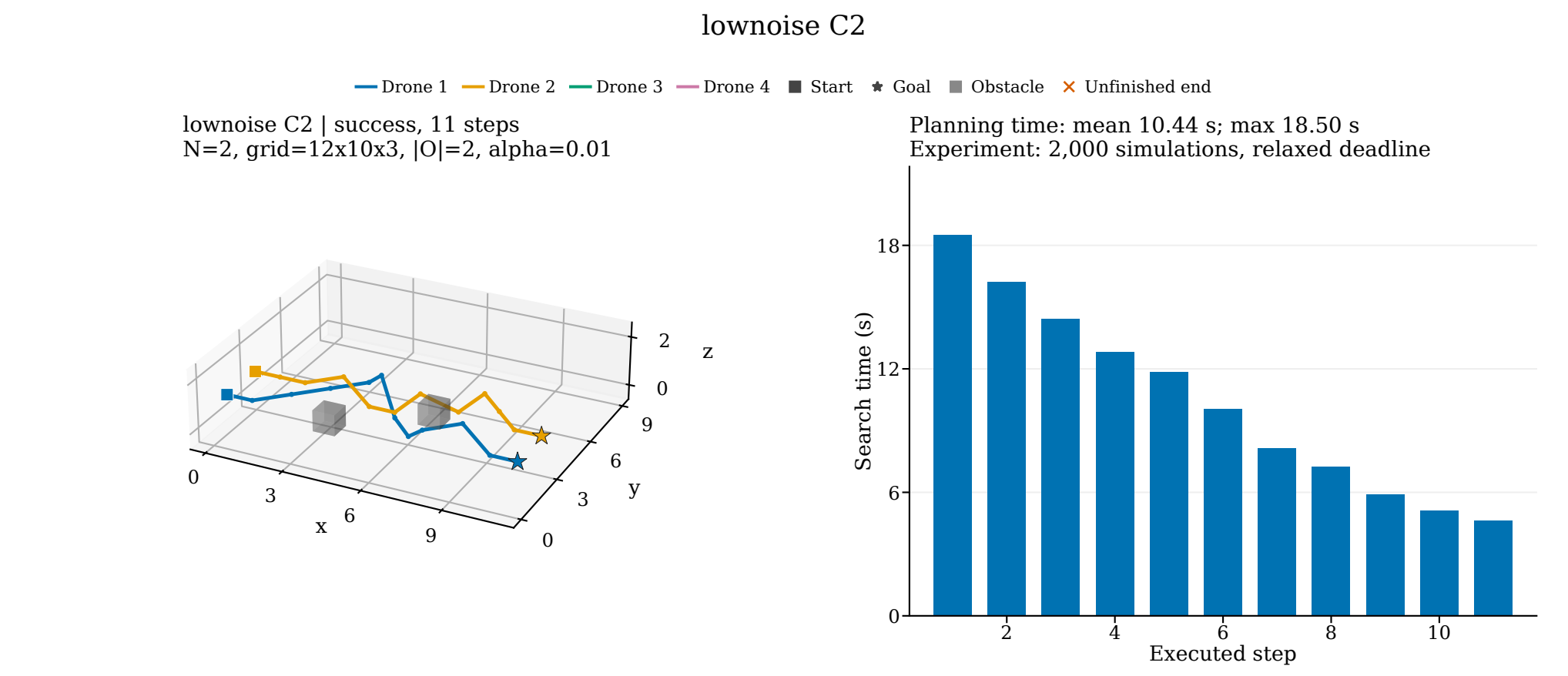
lownoise C0





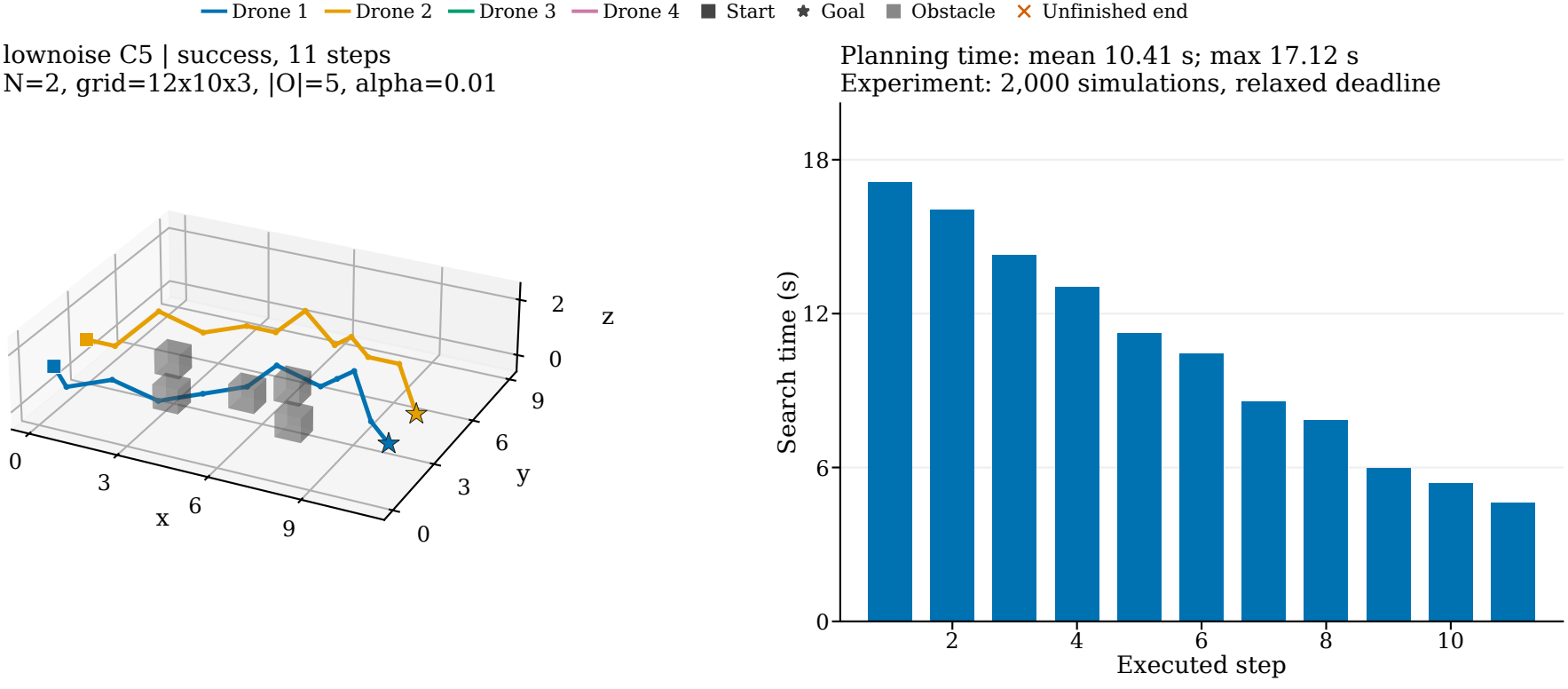
Saved full-solver trace, seed 0: results/trajectories/control_lownoise_C0_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

lownoise C2



Saved full-solver trace, seed 0: results/trajectories/control_lownoise_C2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

lownoise C5

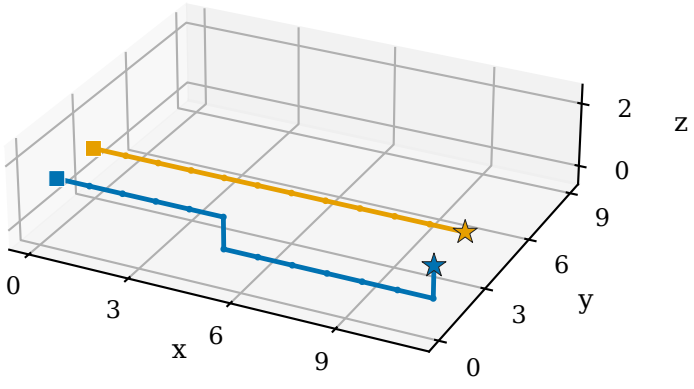


Saved full-solver trace, seed 0: results/trajectories/control_lownoise_C5_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

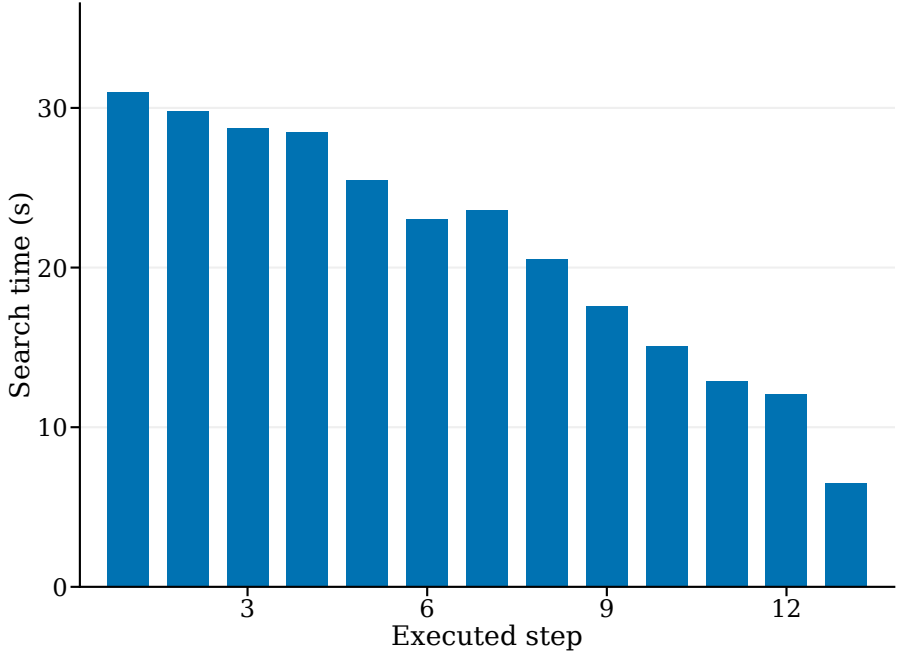
lownoise A6 C0

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

lownoise A6 C0 | success, 13 steps
N=2, grid=12x10x3, |O|=0, alpha=0.01

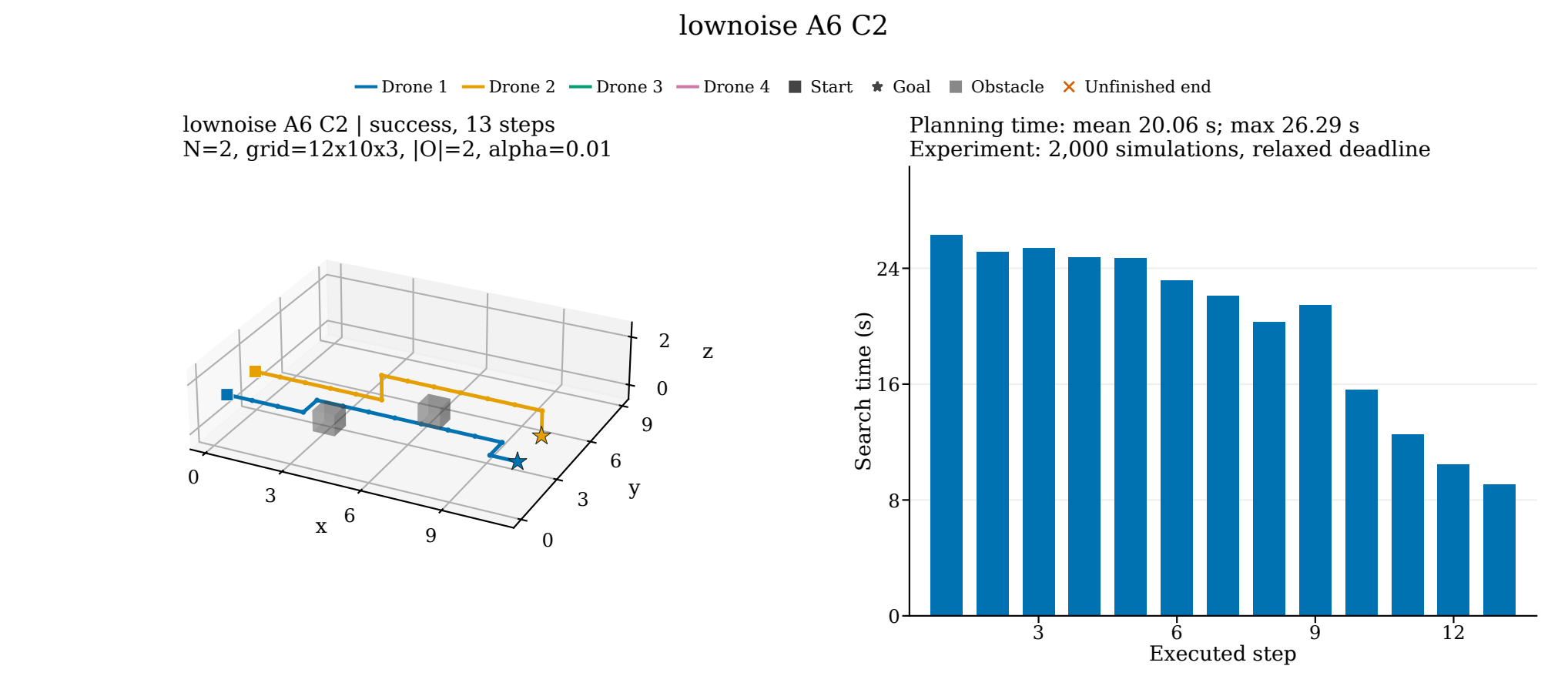


Planning time: mean 21.11 s; max 30.97 s
Experiment: 2,000 simulations, relaxed deadline



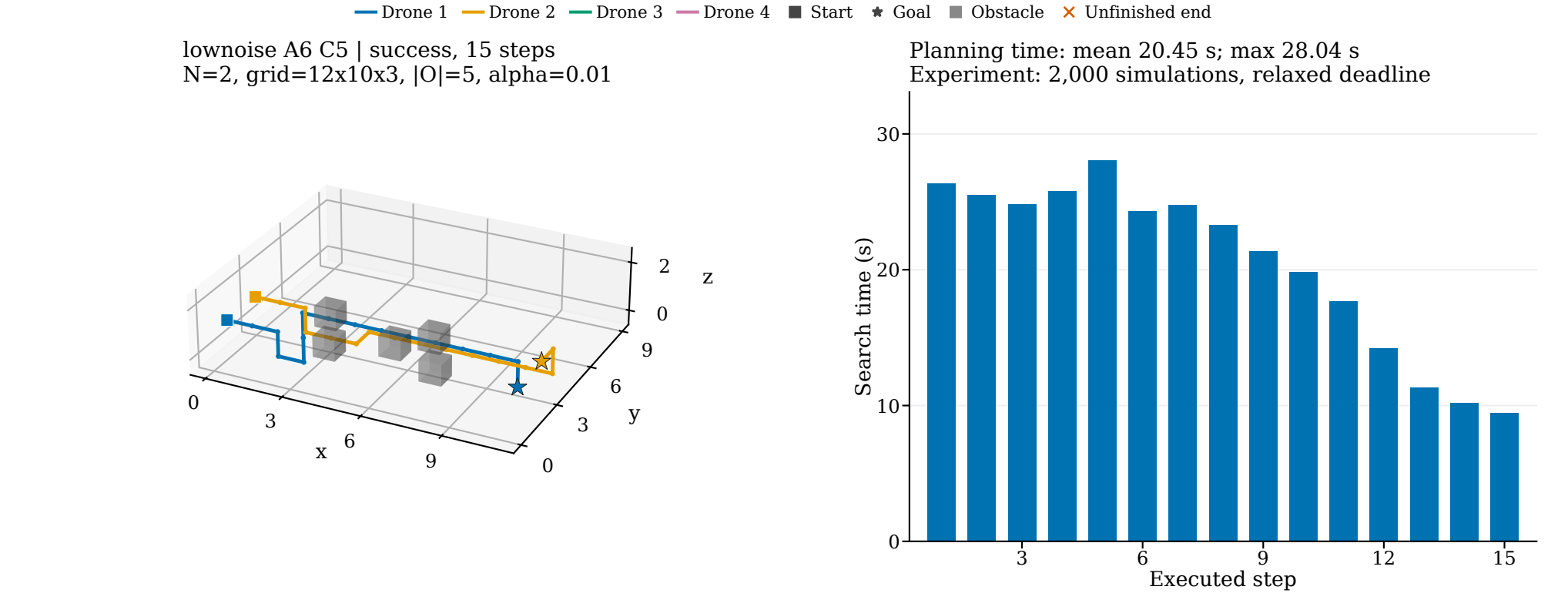
Saved full-solver trace, seed 0: results/trajectories/control_lownoise_A6_C0_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

lownoise A6 C2



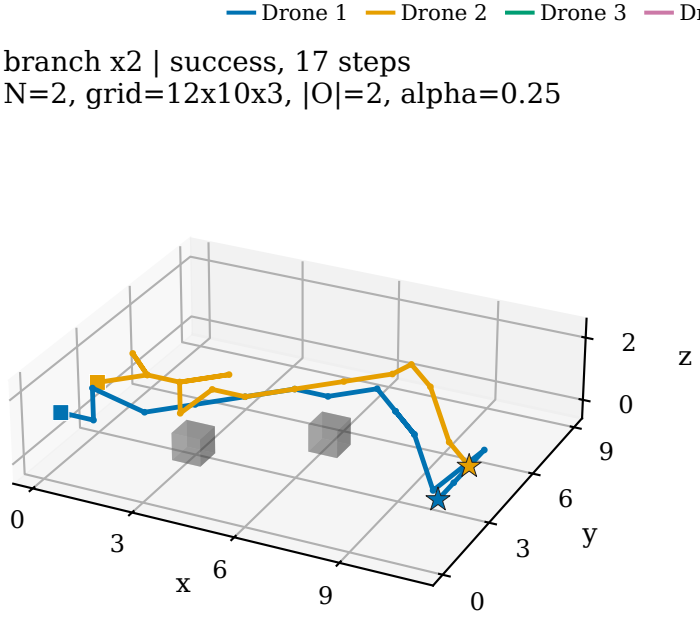
Saved full-solver trace, seed 0: results/trajectories/control_lownoise_A6_C2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

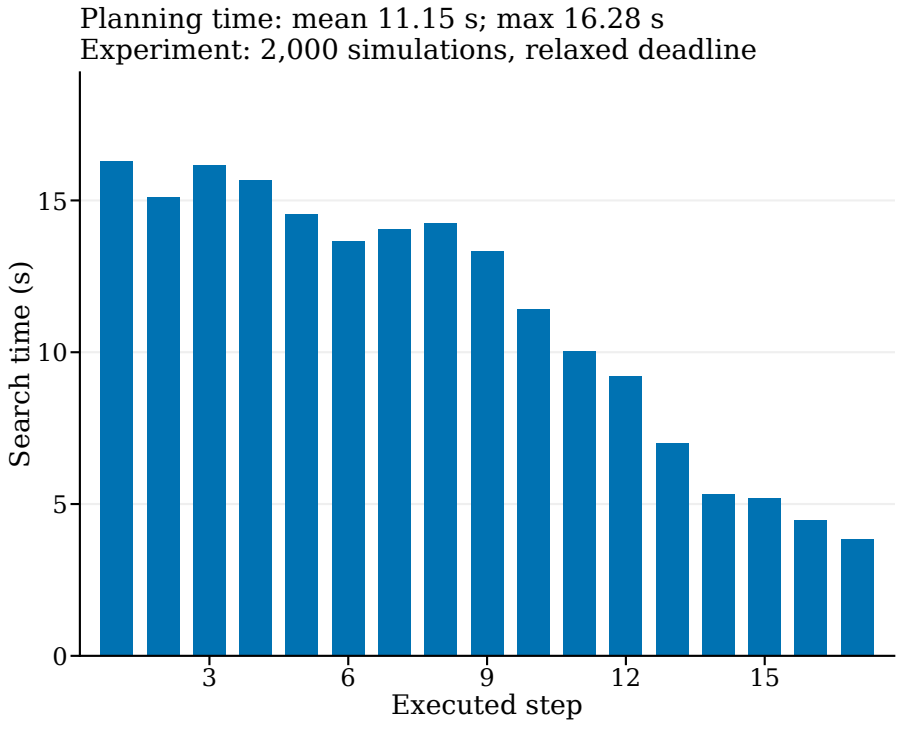
lownoise A6 C5



Saved full-solver trace, seed 0: results/trajectories/control_lownoise_A6_C5_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

branch x2



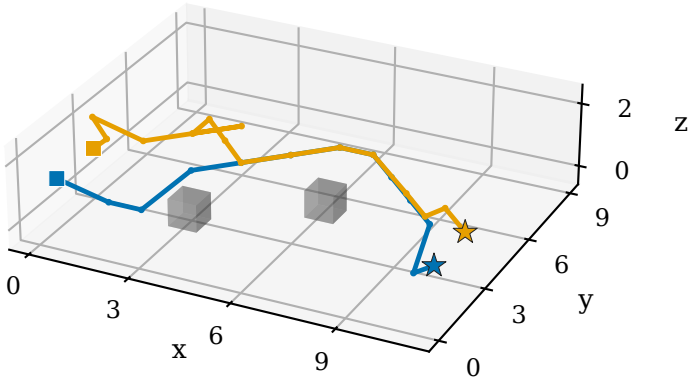


Saved full-solver trace, seed 0: results/trajectories/control_branch_x2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

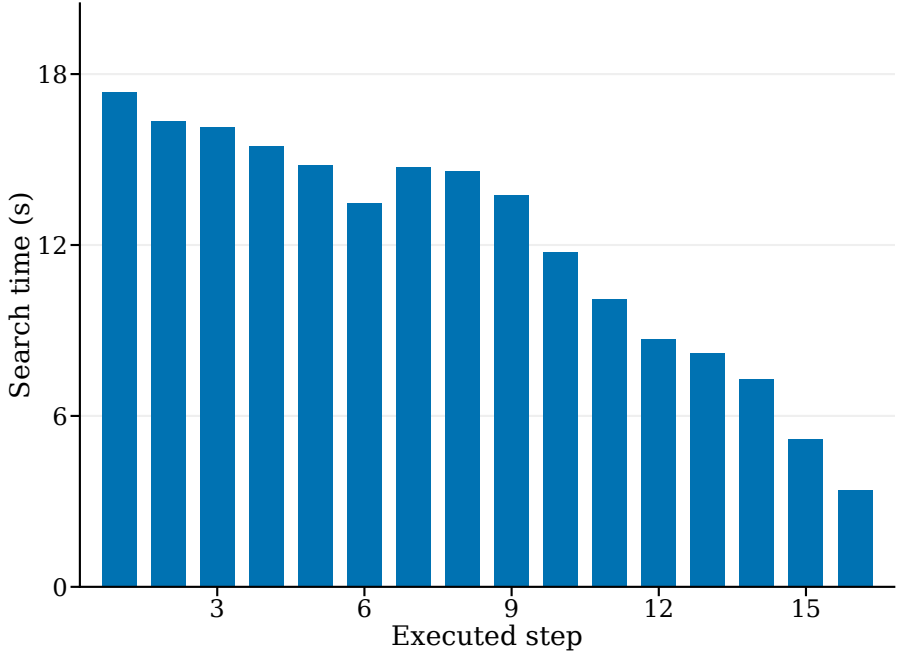
branch x4



branch x4 | success, 16 steps
N=2, grid=12x10x3, |O|=2, alpha=0.25

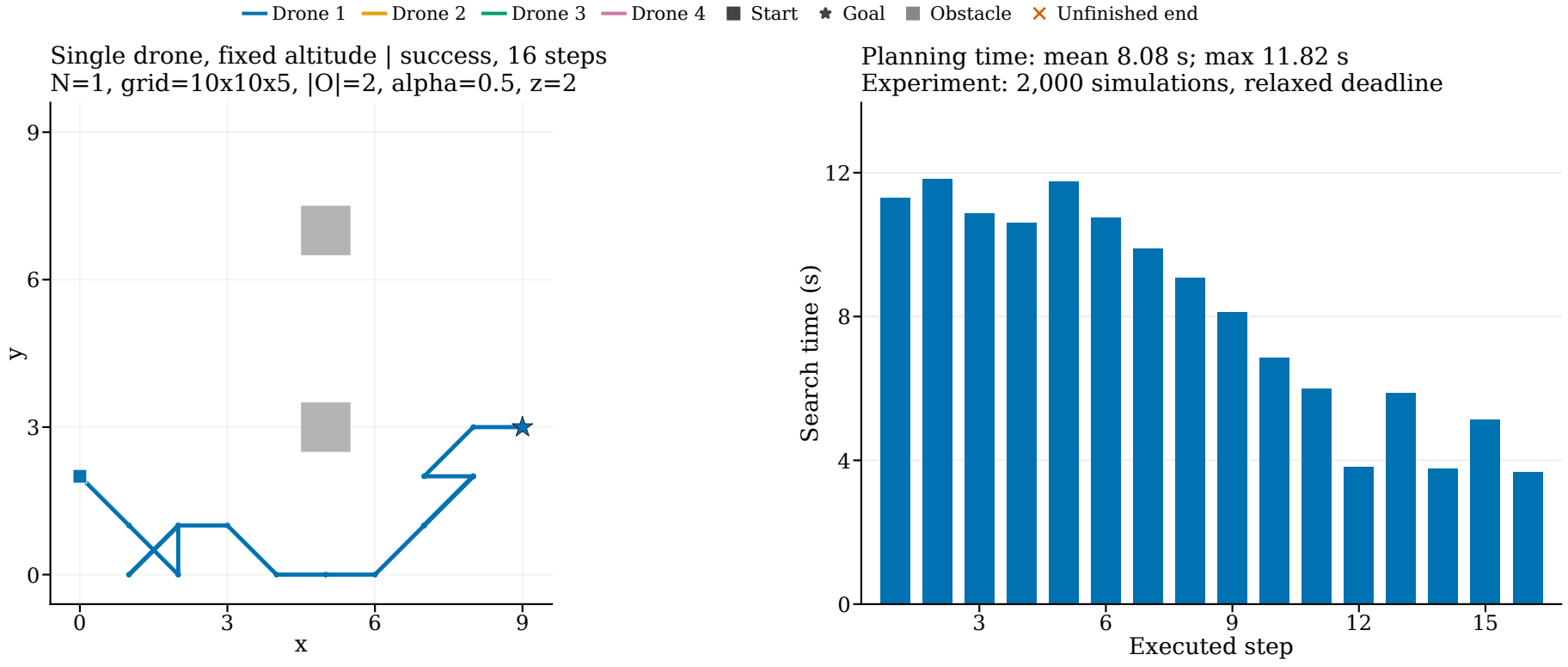


Planning time: mean 11.95 s; max 17.36 s
Experiment: 2,000 simulations, relaxed deadline



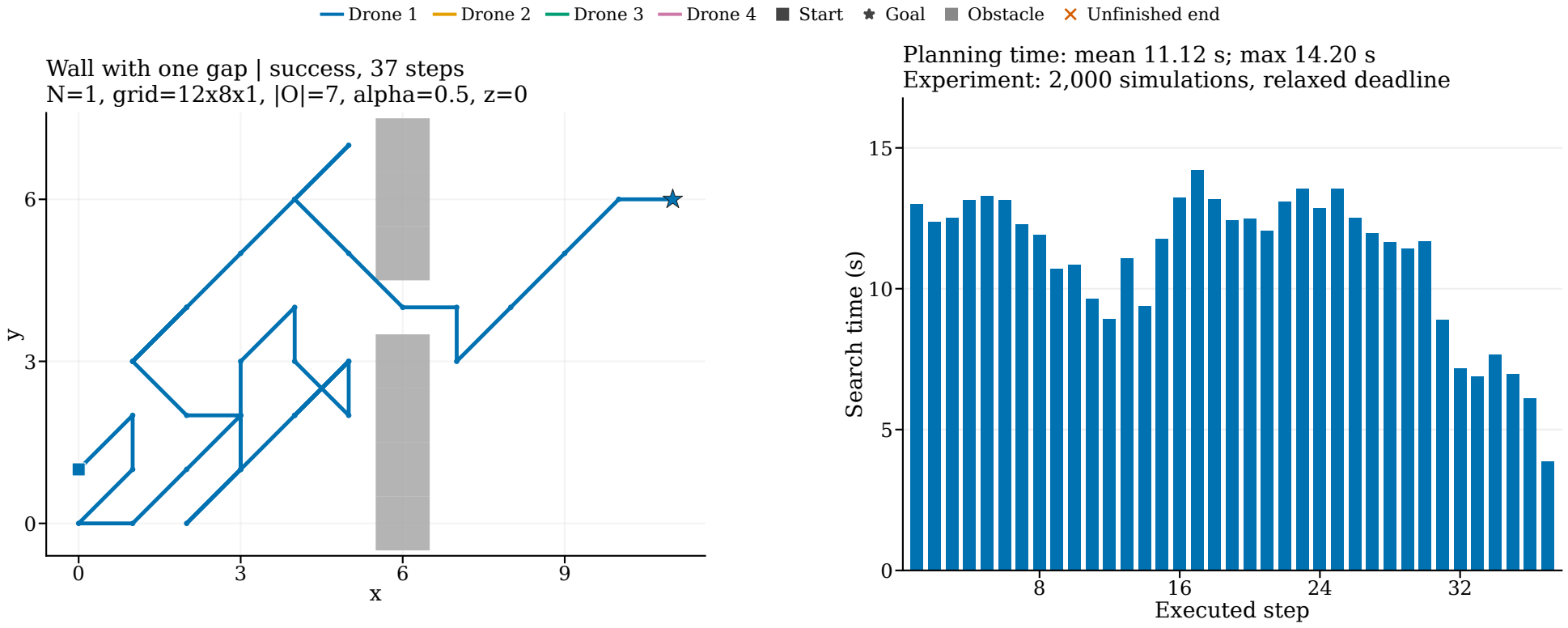
Saved full-solver trace, seed 0: results/trajectories/control_branch_x4_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

supp single example 2d



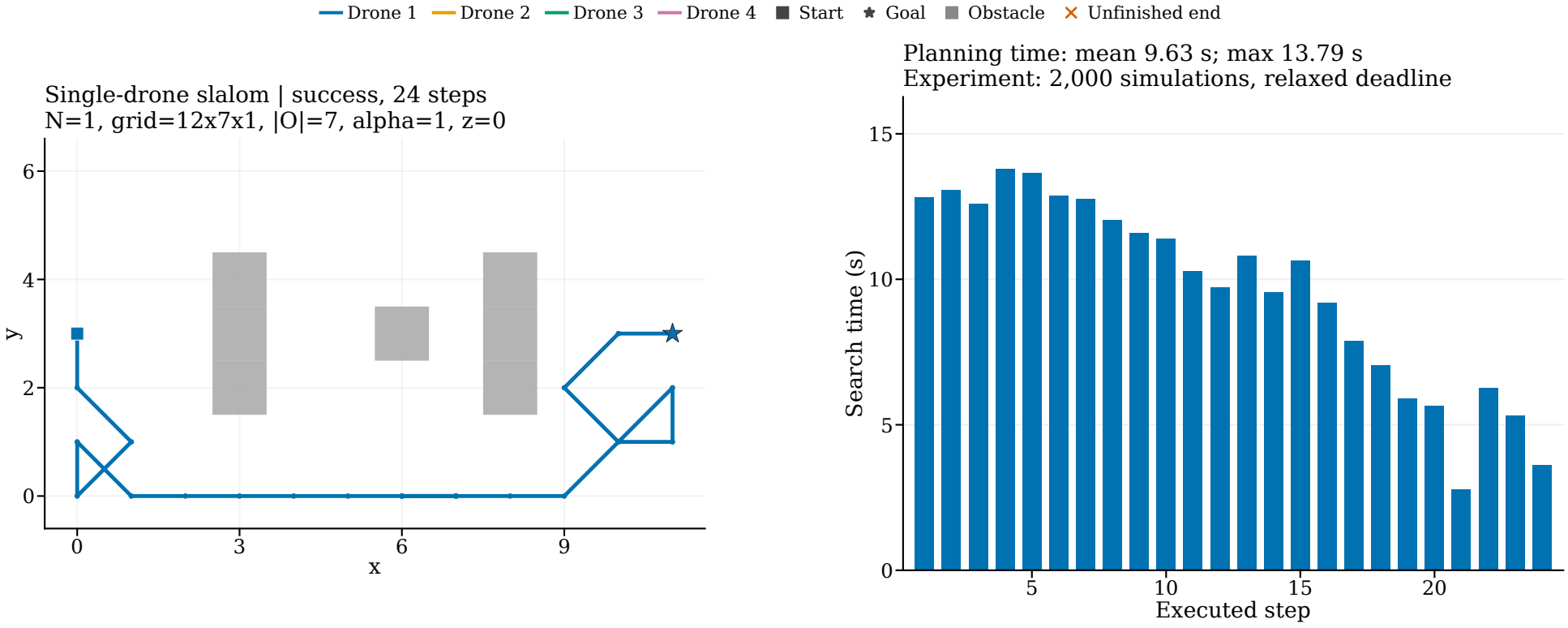
Saved full-solver trace, seed 0: results/trajectories/supplement_supp_single_example_2d_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

supp single wallgap 2d



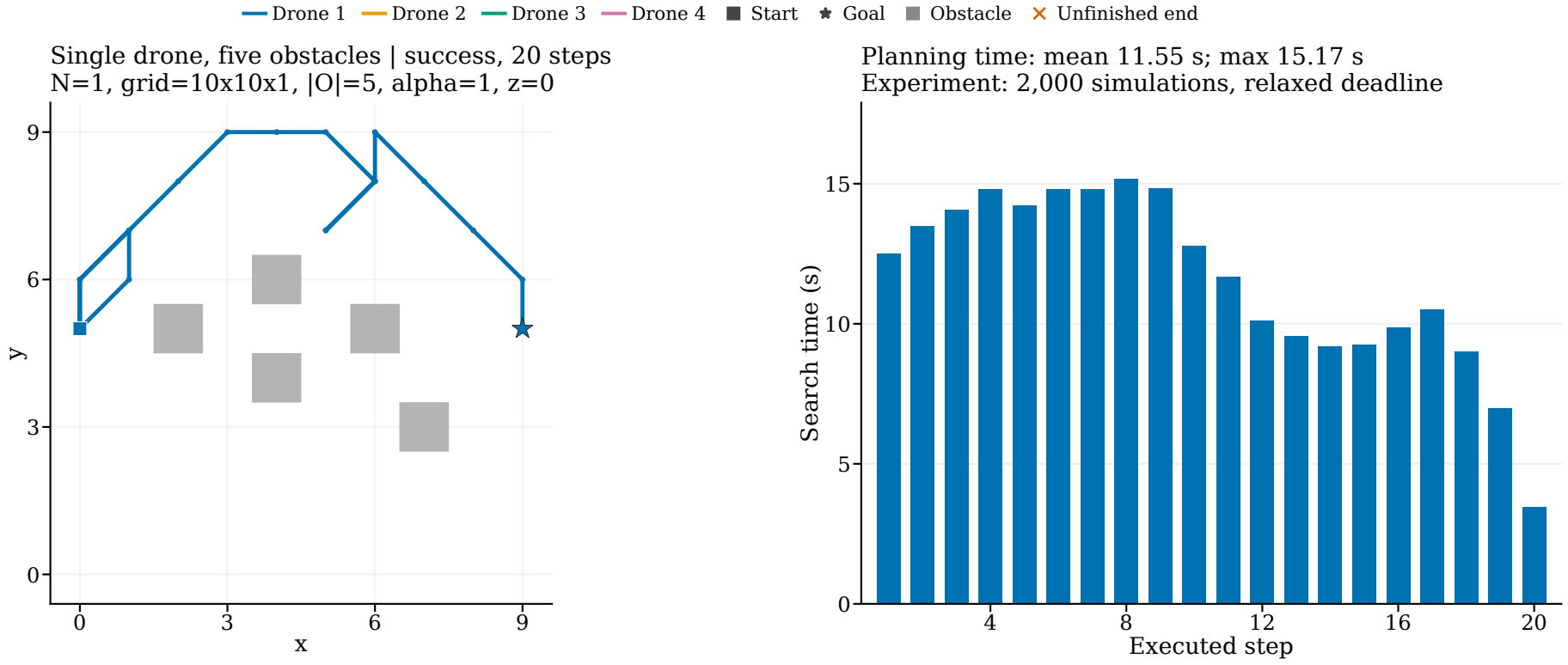
Saved full-solver trace, seed 0: results/trajectories/supplement_supp_single_wallgap_2d_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

supp single slalom 2d



Saved full-solver trace, seed 0: results/trajectories/supplement_supp_single_slalom_2d_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

supp single clutter 2d

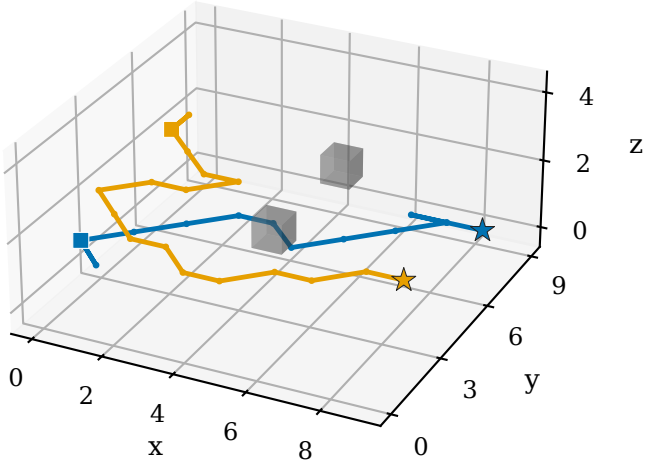


Saved full-solver trace, seed 0: results/trajectories/supplement_supp_single_clutter_2d_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

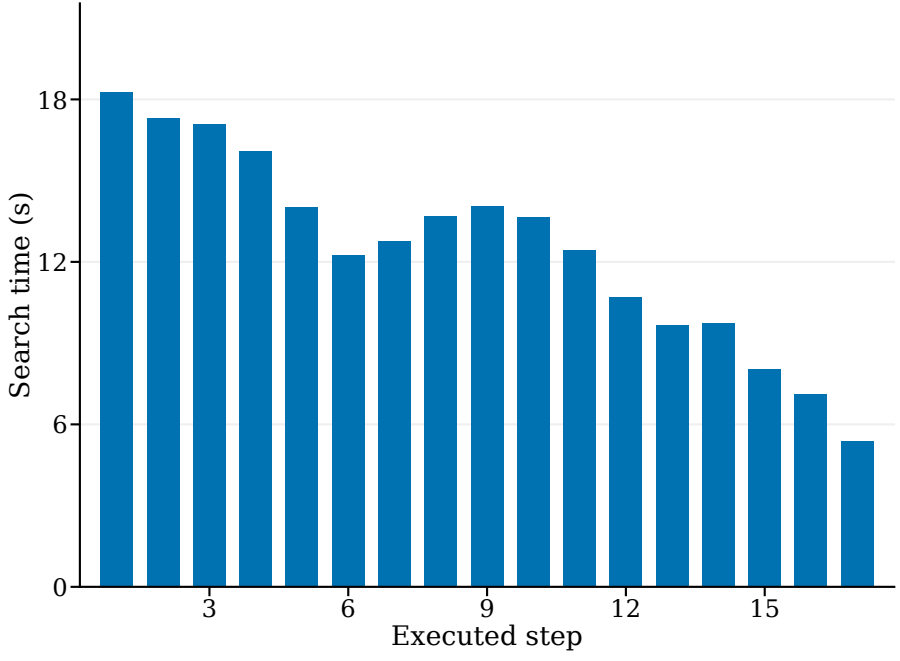
supp layers cross 2d

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

Crossing on separate layers | success, 17 steps
N=2, grid=10x10x5, |O|=2, alpha=0.5

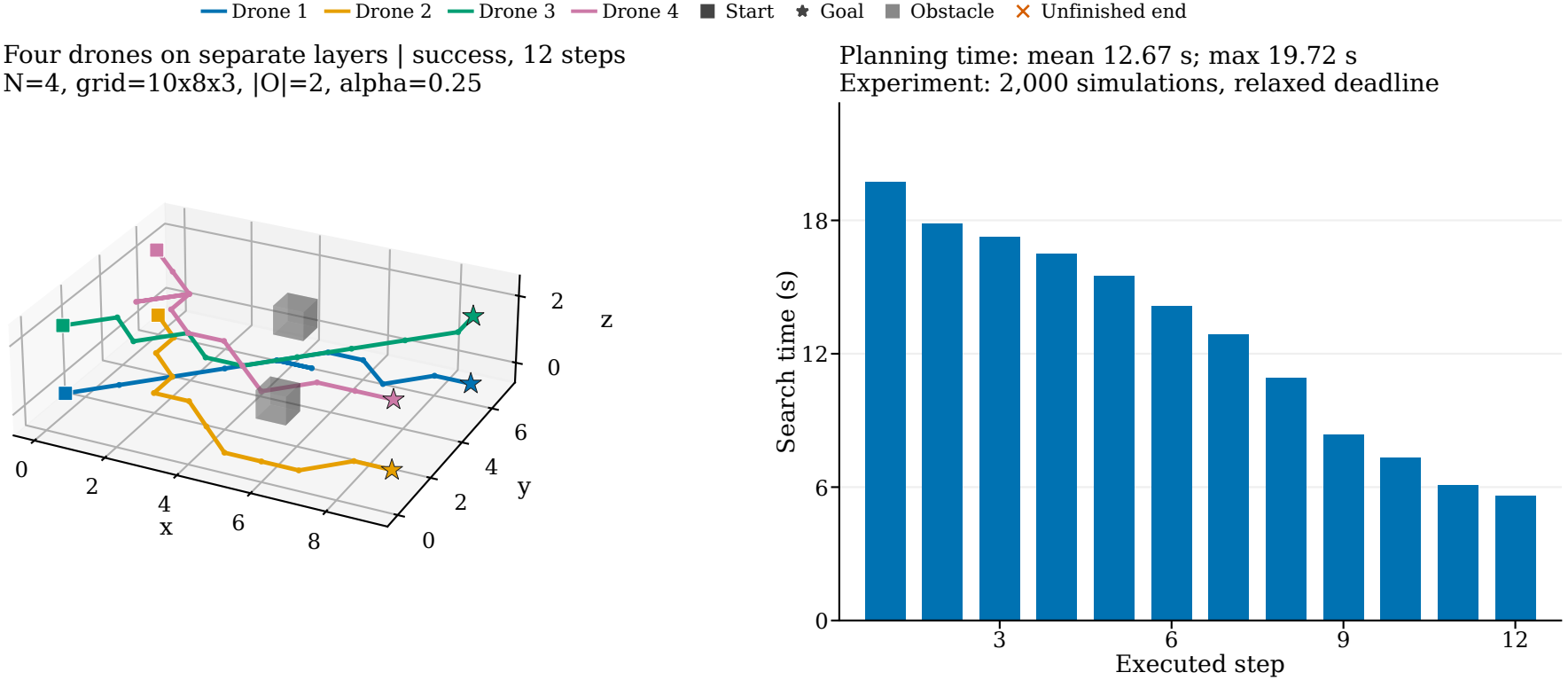


Planning time: mean 12.48 s; max 18.26 s
Experiment: 2,000 simulations, relaxed deadline



Saved full-solver trace, seed 0: results/trajectories/supplement_supp_layers_cross_2d_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

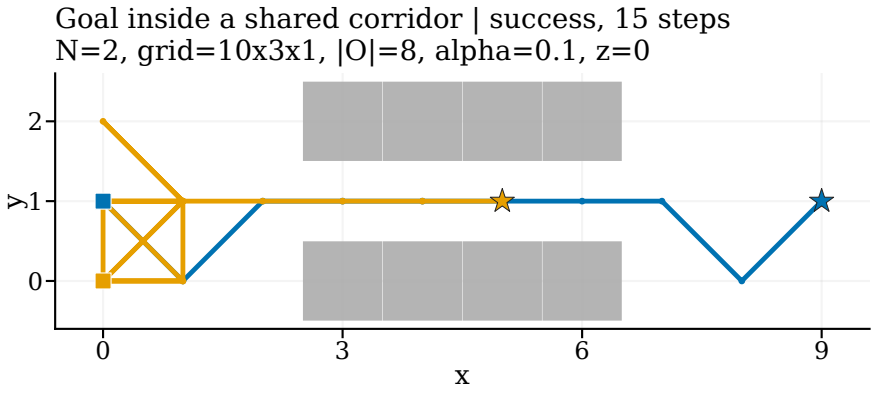
supp layers four 2d

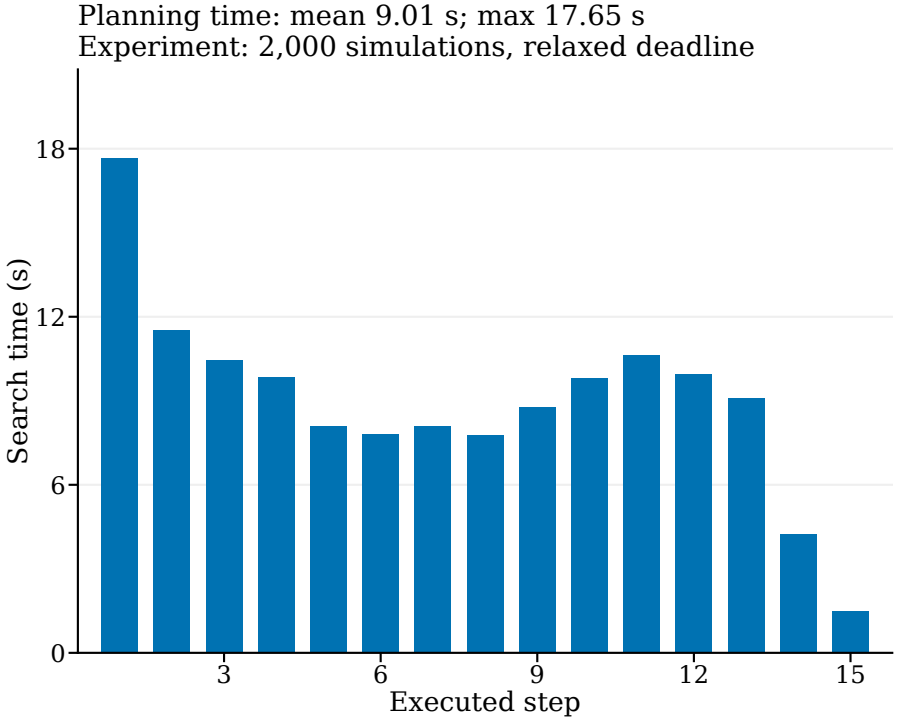


Saved full-solver trace, seed 0: results/trajectories/supplement_supp_layers_four_2d_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

supp yield corridor

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end





Saved full-solver trace, seed 0: results/trajectories/supplement_supp_yield_corridor_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.